

Coding with AI



**[https://github.com/up1/
course-ai-coding-2026](https://github.com/up1/course-ai-coding-2026)**



Coding with AI



Topics

Programming/Coding workflow
Spec-Driven Development (SDD)
Model-Driven Development (MDD)
Coding with AI Agent workshop

Requirement

Planning

Coding
Testing

Validate and Testing

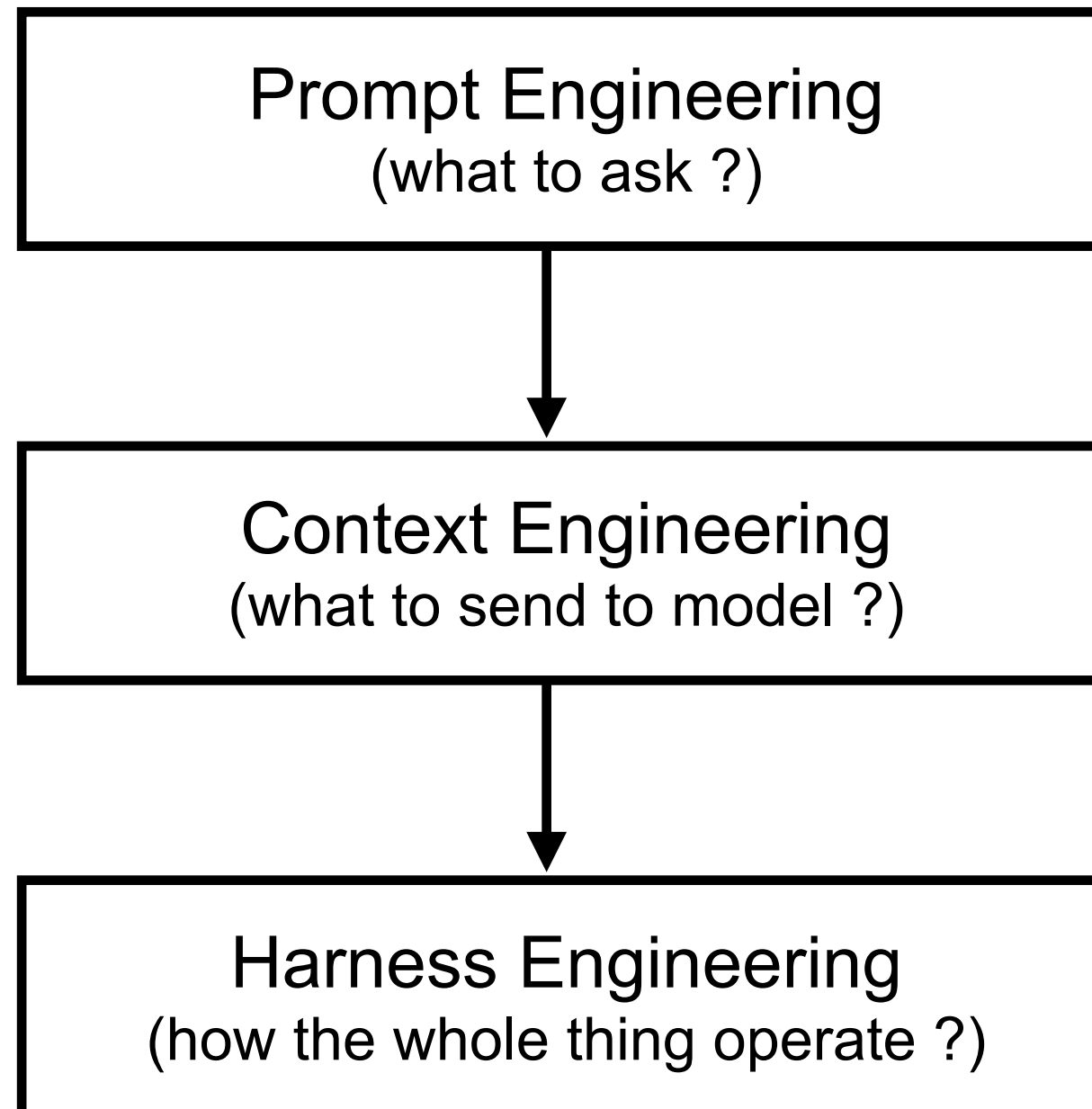


History of Programming



Increase of abstraction





2 Background: From Weights to Context to Harness

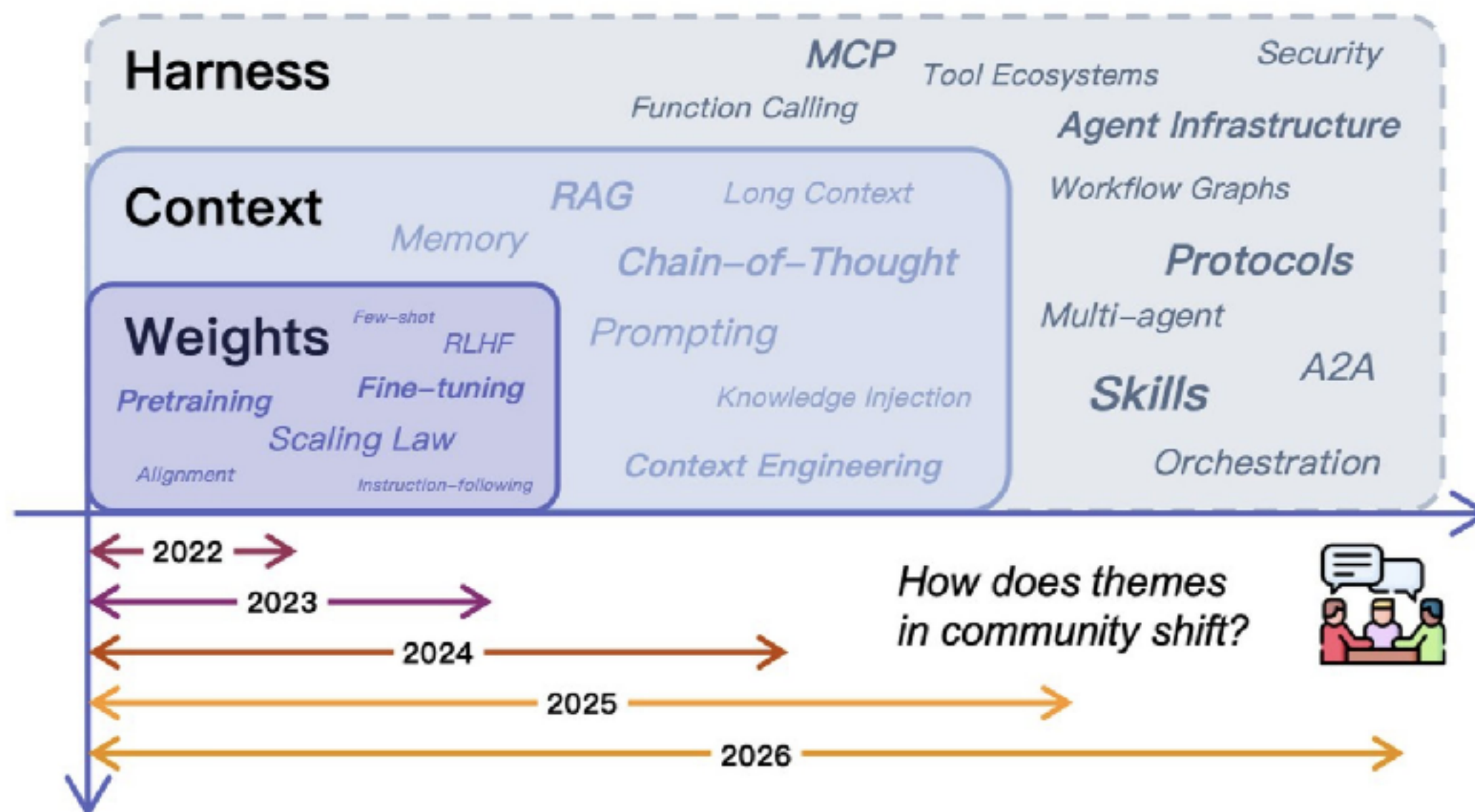
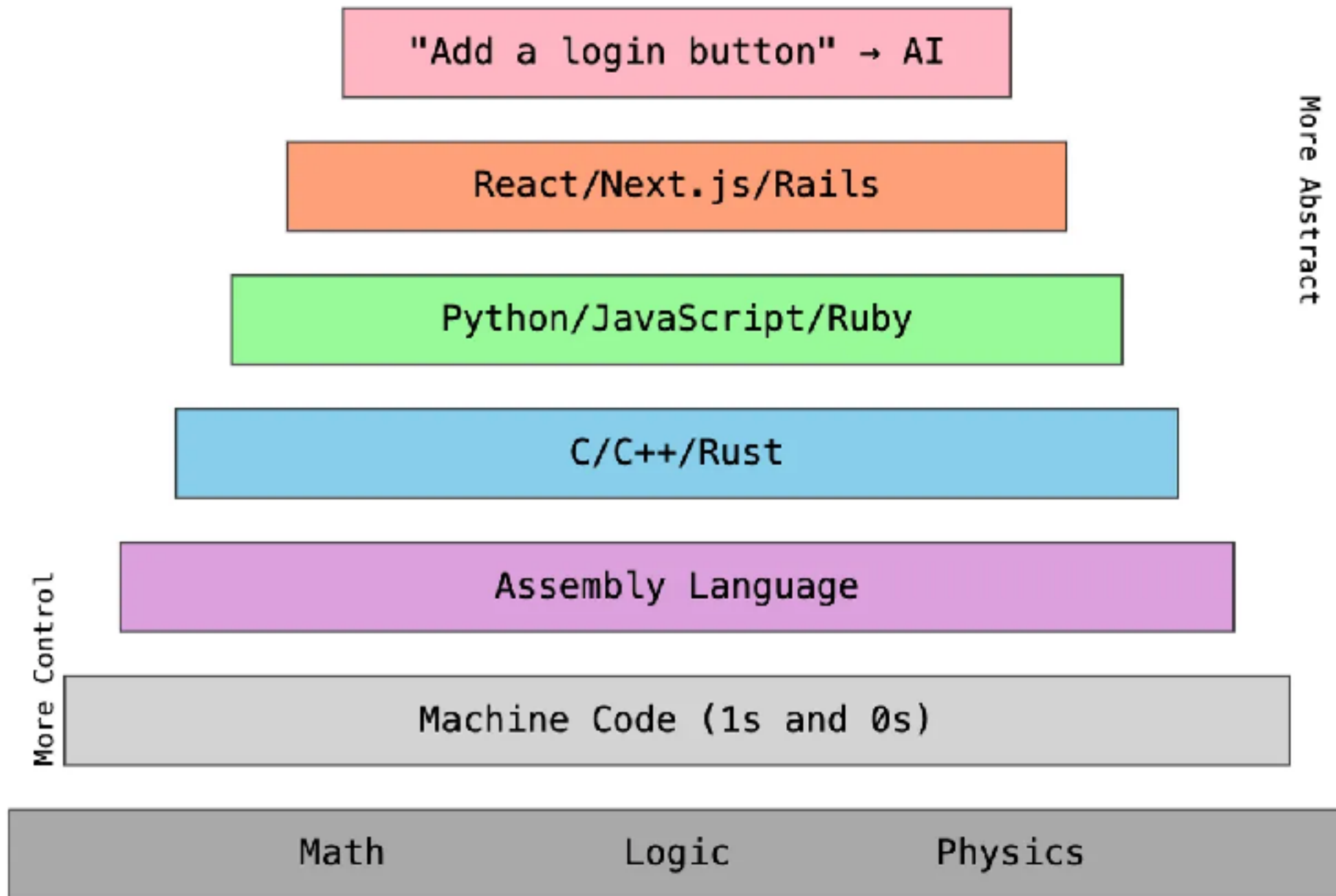


Figure 2 Community theme evolution across three capability layers. The stacked layers—Weights, Context, and Harness—show how the center of gravity in the LLM agent community has shifted outward over time, from parametric knowledge and prompting toward harness-level infrastructure such as tool ecosystems, protocols, skills, and multi-agent orchestration.

<https://arxiv.org/pdf/2604.08224>



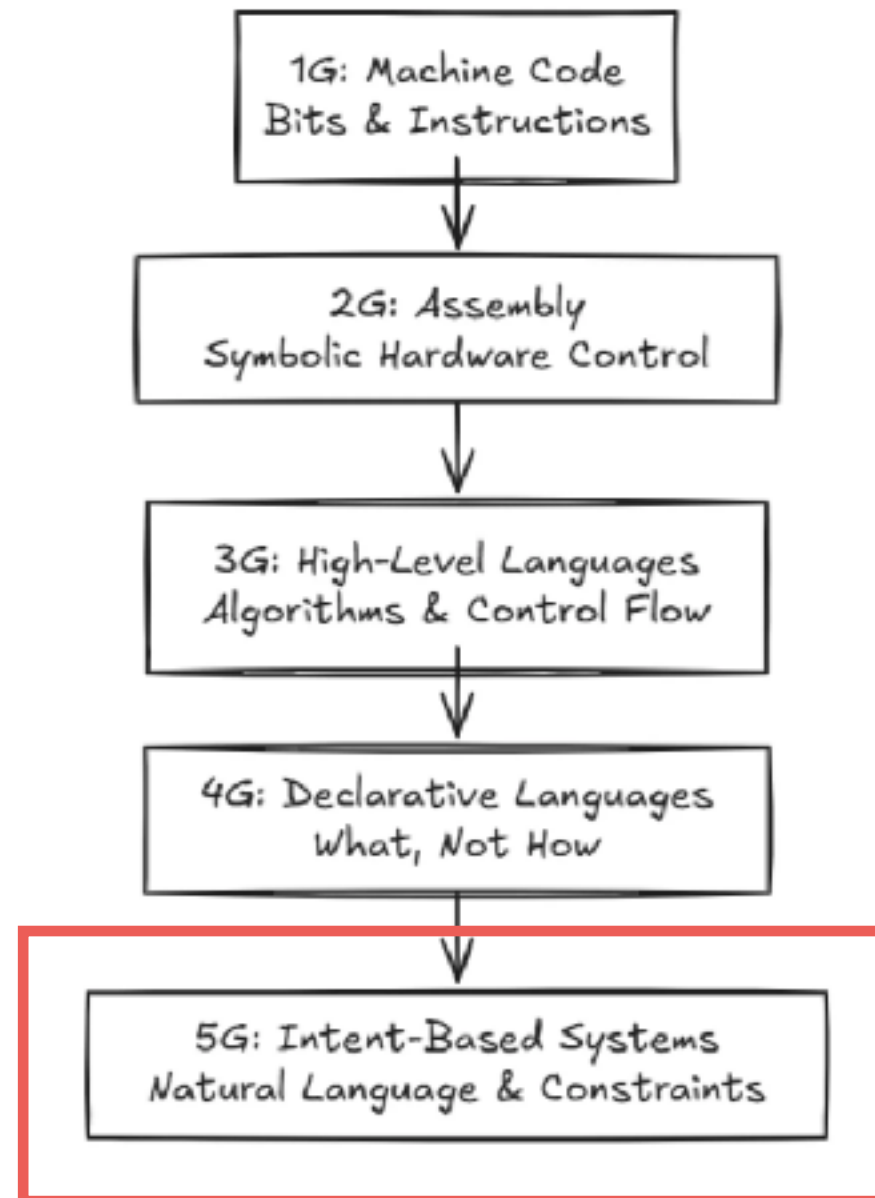
The Modern Tower of Abstraction



<https://cline.bot/blog/from-assembly-to-ai-why-vibe-coding-is-just-another-chapter-in-our-abstraction-story>



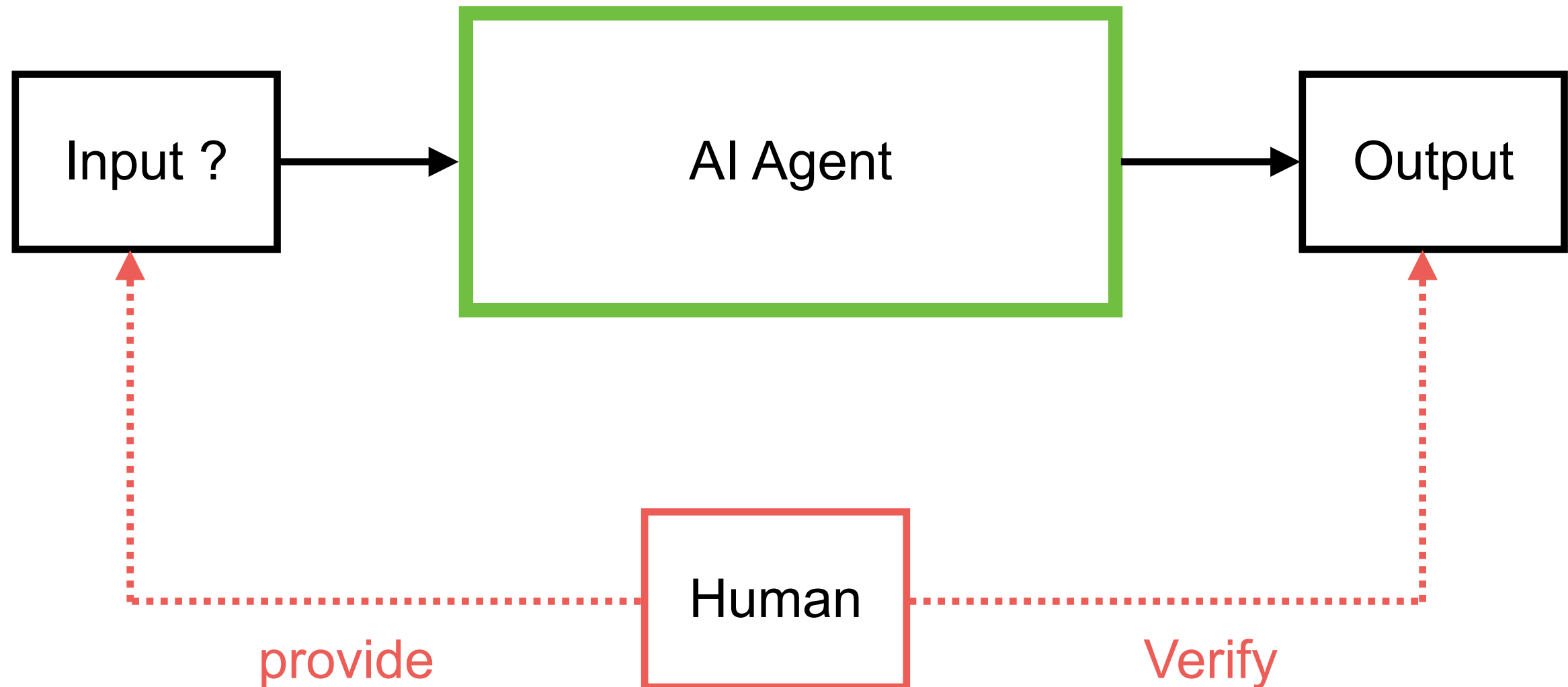
Generation of Programming languages



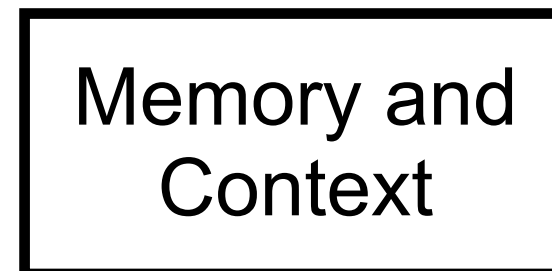
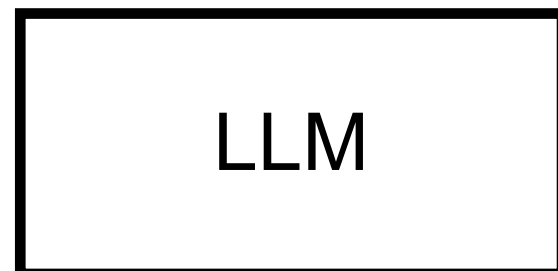
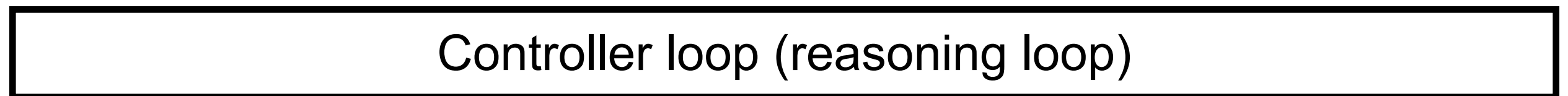
<https://medium.com/@ajuatahcodingarena/generations-of-programming-languages-bed30d19ea8e>

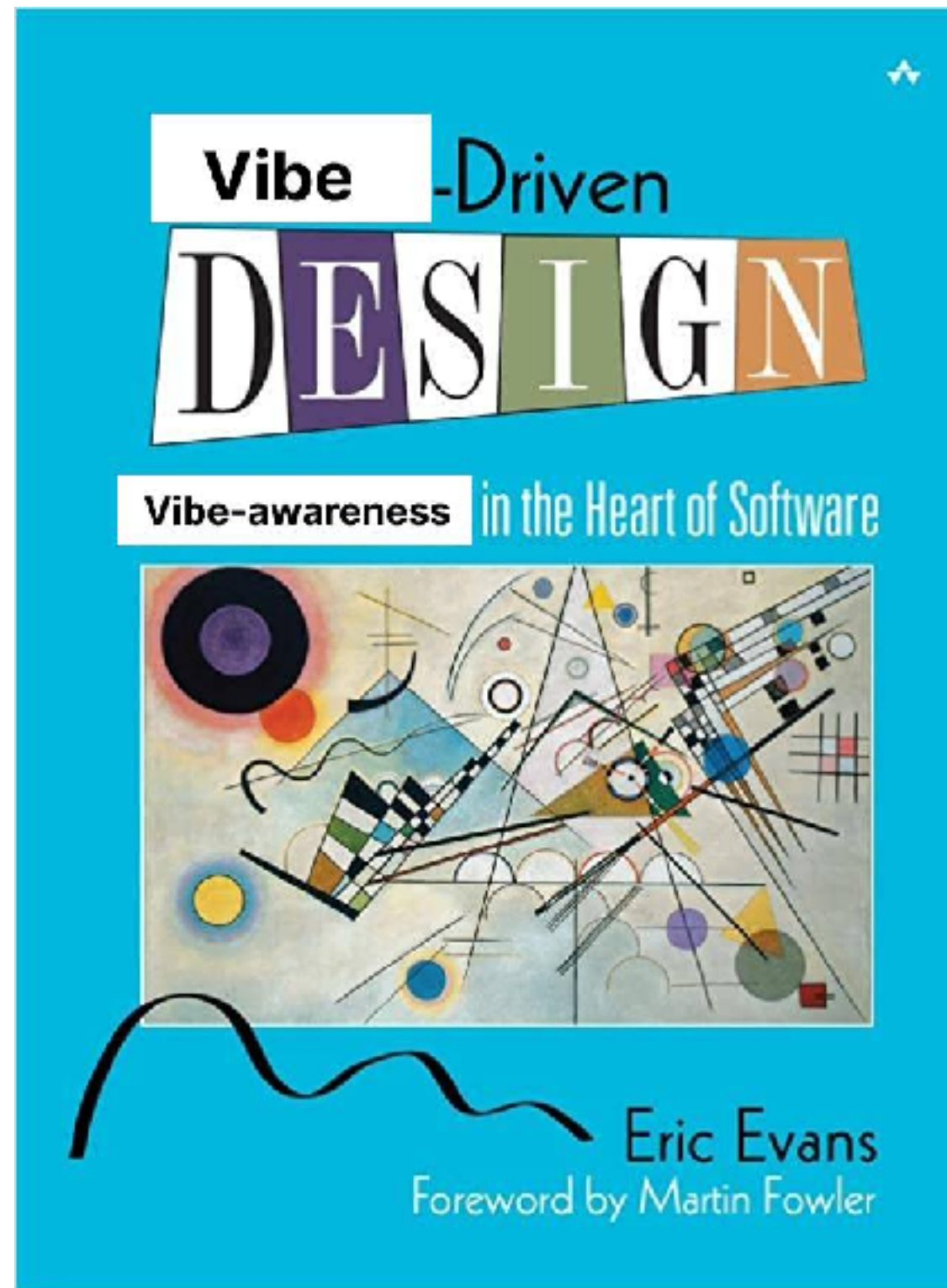


Coding with AI Agent



AI Agent for Coding





"codex is better"

"claude.md"

"i just vibecoded"

"skills"

"ralph"

"agentic"

"obsidian"

"mcp"

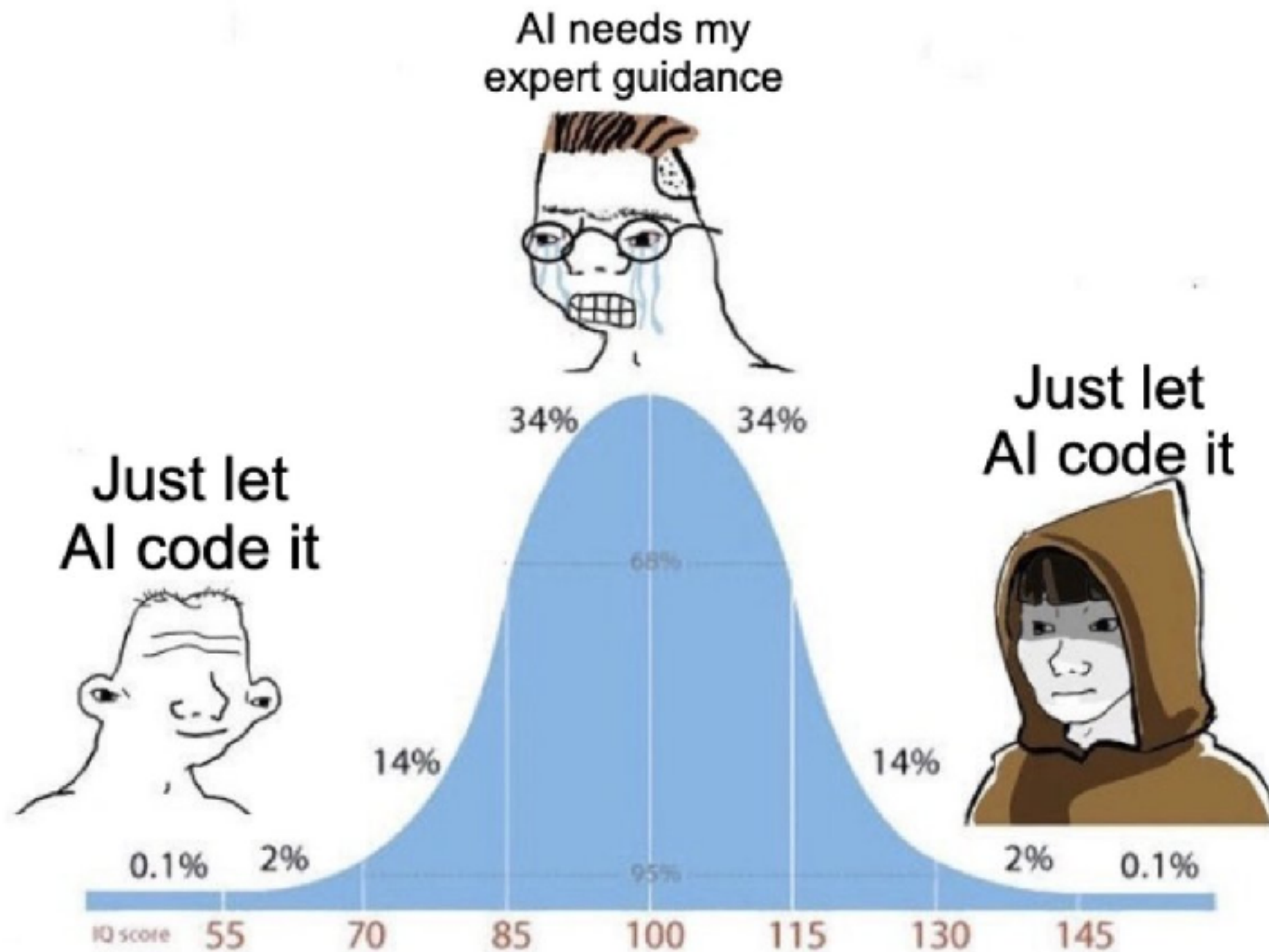
"mac mini"

"clawdbot"

"agent sdk"

"remotion"







Andrej Karpathy ✓

@karpathy



There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

6:17 AM · Feb 3, 2025 · 5.3M Views

<https://x.com/karpathy/status/1886192184808149383>



Vibe Coding

Software development practice that uses artificial intelligence (AI) to generate functional code from **natural language prompts**, accelerating development, and making app building more accessible, especially for those with limited programming experience

<https://cloud.google.com/discover/what-is-vibe-coding>



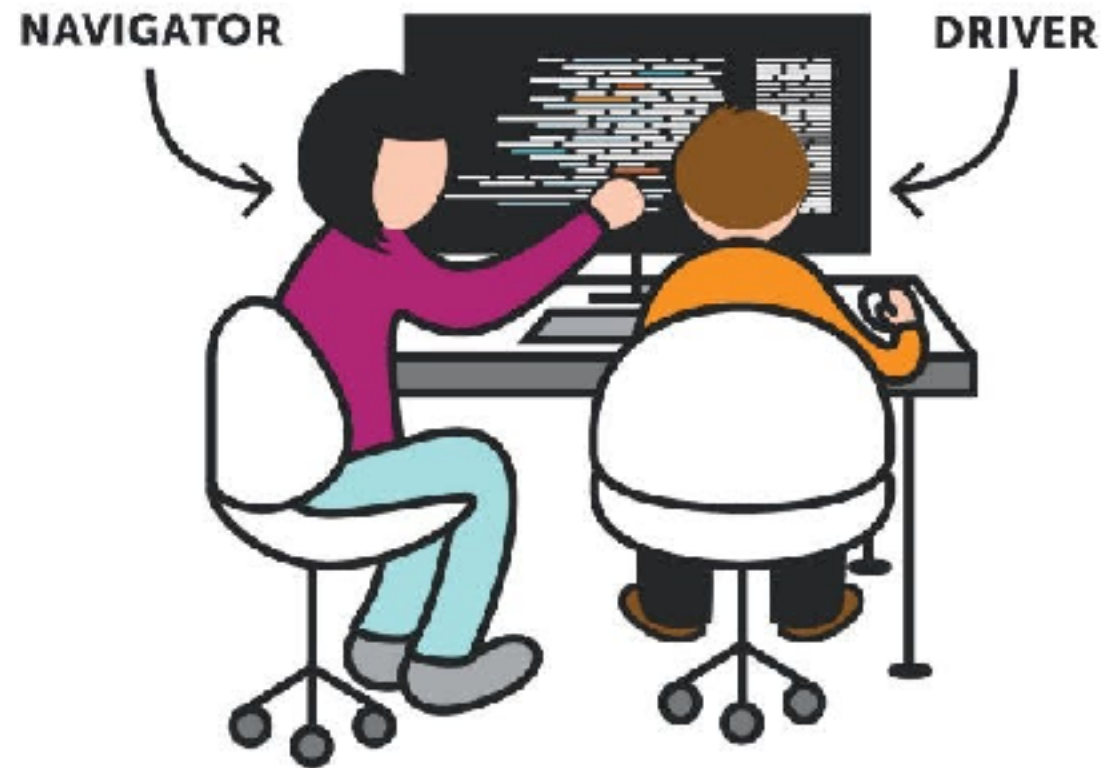
Types of Vibe coding ?

Pure vibe coding

Responsible AI-assist development



PAIR PROGRAMMING



Reviews, Tests and Understands
with experience



AI draft

Human craft



Workflows ?

Application level

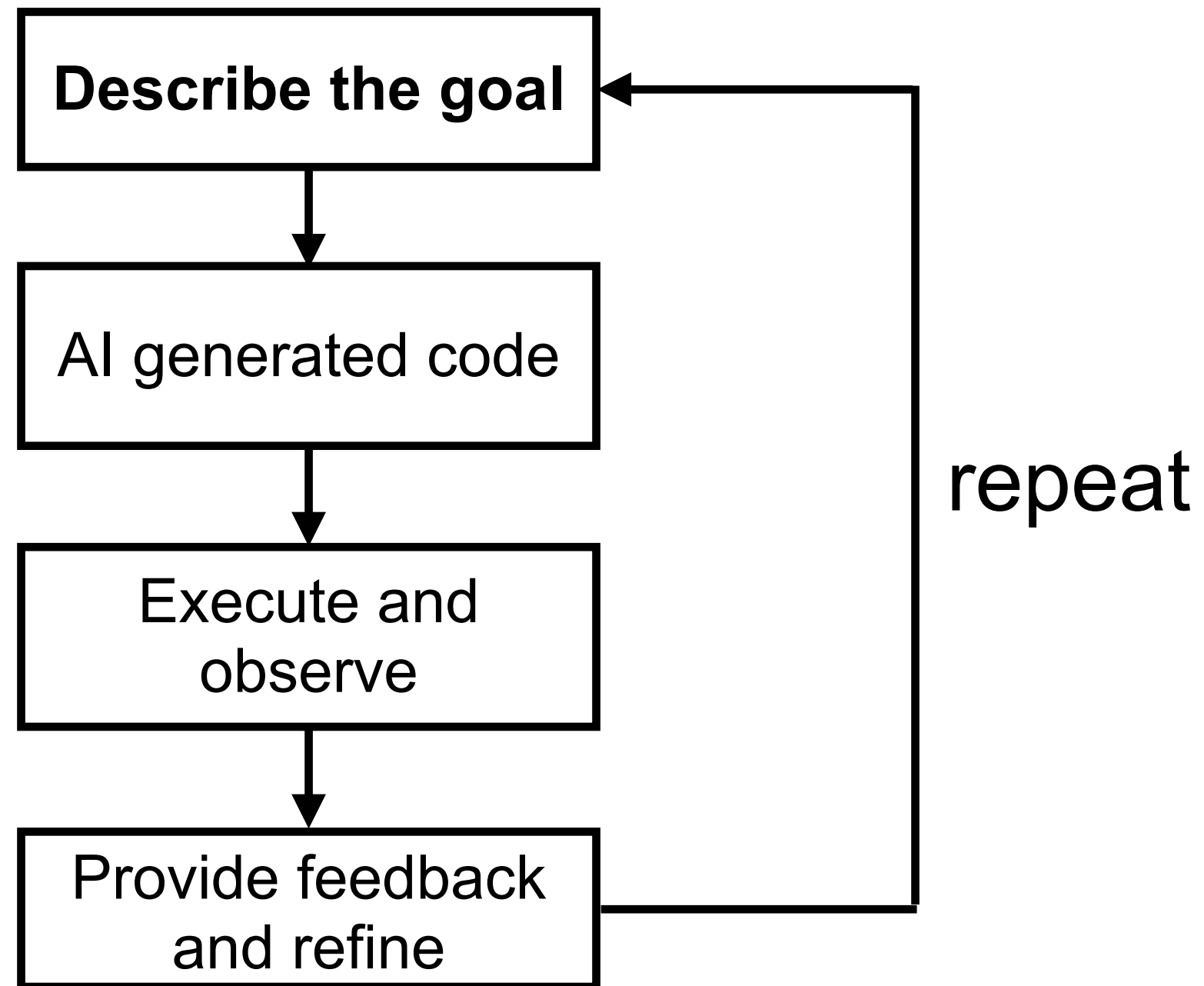
Code level

<https://cloud.google.com/discover/what-is-vibe-coding>



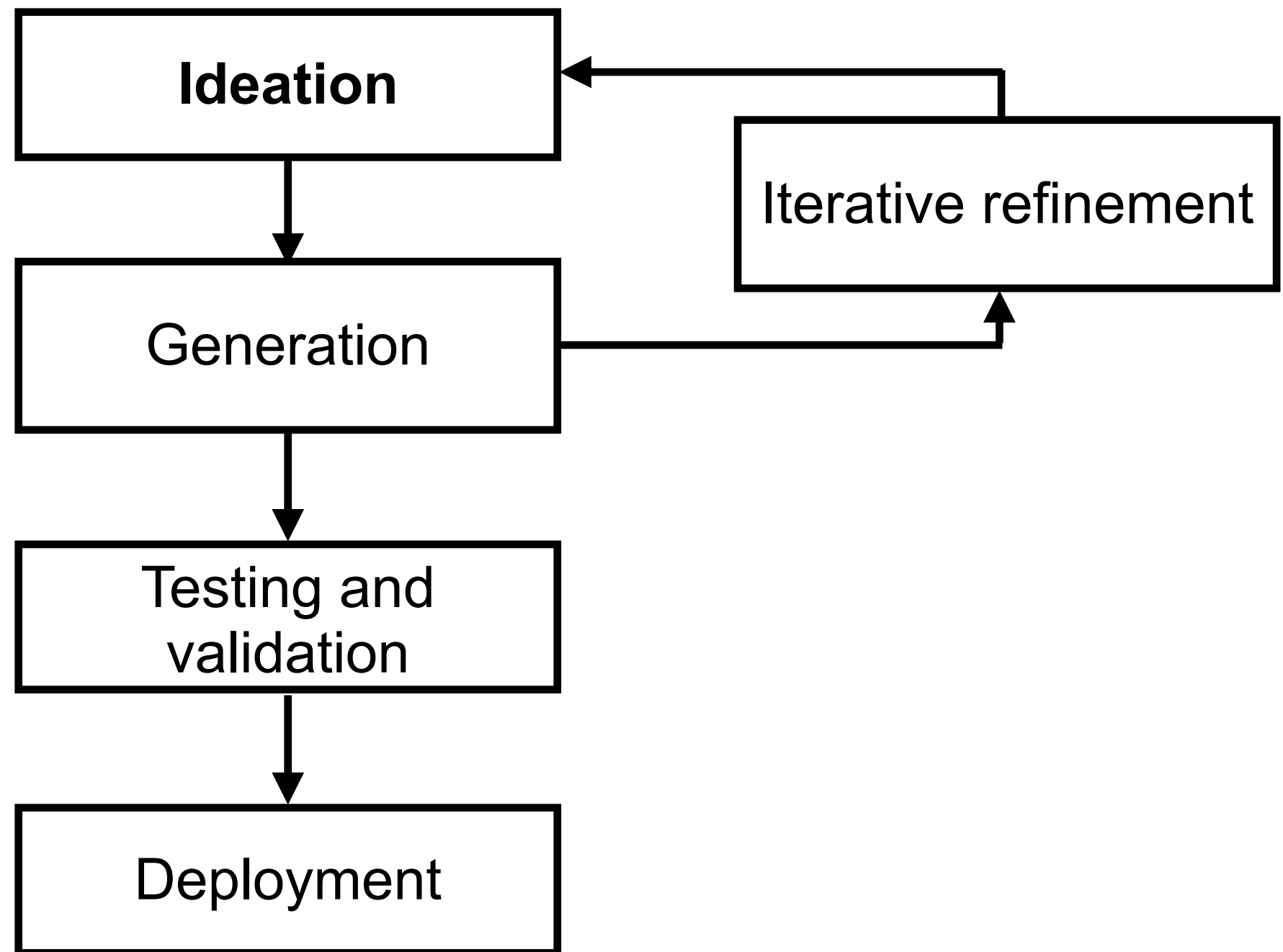
Code level workflow

Conversational loop to create a specific code



Application level workflow

Talking about high-level idea from concept to deployment



AI Agent for Coding

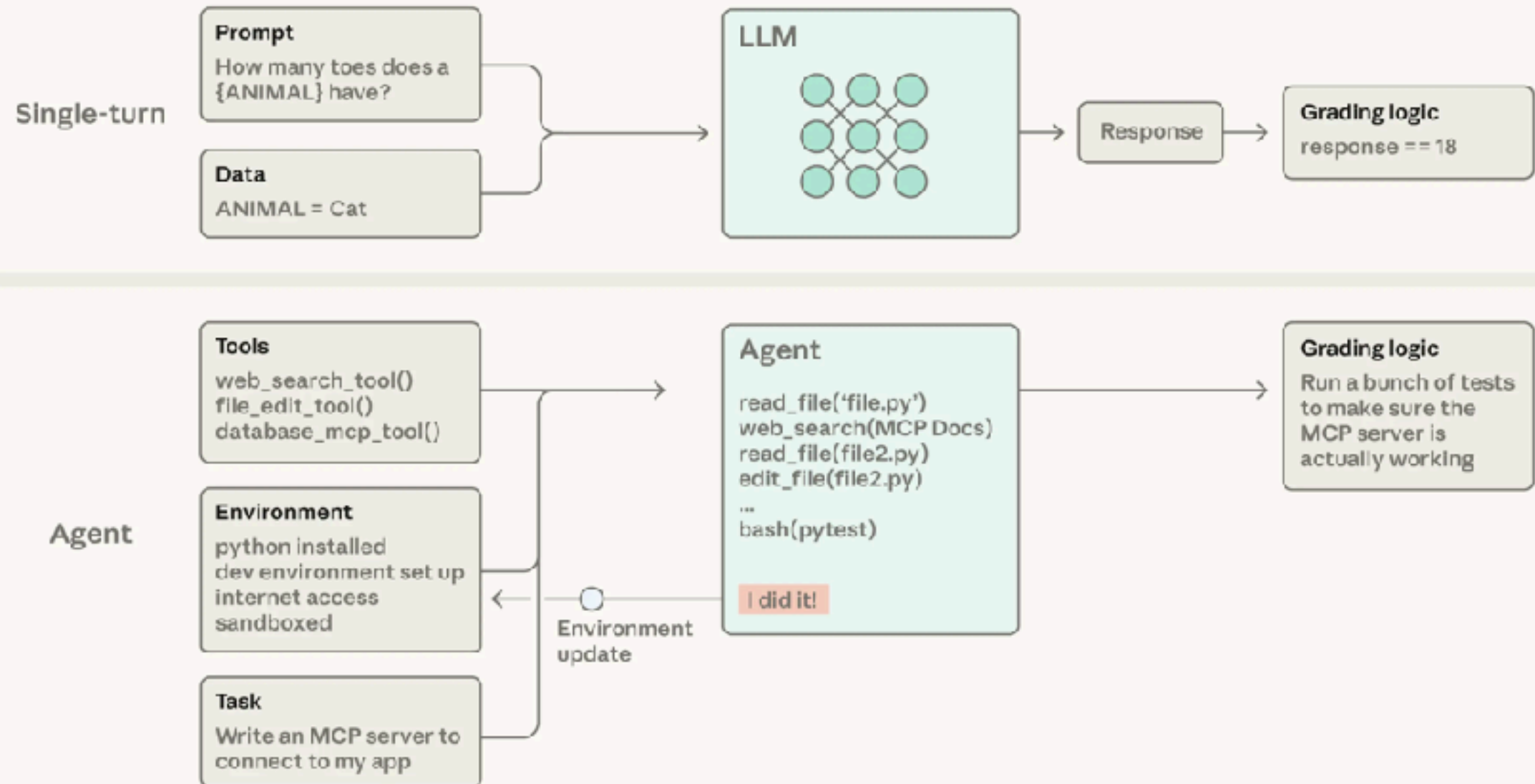


AI Agent for Coding



AI Agent for Coding

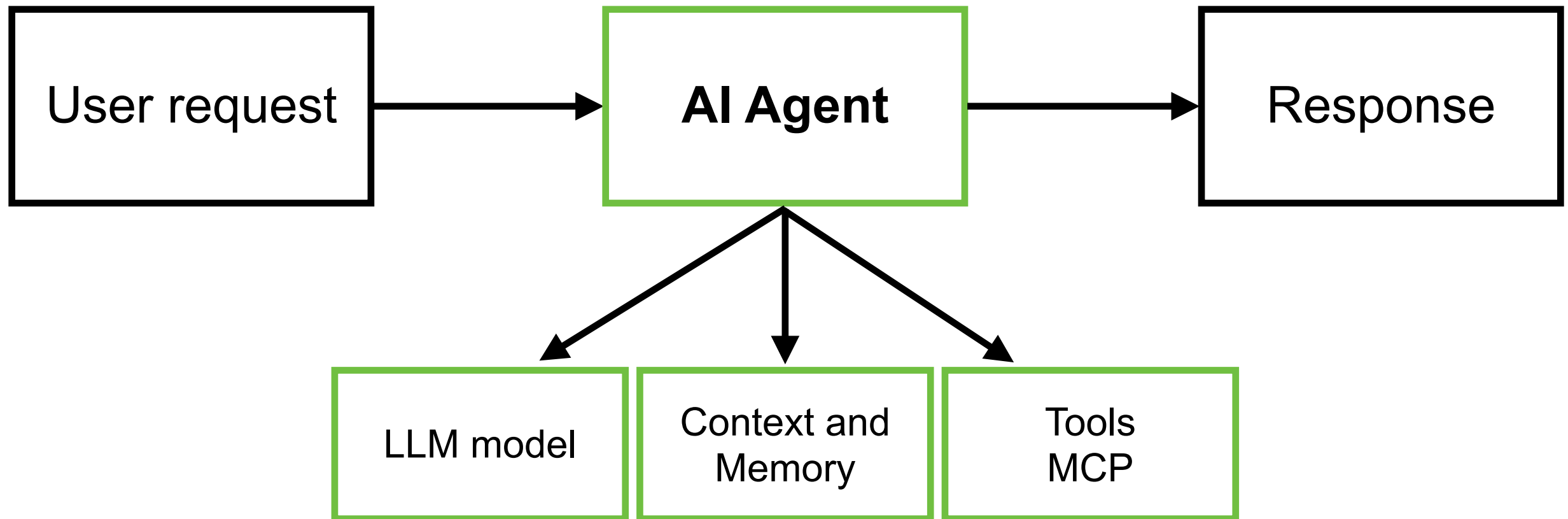
Comparison: Single-Turn vs Agent Evaluations



<https://www.anthropic.com/engineering/demystifying-evals-for-ai-agents>



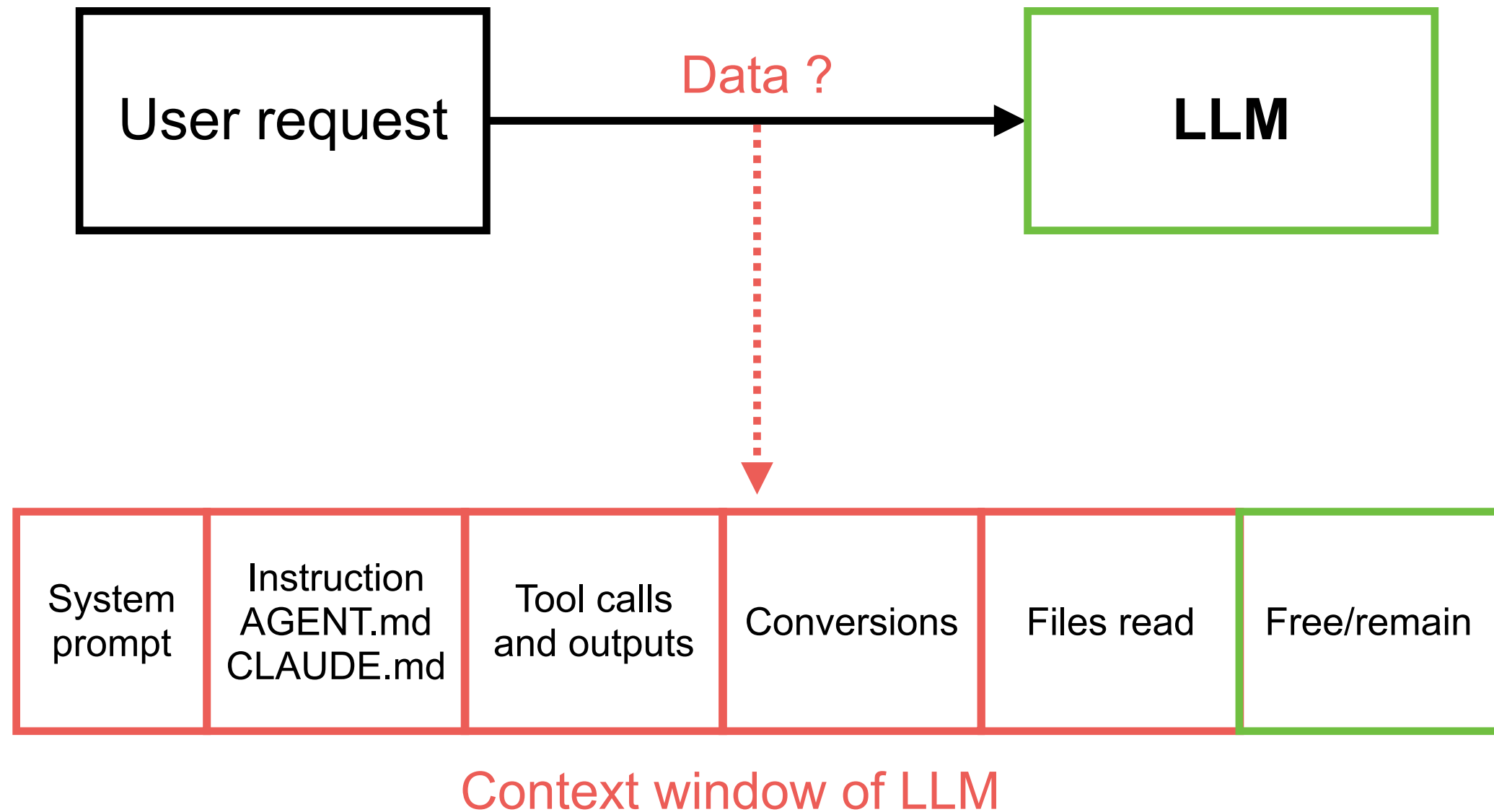
Simple AI Agent



How to manage context or knowledge for Agent ?



Context Window ?



Types of Knowledge/Context ?

Shared context

Static Knowledge

Prompt engineering
Global Instructions
AGENTS.md
CLAUDE.md
Rules

Task context

Dynamic Knowledge

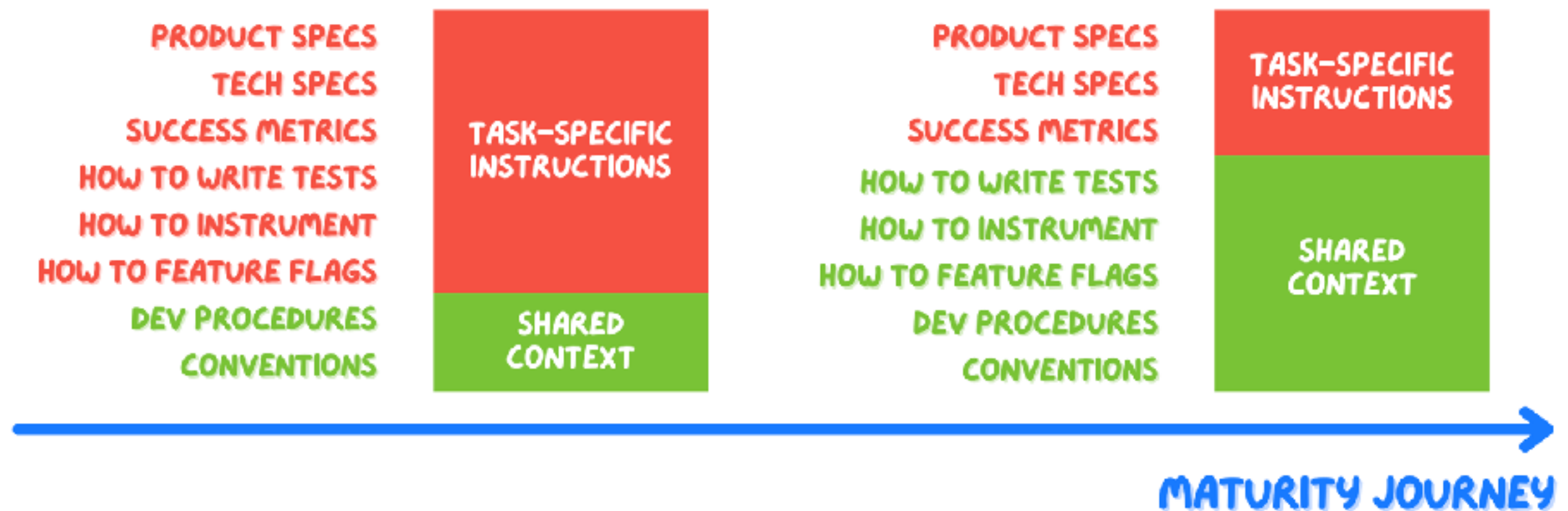
Memory
Graph/Vector database
MCP tools
Specific Instructions
Agent SKILLS

Sub-agent and Modes

Context engineering ?



Types of Knowledge/Context ?



<https://refactoring.fm/p/context-capabilities-and-tech-hubs>



Vocabulary before start !!



Vocabulary before start !!

Rules

Prompts

MCP servers

Instructions

Commands

Subagents

Modes

Hooks

Agent Skills



Rules and instructions

Static context

Rules are files that contain essential context
Codebase, best practices, common pitfalls
Business requirements

Prevent LLM models from hallucinating
and to ensure specific standards



Commands and Prompts

Static context

Text instructions sent to the agent

Repeat workflow that can be triggered on demand

Allow teams to share standardized workflows
And run complex tasks conditionally

git add -> generate commit message -> git commit



Subagents and Modes

Specific persona or task to specialized agent
Agent, Plan, Ask, Edit mode
Work AI as a team

Make the agent's behavior more deterministic, reliable
and discoverable



Agent Skills

Dynamic context

Create workflow + resources + script

Skills do not bloat the initial context window

Only loaded when the agent determines (relevant with task)

Provide an open standard for sharing capabilities with a team without hitting context limits

<https://agentskills.io/>



Hooks

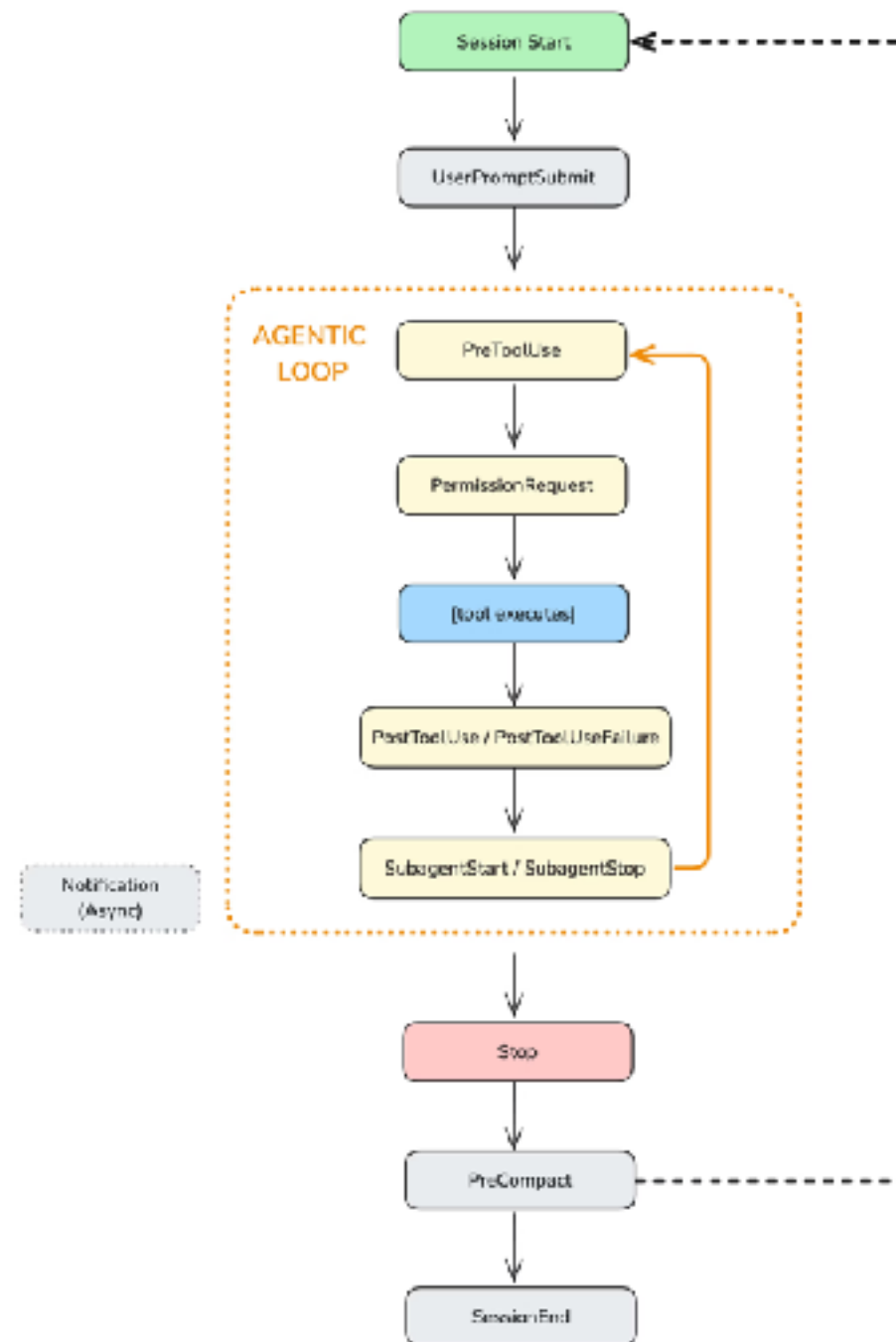
Customize and extent behavior of agent
Triggers for actions before/after conversation

Setup environment, run test, cleanup resources,
Notification, automated pre-commit check ...

<https://code.claude.com/docs/en/hooks>



Hooks in Agent's Lifecycle

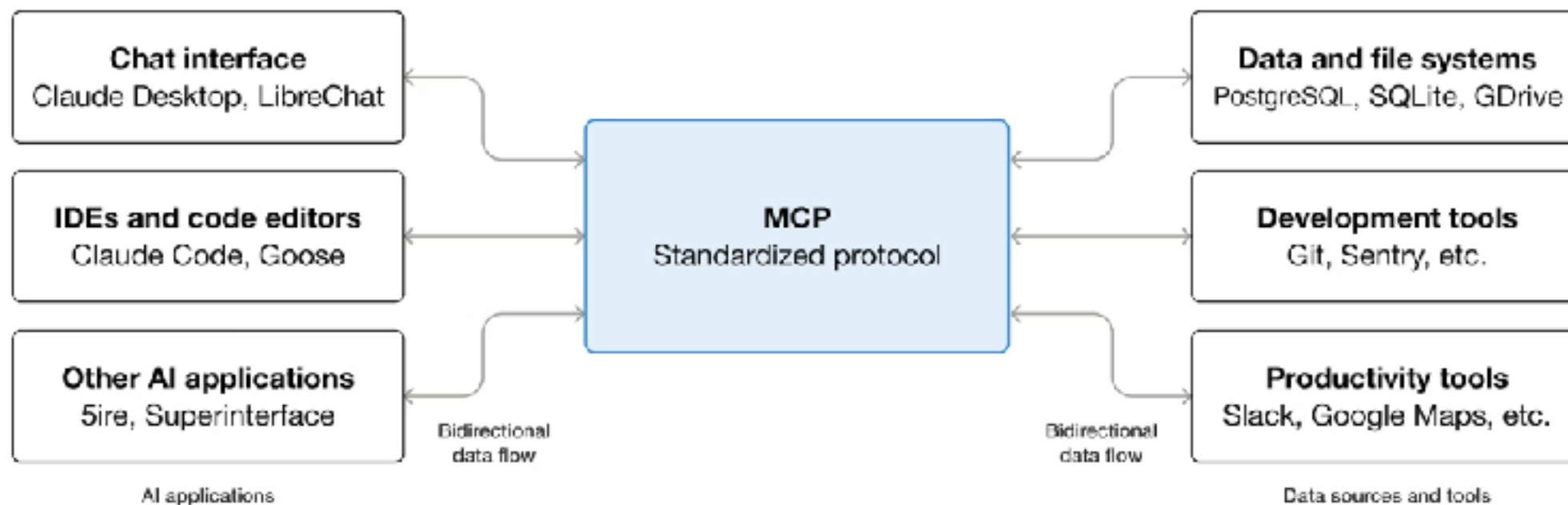


<https://code.claude.com/docs/en/hooks>



MCP (Model Context Protocol)

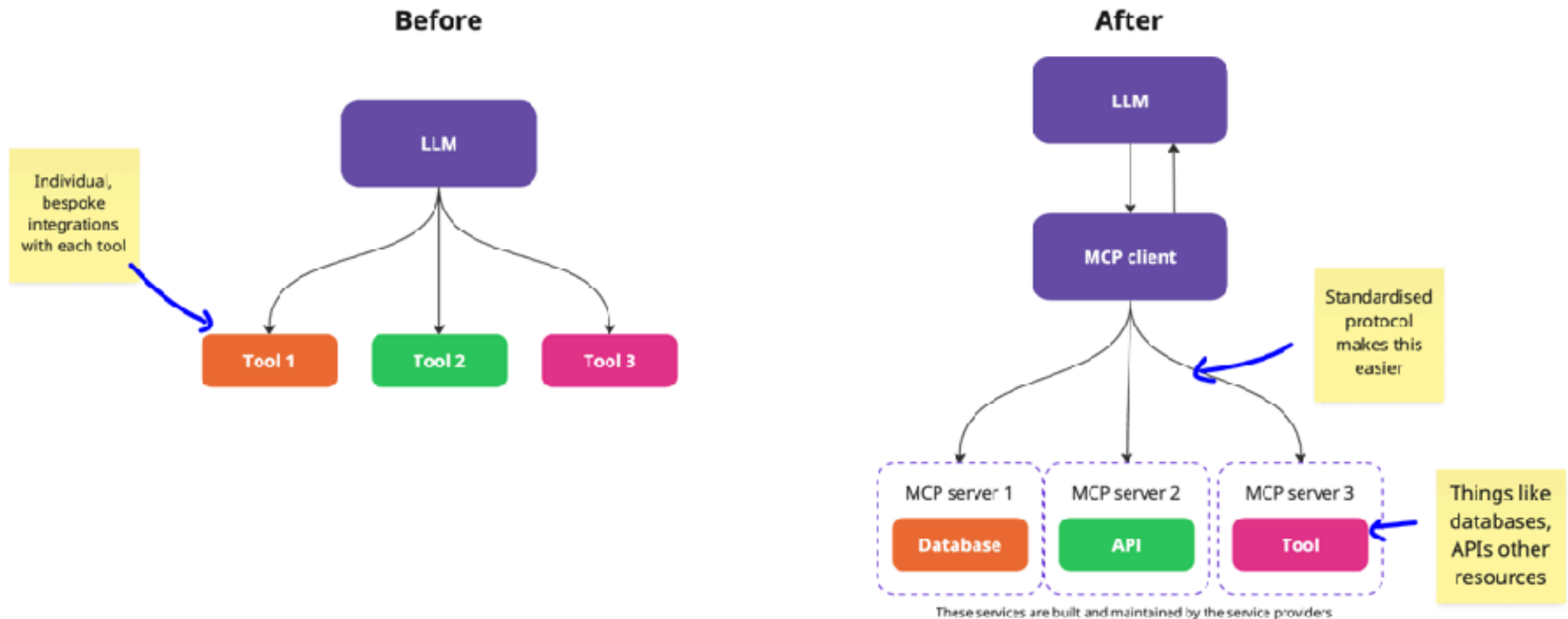
Open standard for connecting AI app to external system



<https://modelcontextprotocol.io/>



MCP (Model Context Protocol)



<https://departmentofproduct.substack.com/p/mcp-explained-a-simple-guide-for>

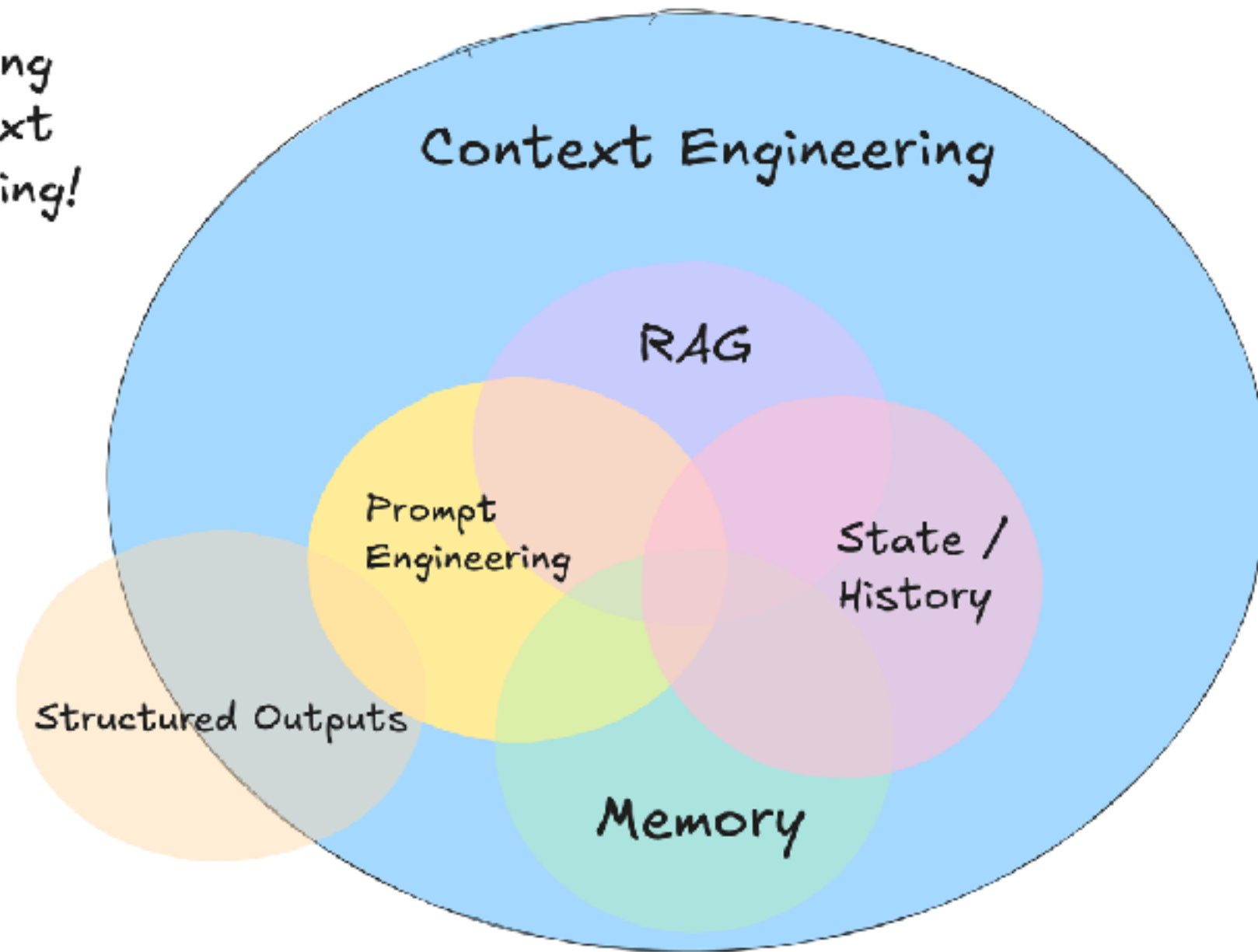


How to manage context ?



Context Engineering

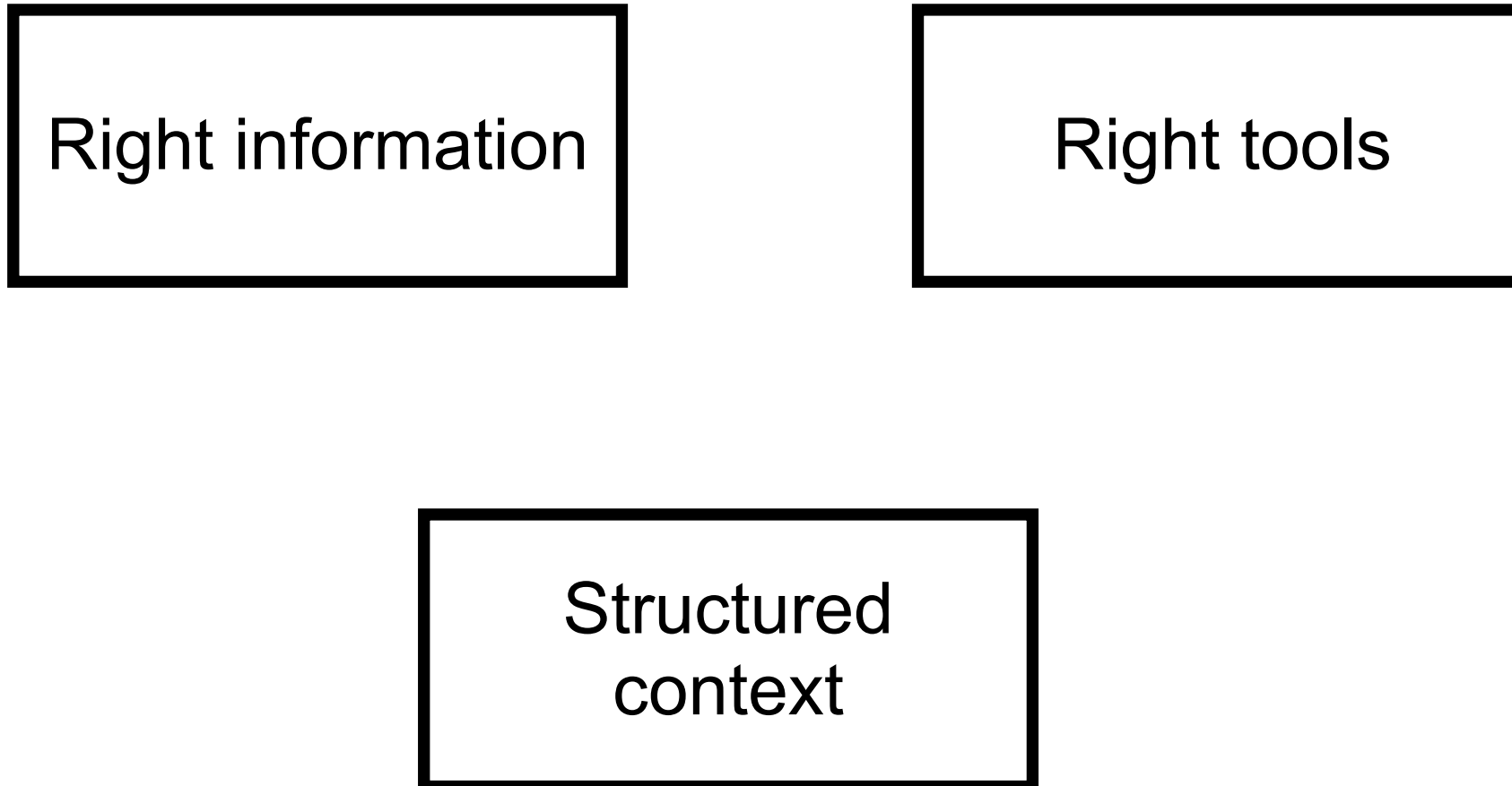
Everything
is Context
Engineering!



<https://www.promptingguide.ai/guides/context-engineering-guide>



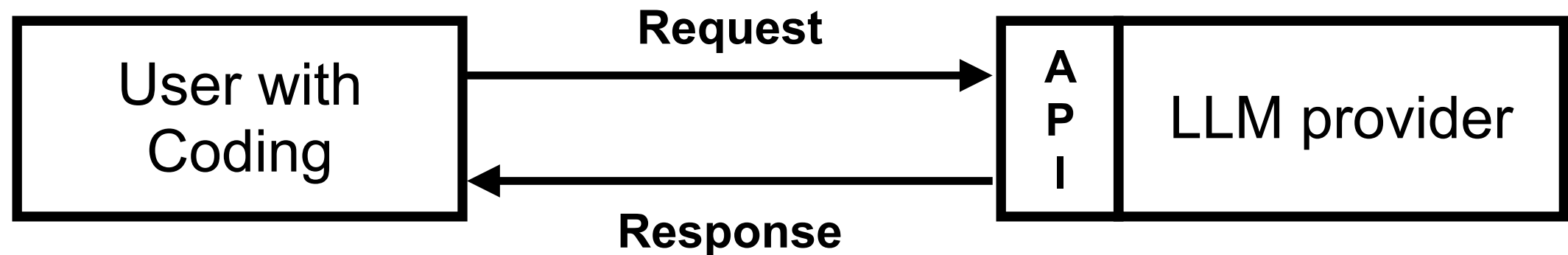
Context Engineering



Context window size ?

Amount of text a model can process and remember
at once time

Measure with **token**



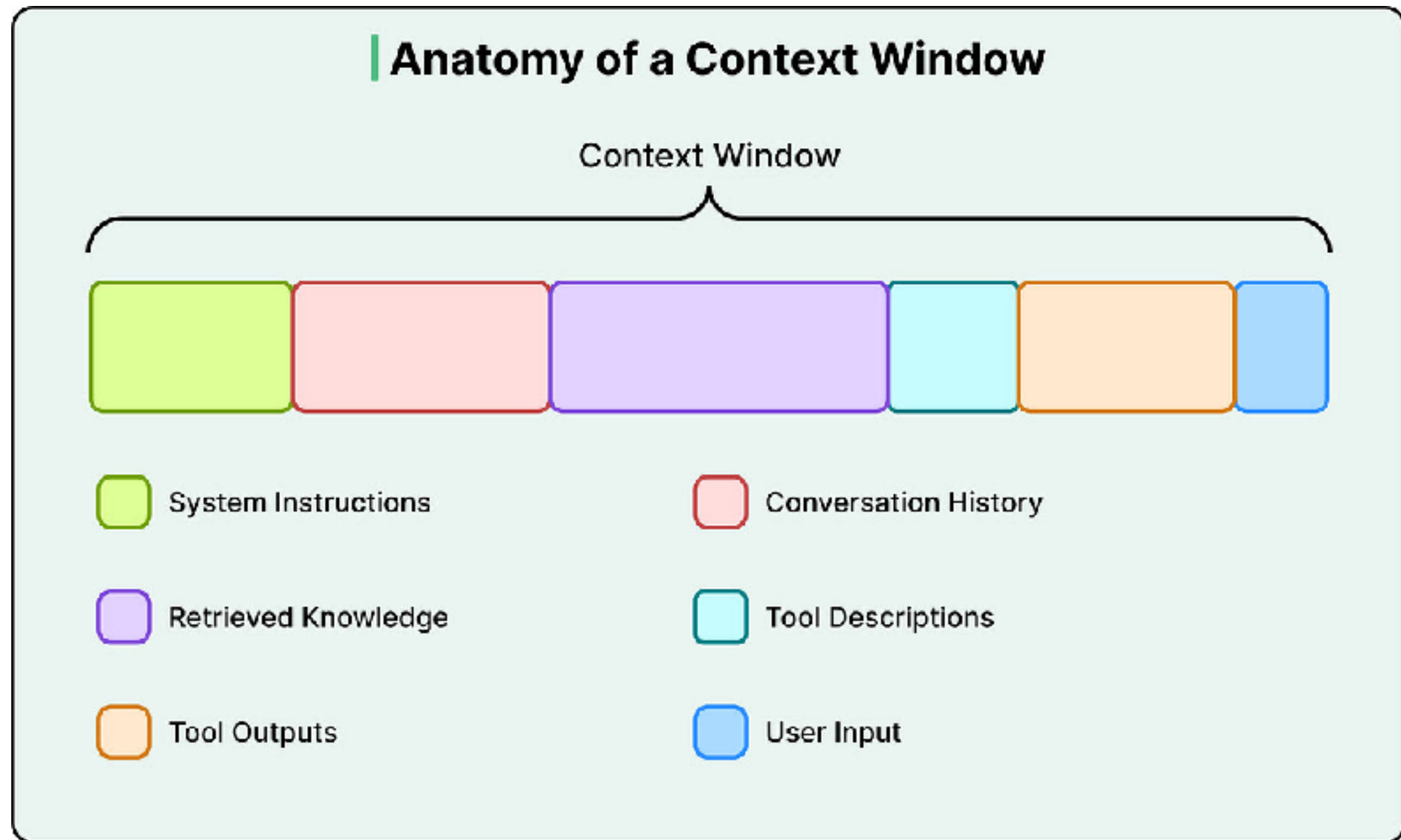
Context window size of models

Different models may have different window size

Model name	Context window size
GPT 4.1	1M
Gemini 1.5, 2.5	1M
Llama 4	1M
Claude Sonnet 3.7	200,000
o4	200,000



Anatomy of Context Window

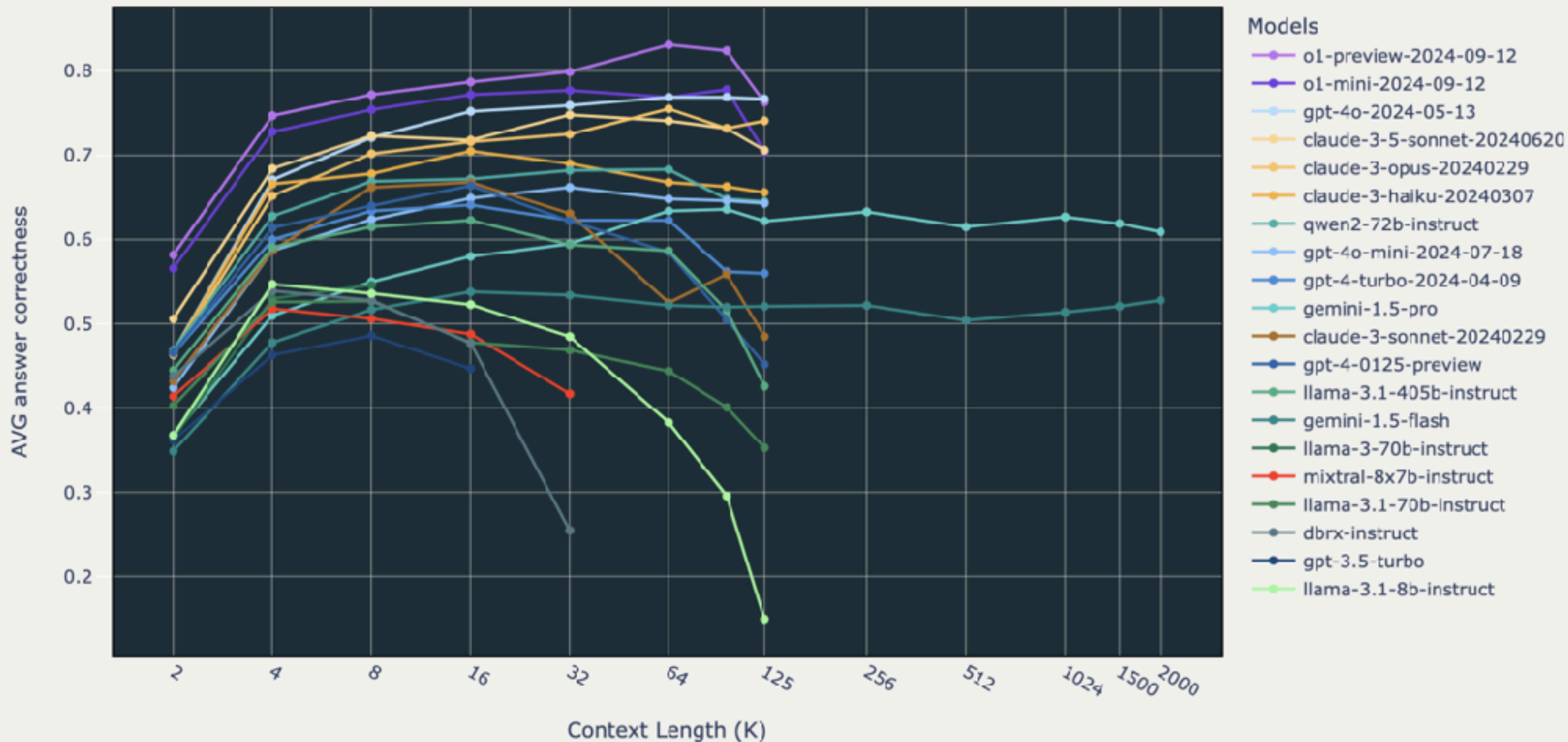


Long context == Better result ?



Long context LLM is better !!

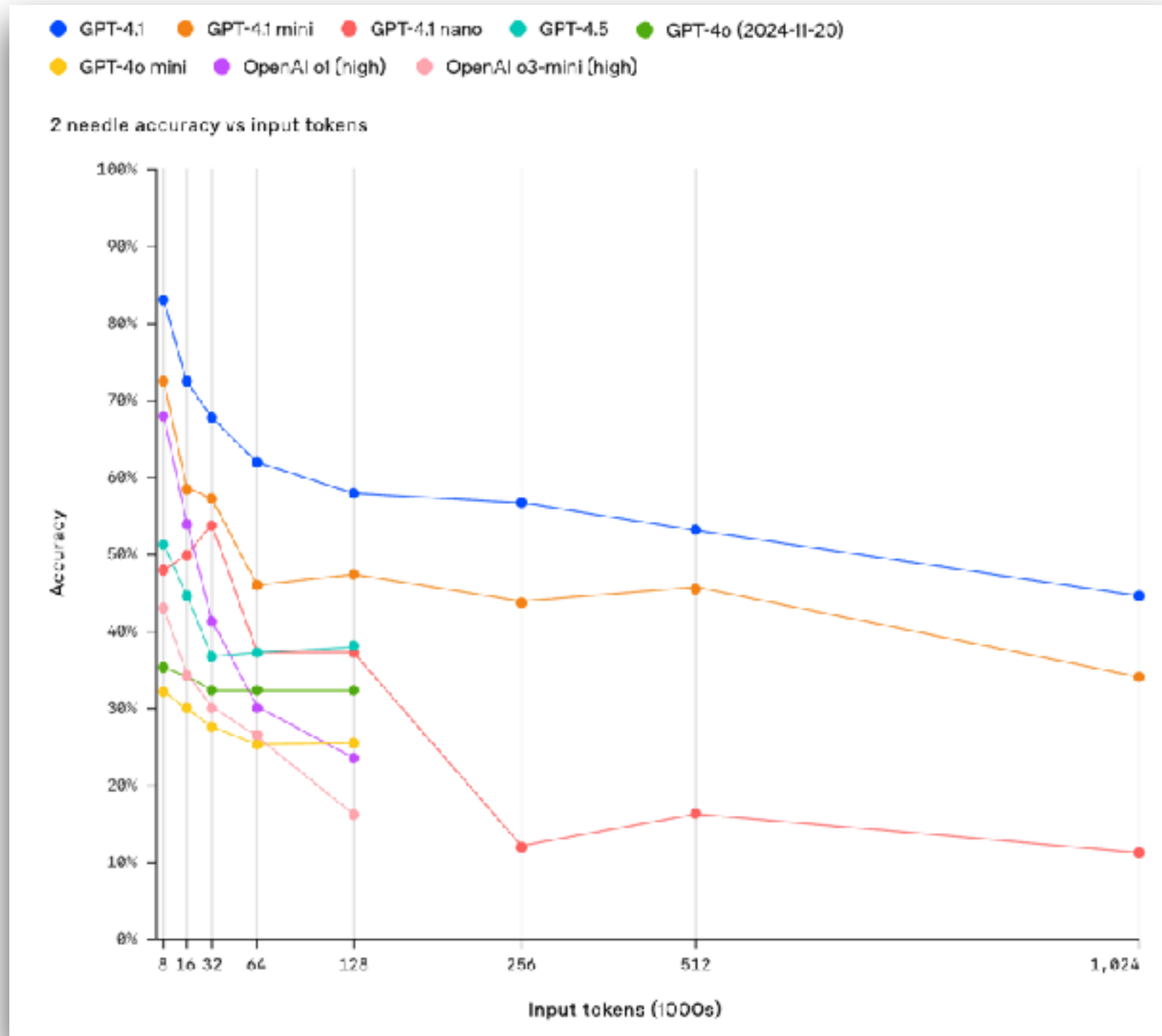
Long Context RAG Performance of LLMs



<https://arxiv.org/abs/2411.03538>



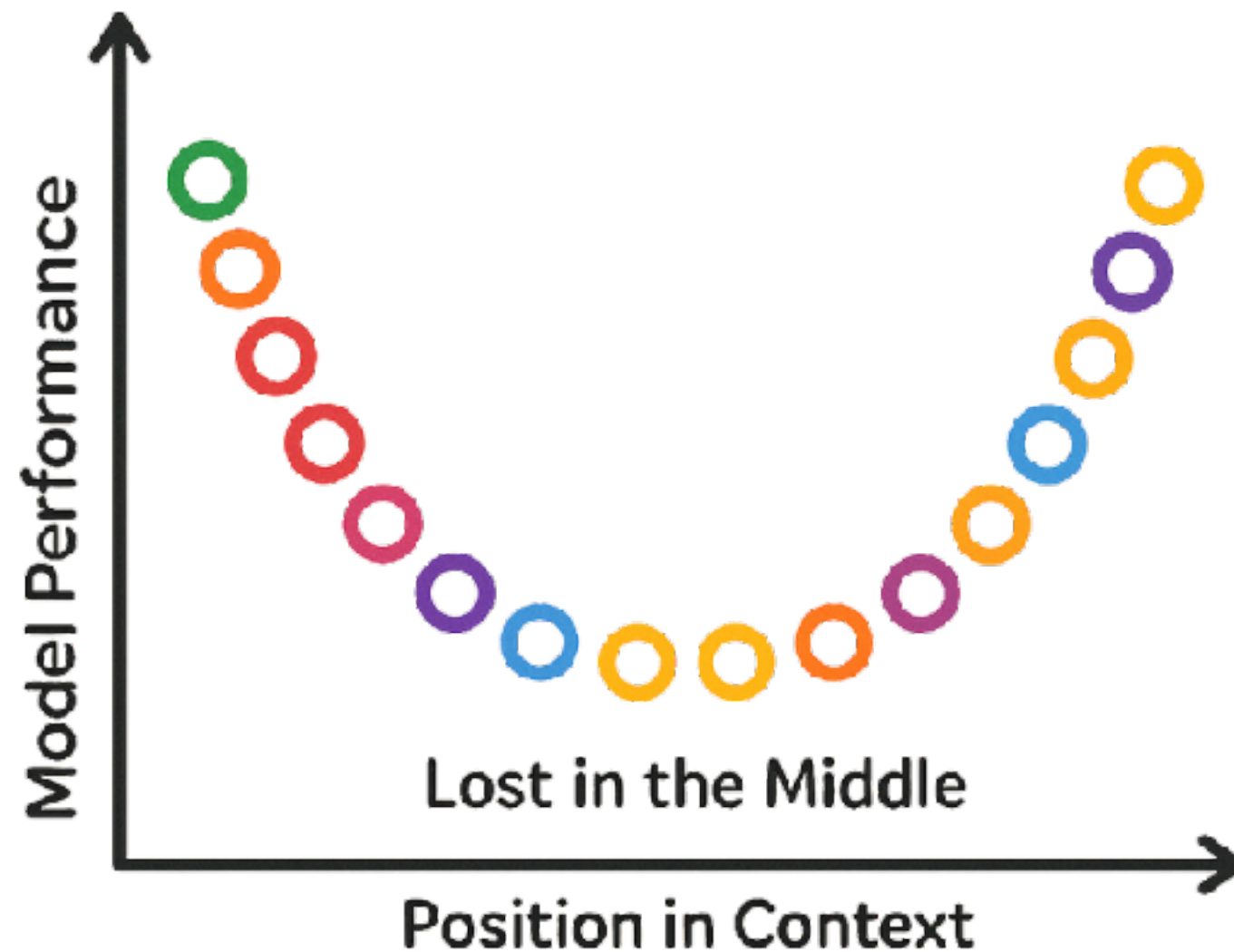
Long context LLM is better !!



<https://openai.com/index/gpt-4-1/>



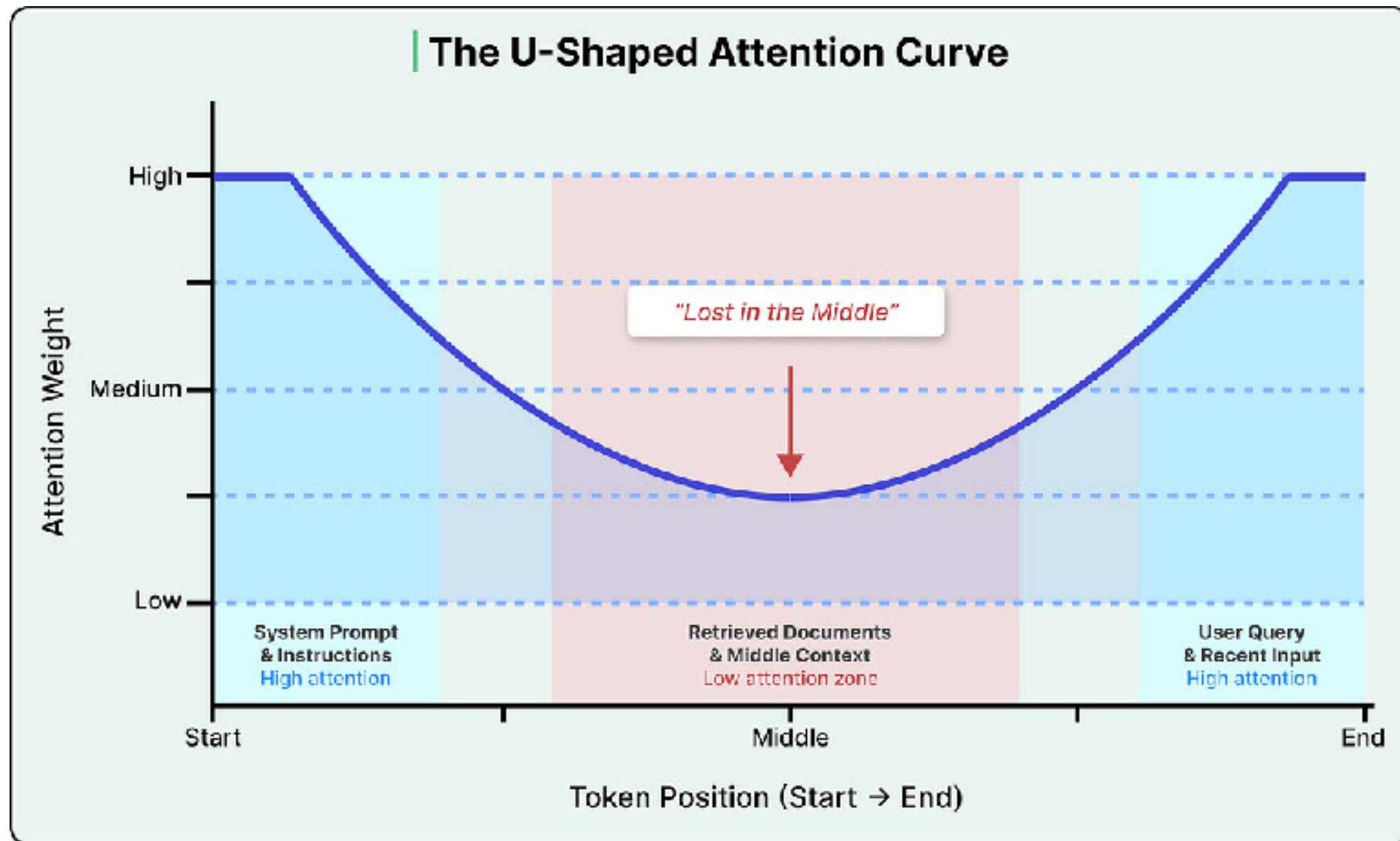
Loss in the middle of LLM



<https://arxiv.org/abs/2307.03172>



Loss in the middle of LLM



<https://arxiv.org/abs/2307.03172>



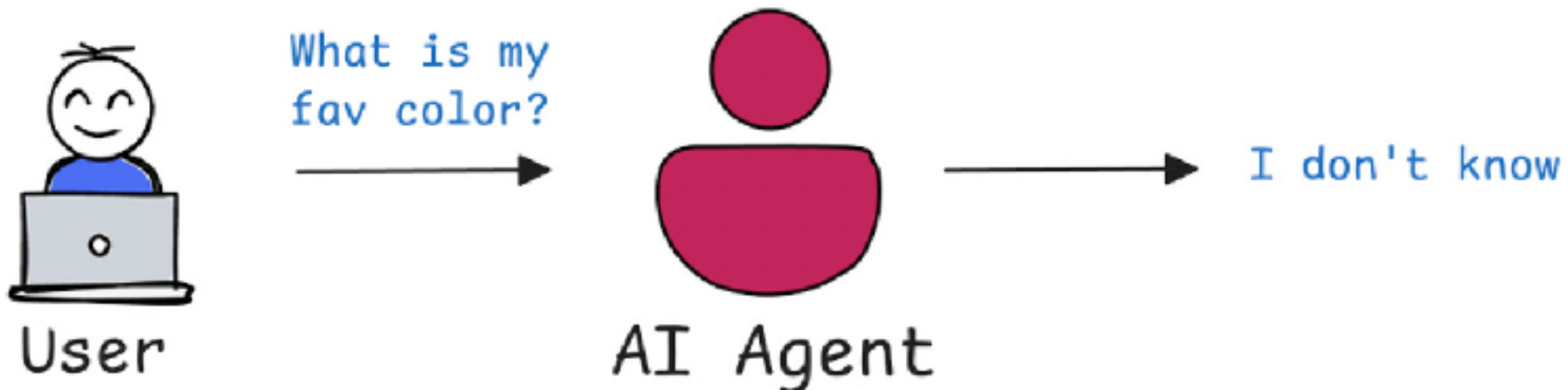
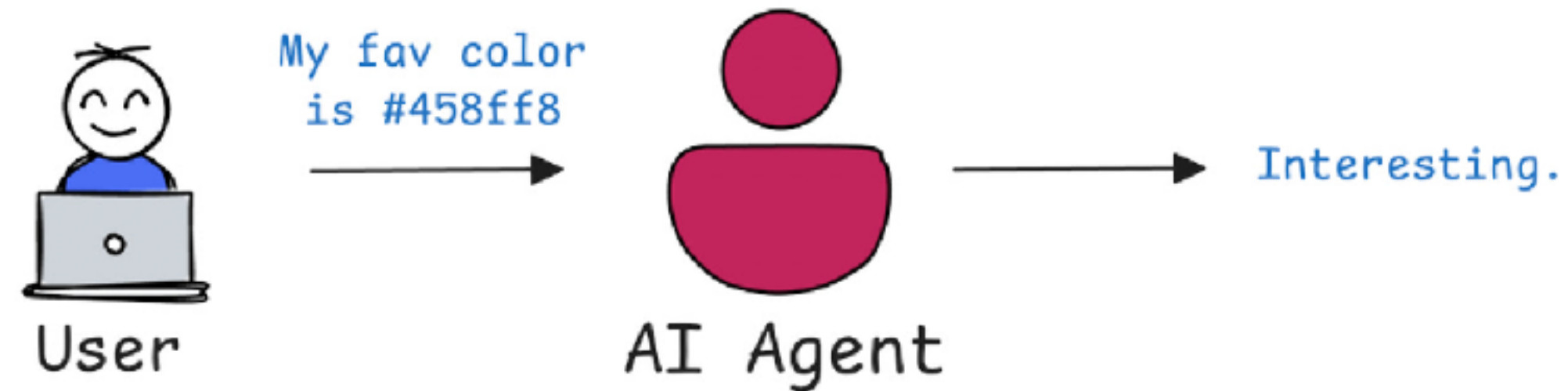
Better context == Better result ?



Agent Memory Memory Bank



Interaction without memory

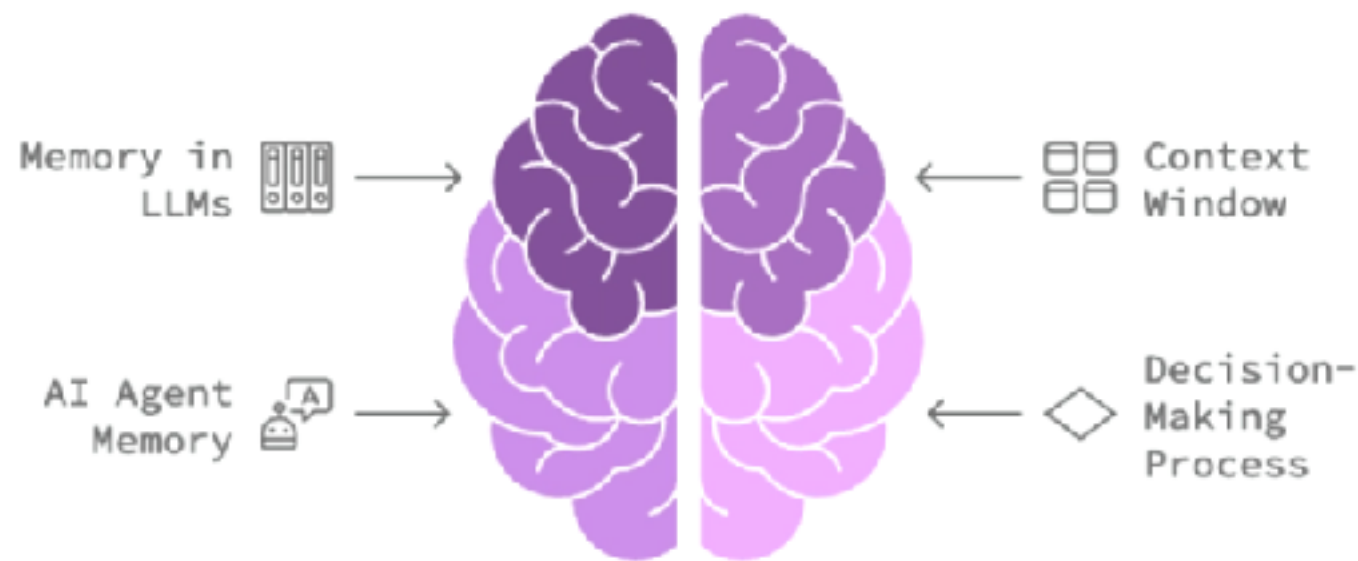


Why AI Agent memory important ?

Enable personalized and context-aware interactions

Improve decision-making and adaptability

Enhance LLM's performance and response



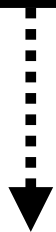
<https://www.falkordb.com/blog/ai-agents-memory-systems/>



Types of Memory

To make intelligent decisions
Agents combine different types of memory

Short-term



Current goals and steps
File system (markdown, JSON)
In-memory database

Episodic



Logs of past interactions
Success and failure

Semantic

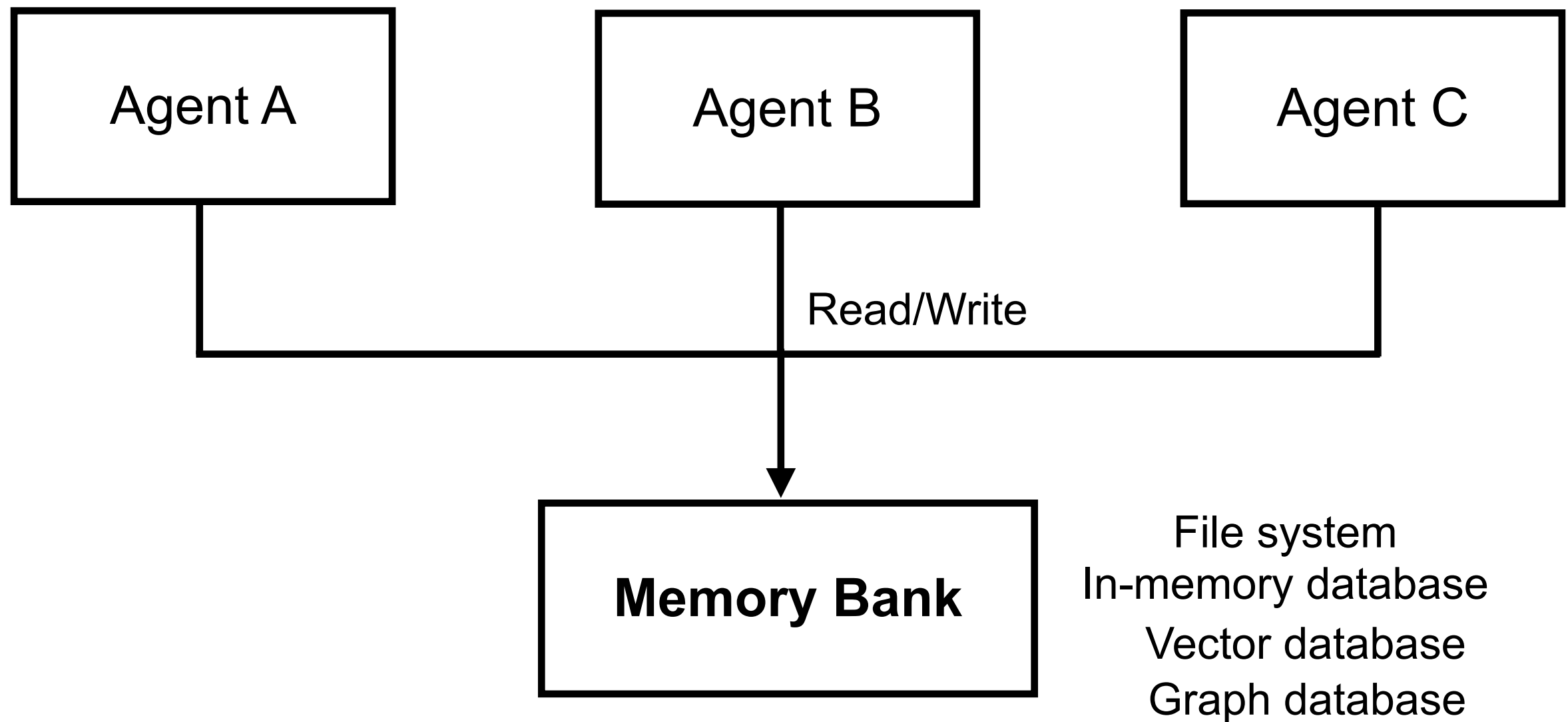


Long-term knowledge
Fact and strategies
RAG
Vector and graph database

<https://bhavishyapandit9.substack.com/p/how-memory-works-in-agentic-ai-a>

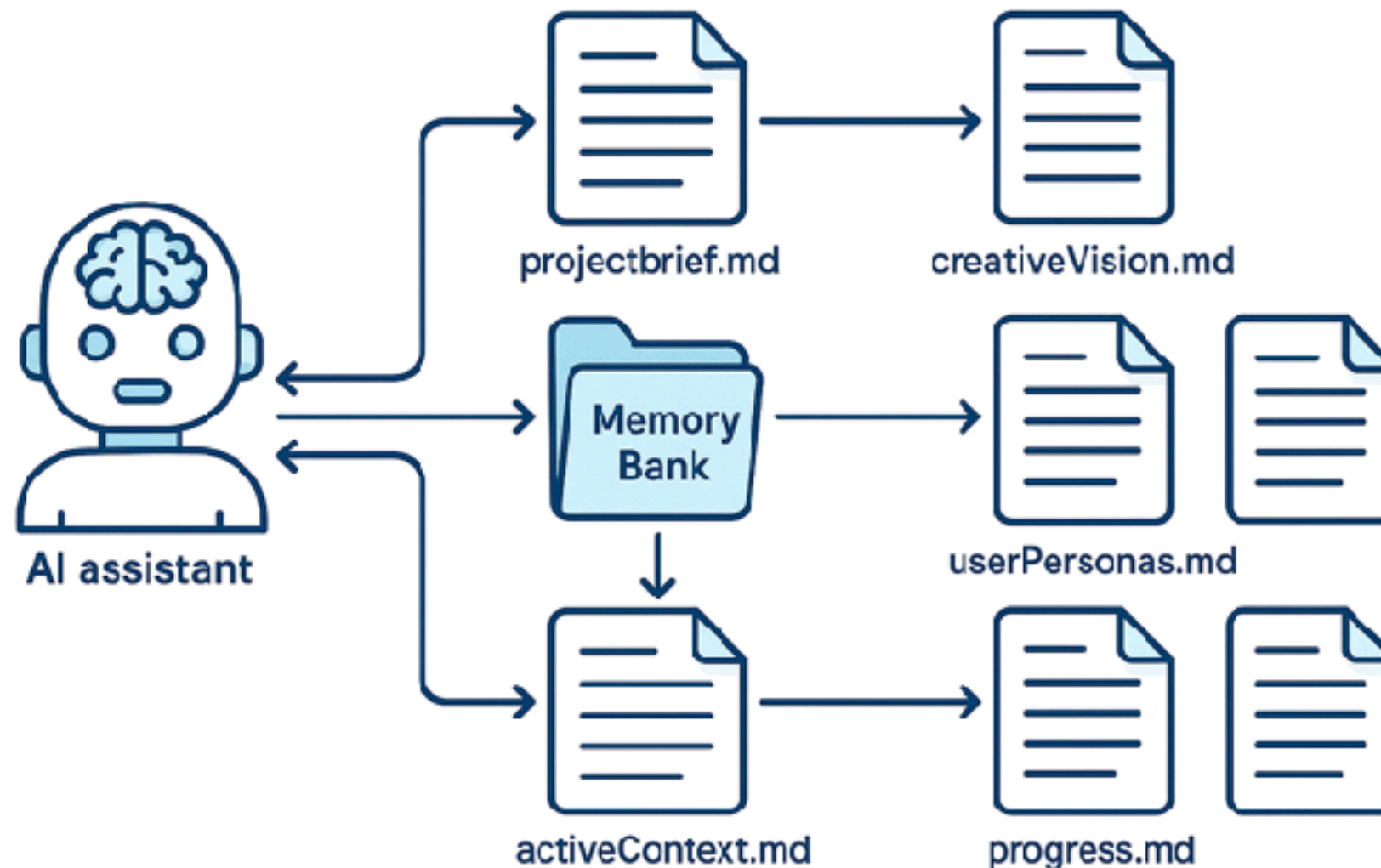


Shared memory between Agent



Cline Memory Bank

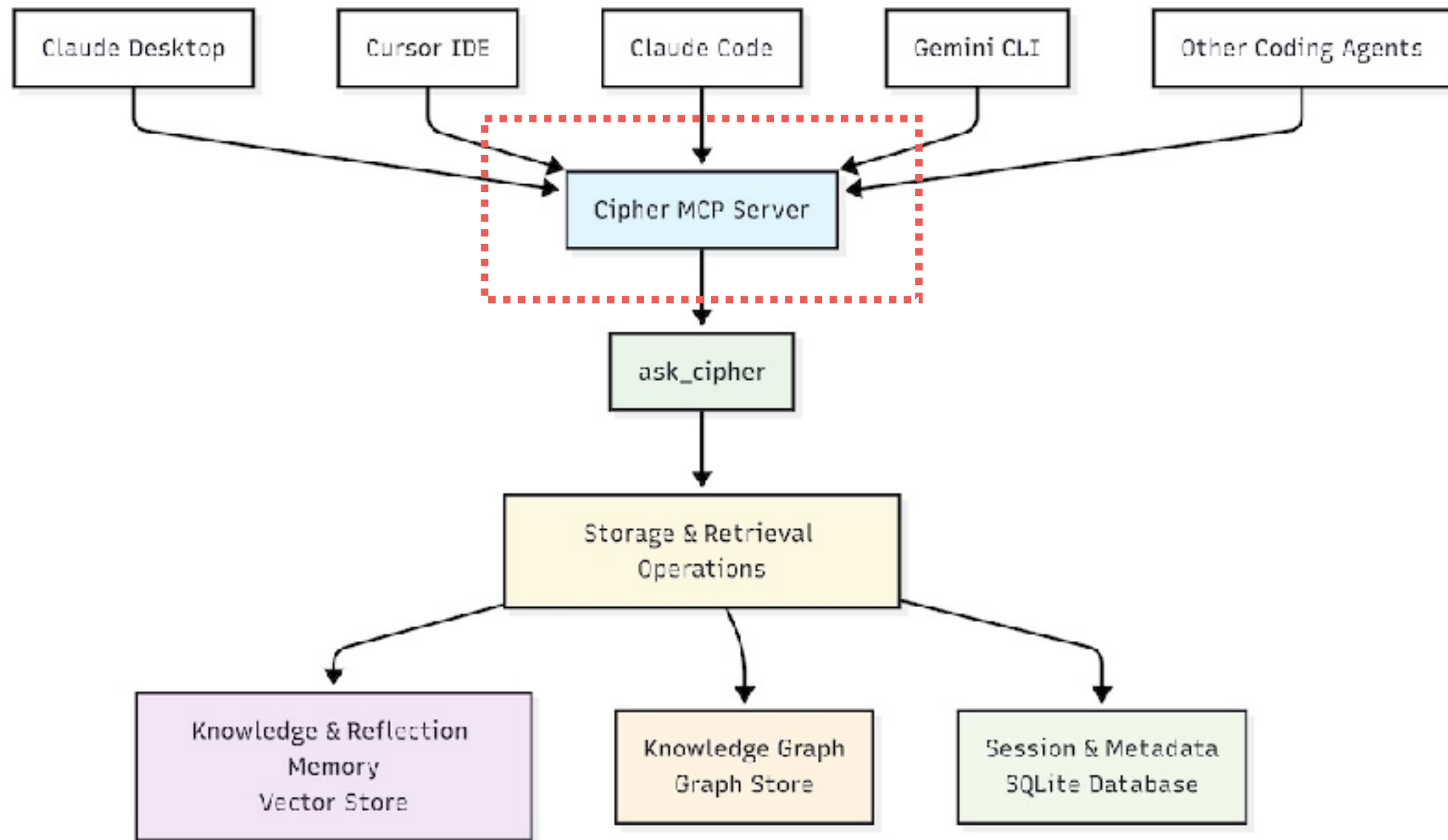
Working with file system



<https://docs.cline.bot/prompting/cline-memory-bank>



Memory Bank via MCP



<https://www.somkiat.cc/share-memory-for-ai-ide/>



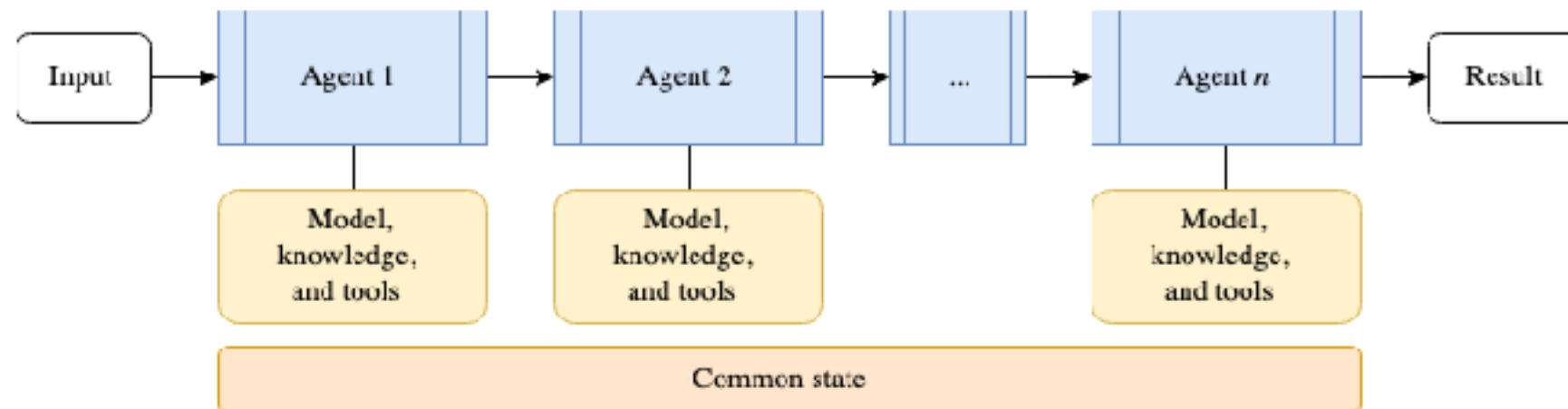
Sub-agent and Multi-agent !!

Reduce size of knowledge !!



Monolith to Multi-agent

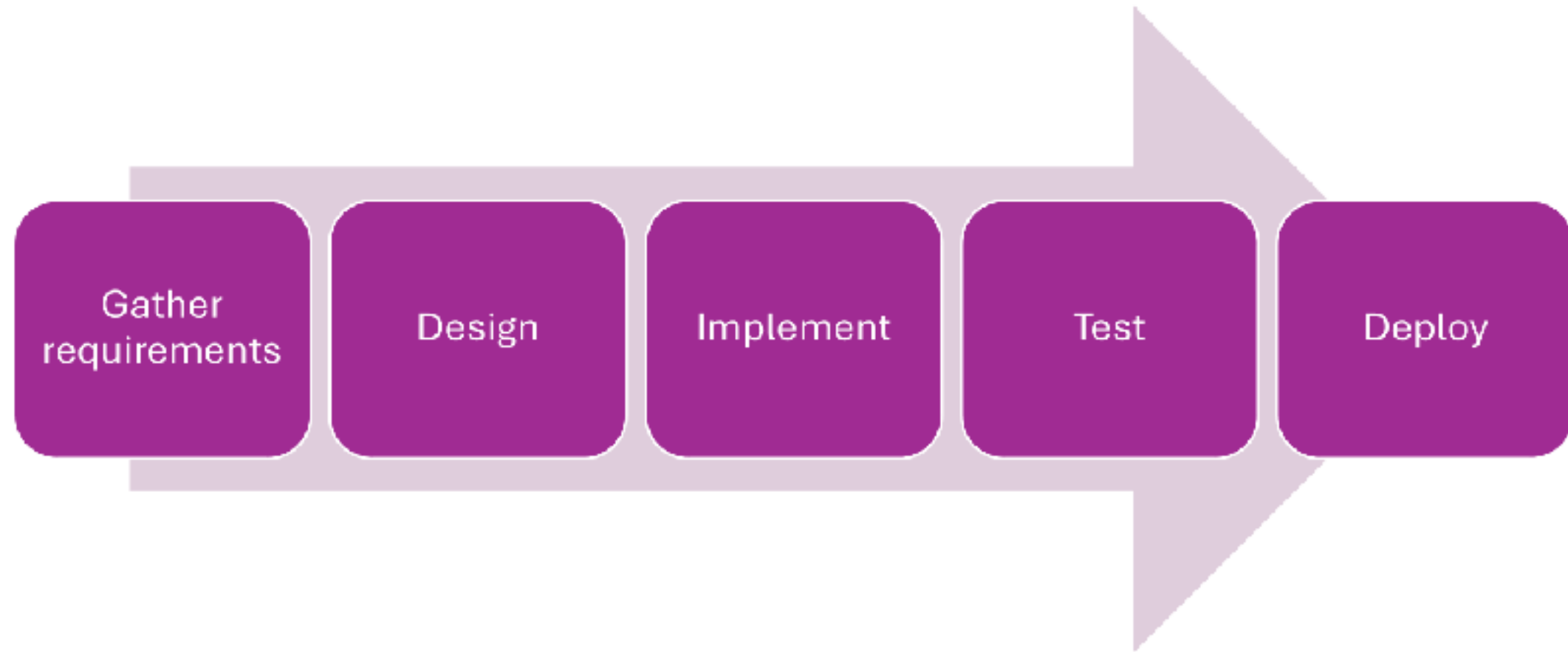
Specialization
Scalability
Maintainability
Optimization



<https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns>



Design your Agents ?



<https://devblogs.microsoft.com/dotnet/introducing-microsoft-agent-framework-preview/>



Design your Agents ?

Each step may contain subtasks
Different specialists have different steps
Progress isn't always linear

Bugs found during testing may send back to
implementation



Design your Agents ?

Full Stack Agent !!

Frontend

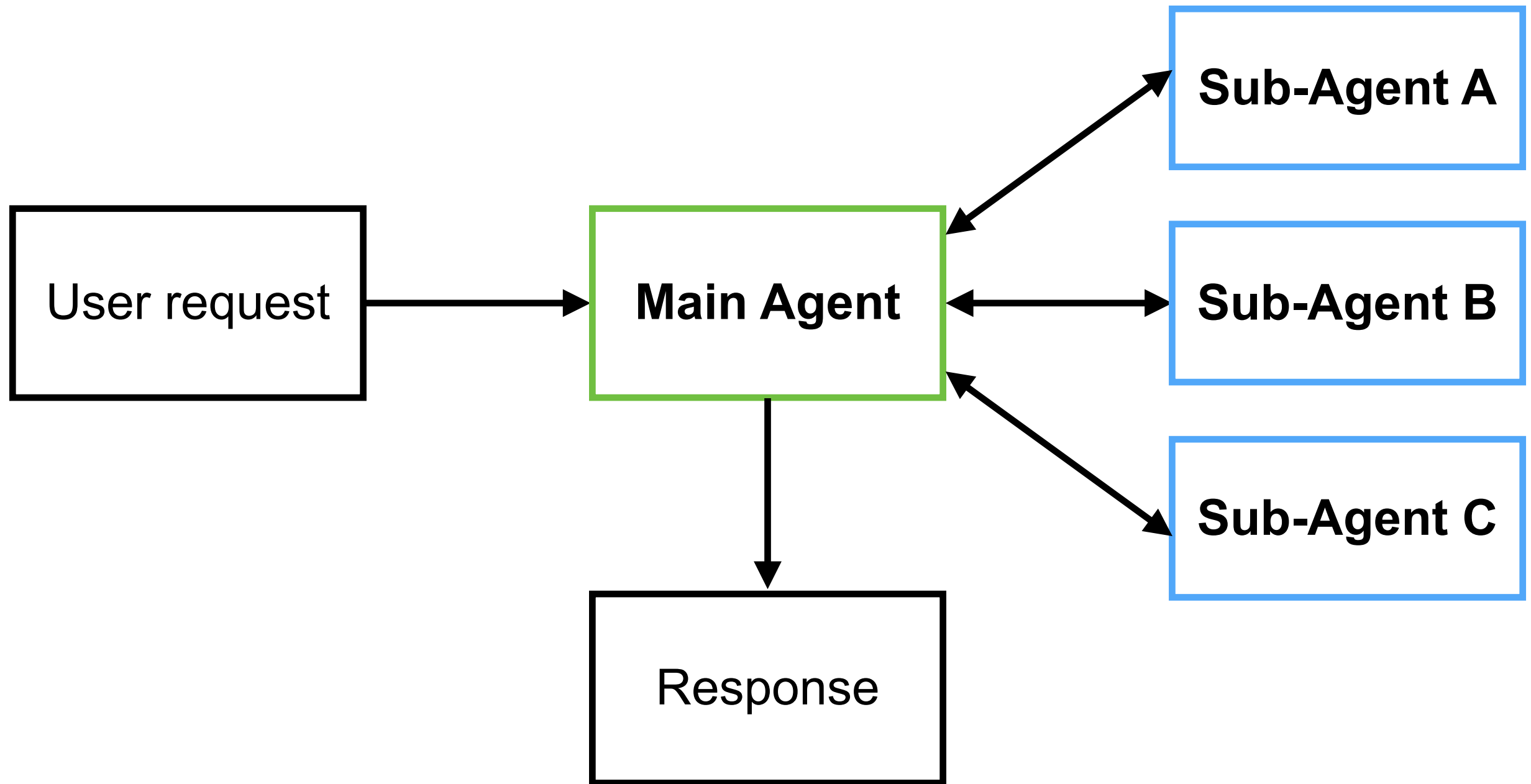
Backend

Database



Multi AI Agents !!

Sub-agent with **centralized** orchestration

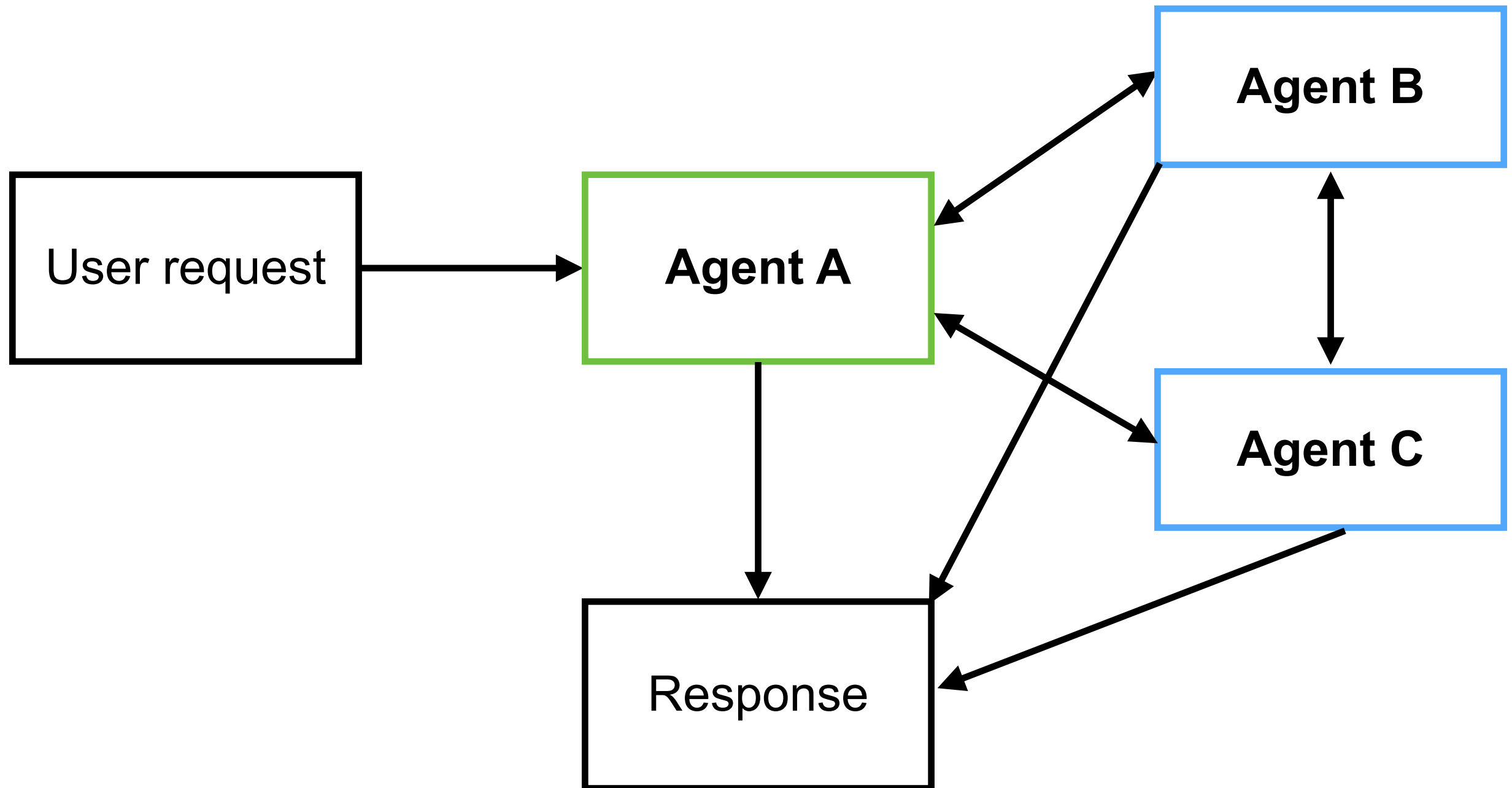


<https://www.blog.langchain.com/choosing-the-right-multi-agent-architecture/>



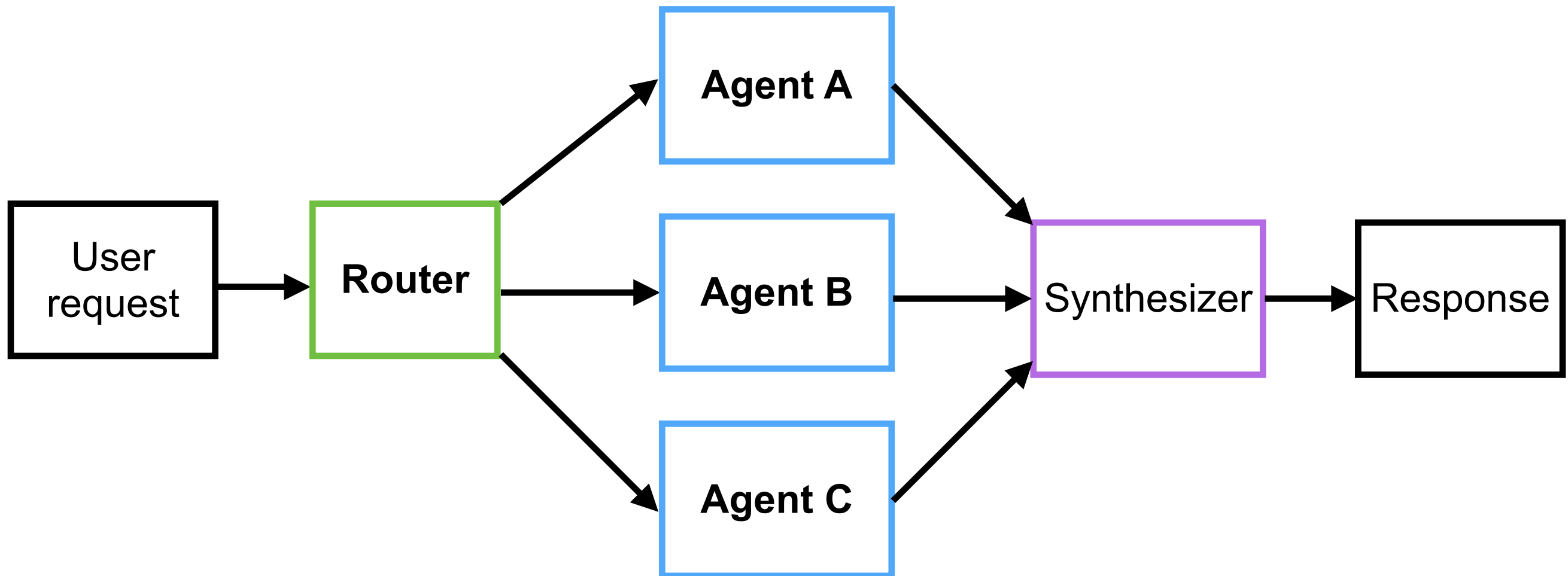
Multi AI Agents !!

State-driven transitions based-on conversation context



Multi AI Agents !!

Router and parallel run



Prompt vs Agent Skills ?



Command

<https://agentskills.io/>



Agent Skills

Moving from a paradigm of
“prompt” to “programming by instruction”

Agent loads specialized prompts and knowledge on-demand

Specialized
task

Reduce
repetition

Compose
capabilities

Efficient
loading

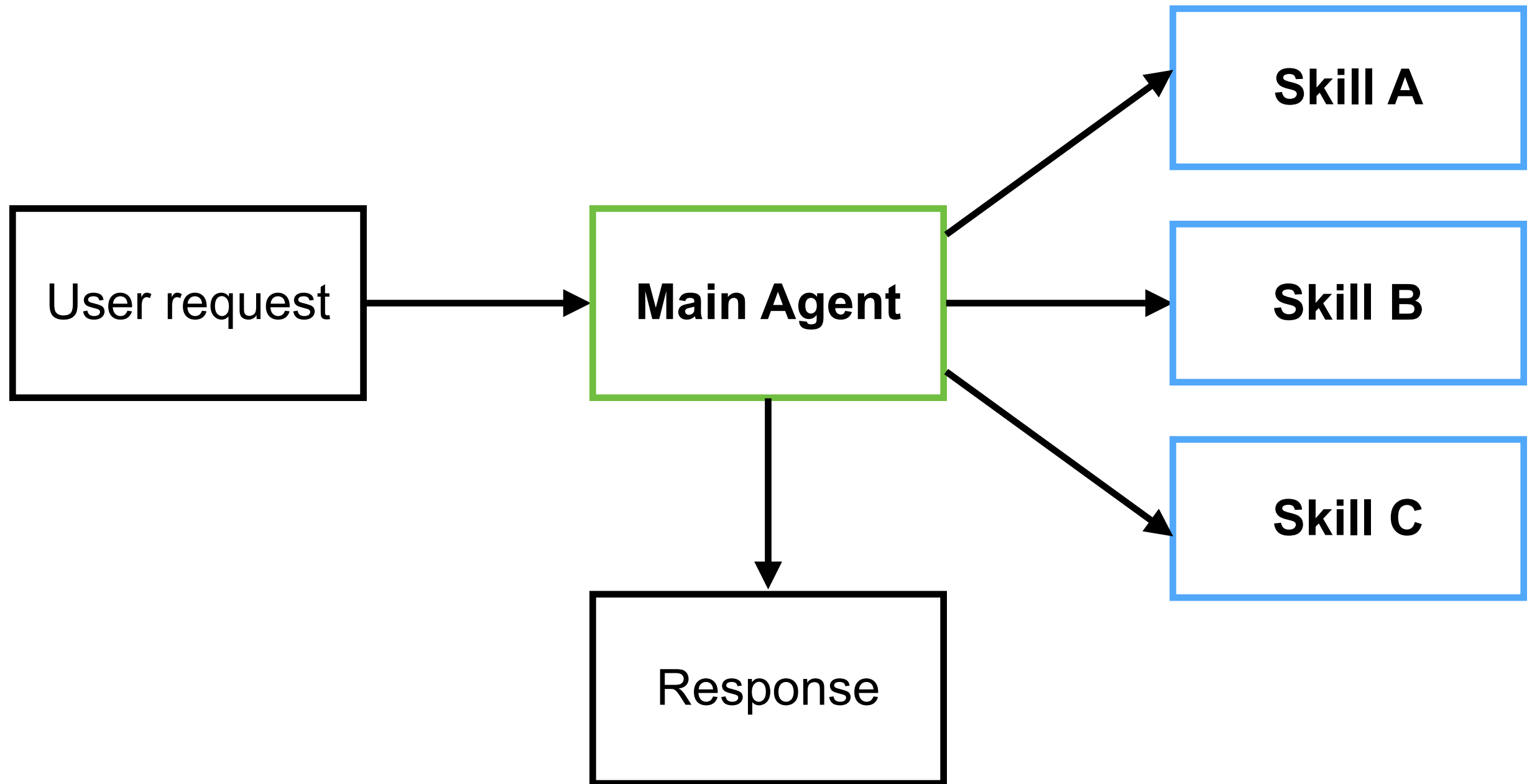
Interoperability

<https://agentskills.io/>



Agent with Skills

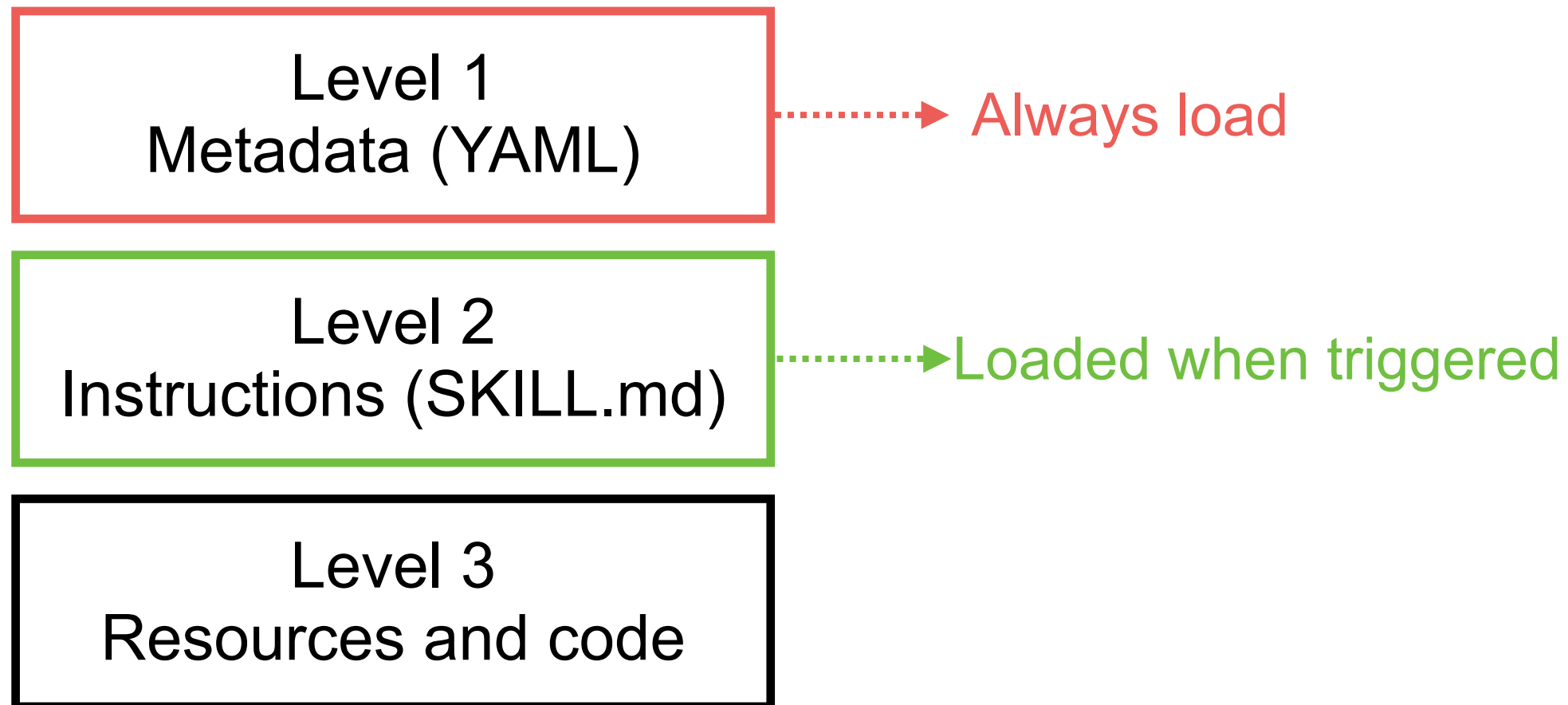
Load skills on-demand from context



<https://www.blog.langchain.com/choosing-the-right-multi-agent-architecture/>



Structure of Agent Skills



<https://agentskills.io/>



Prompt vs Skill

Features	Prompt	Agent Skills
Persistence	Ephemeral (chat-specific)	Persistent (Useable cross conversation)
Activation	Manual by user	Automatic (Triggered by context)
Structure	Unstructured text/ instruction	Structured markdown File system-based
Context load	Full instruction	Load only when needed
Token cost	Higher (Repeated usage)	Lower (Load on-demand)
Use case	One-time, creative quick query	Repeatable workflows Specialized tasks

<https://agentskills.io/>



Awesome Agent Skills

The screenshot shows the Agent-Skills.md website interface. At the top left is the logo 'Agent-Skills.md' and a 'Submit' button. The main heading is 'Find awesome Agent Skills'. Below it, a description states: 'Agent-Skills.md is a agent skills marketplace, to find the right agent skills for you.' A search bar contains the text 'c#'. Below the search bar, three skill cards are displayed:

- csharp-developer**
Use when building C# applications with .NET 8+, ASP.NET Core APIs, or Blazor webapps. Invoke for Entity Framework Core, minimal APIs, async patterns, CQRS with MediatR. Keywords: C#, .NET, ASP.NET...
Tags: c#, .net, aspnet-core, blazor
Author: Jeff Allan (24 stars, 5 forks)
- async-expert**
Expert in asynchronous programming patterns across languages (Python asyncio, JavaScript/TypeScript promises, C# async/await, Rust futures). Use for concurrent programming, event loops, async patterns...
Tags: asynchronous-programming, concurrency, performance-optimization, error-handling
Author: martinholovsky (9 stars, 2 forks)
- xUnit Test Framework**
Execute and generate xUnit tests for C#/.NET projects with FluentAssertions and Moq support.
Tags: dotnet, xunit, test-framework, moq
Author: FortiumPartners (5 stars, 1 fork)

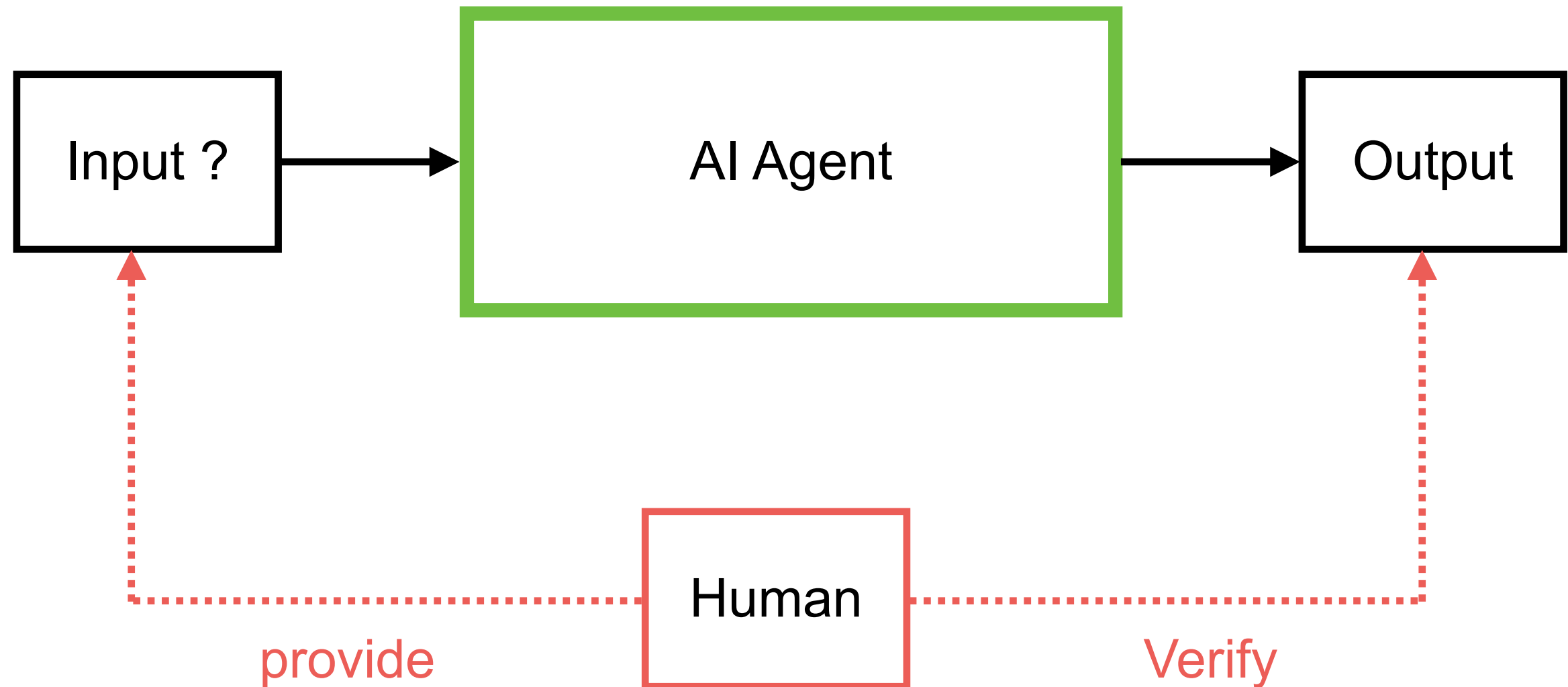
<https://agent-skills.md/>



How to provide inputs ?



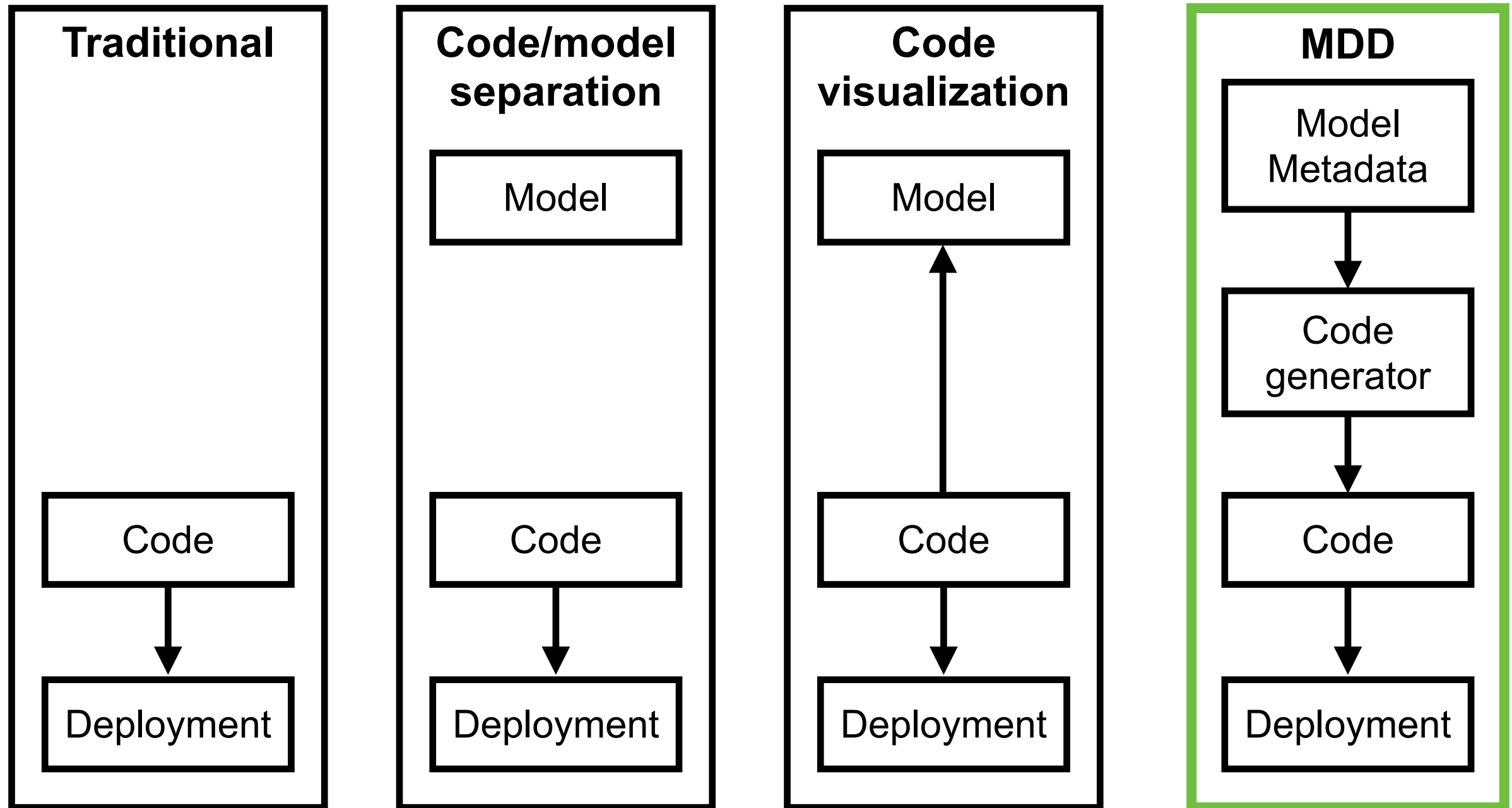
How to provide inputs ?



Model-Driven Development (MDD)



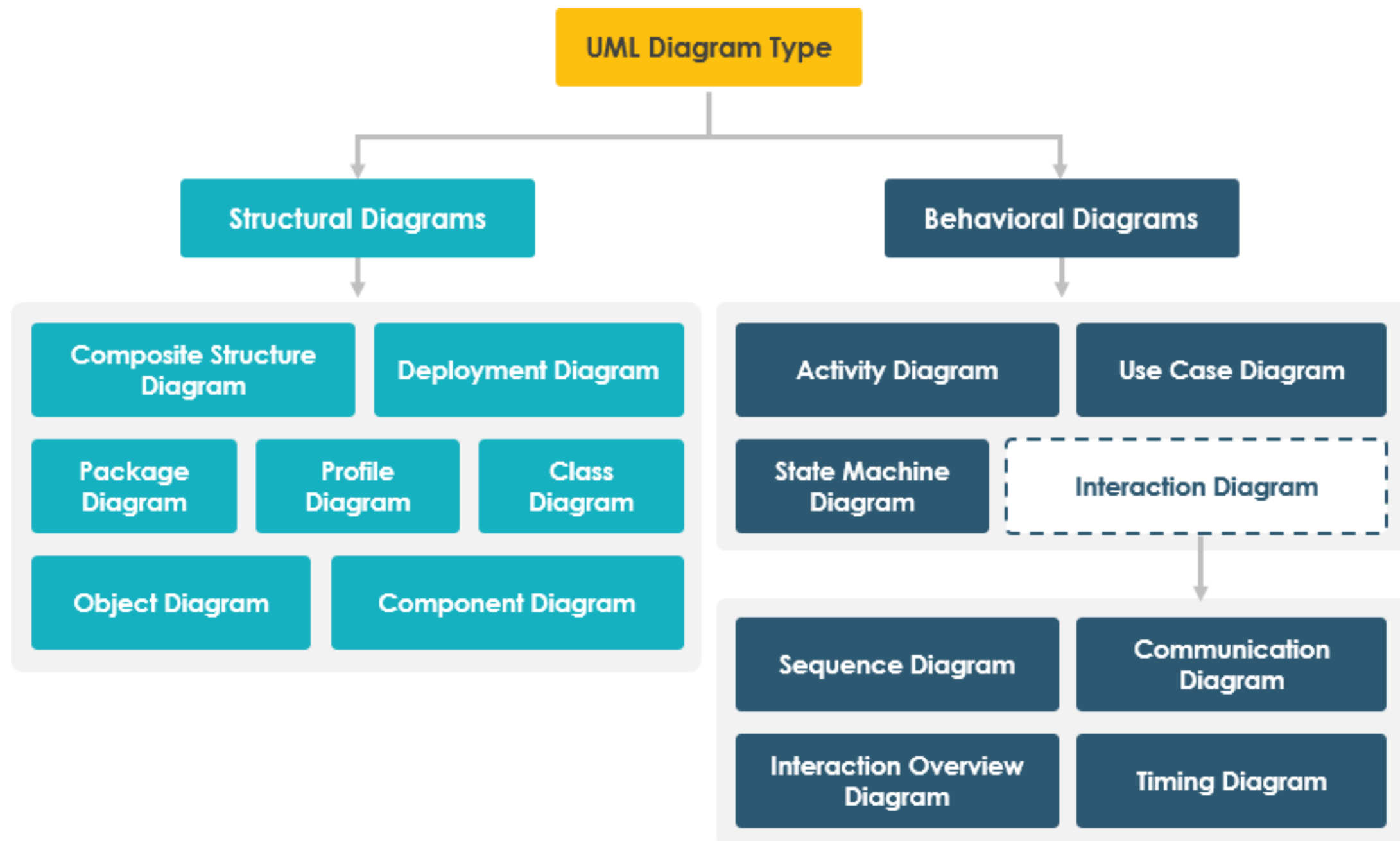
Traditional coding to MDD



<https://arxiv.org/abs/2410.18489>



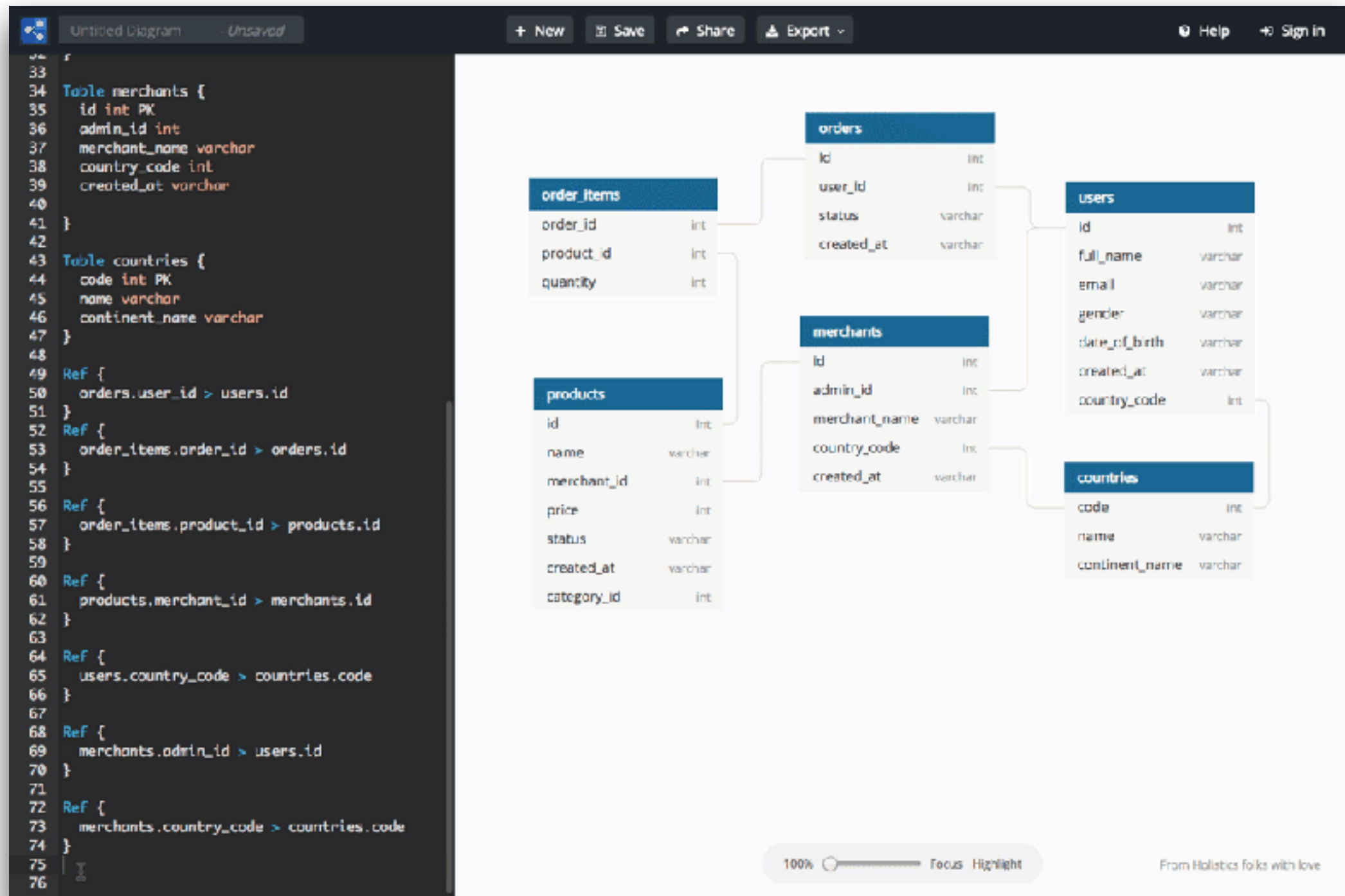
UML Diagrams !!



<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/behavior-vs-structural-diagram/>



ER Diagrams for Database !!



<https://dbml.dbdiagram.io/home/>



Mermaid Diagram

Diagram as a Code (AI-friendly)

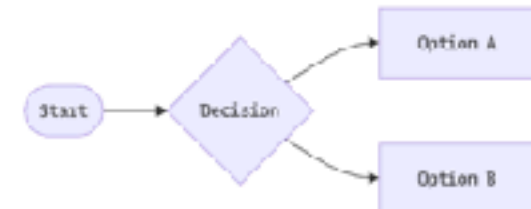
Mermaid Diagramming and charting tool

JavaScript based diagramming and charting tool that renders Markdown-inspired text definitions to create and modify diagrams dynamically.

Try Editor

Get started

```
1 flowchart LR
2   A(["Start"])
3   A --> B["Decision"]
4   B --> C["Option A"]
5   B --> D["Option B"]
```



<https://mermaid.js.org/>



Swagger or OpenAPI

Standardized specification for describe REST APIs

The screenshot displays the Swagger.io homepage. On the left, a hero section titled 'The Future of AI Relies on API Quality' features a green highlight on 'API Quality'. Below the title, text states: 'Swagger enables design, governance, and testing across the full AI-enabled API lifecycle, ensuring quality at every step.' and 'Build APIs ready for humans, LLMs, agents, and continuous innovation.' Two buttons are present: 'Find your tool' and 'Read the docs →'. At the bottom left, it says 'TRUSTED BY'. On the right, a preview of a REST API specification for a 'pet' API is shown. The header for this section is 'pet Everything about your Pets'. The API endpoints are listed in a table with color-coded rows: POST for image upload, POST for adding a new pet, PUT for updating a pet, GET for finding pets by status, GET for finding a pet by ID, POST for updating a pet with form data, and DELETE for deleting a pet. Each row includes the HTTP method, the endpoint path, a brief description, and a lock icon.

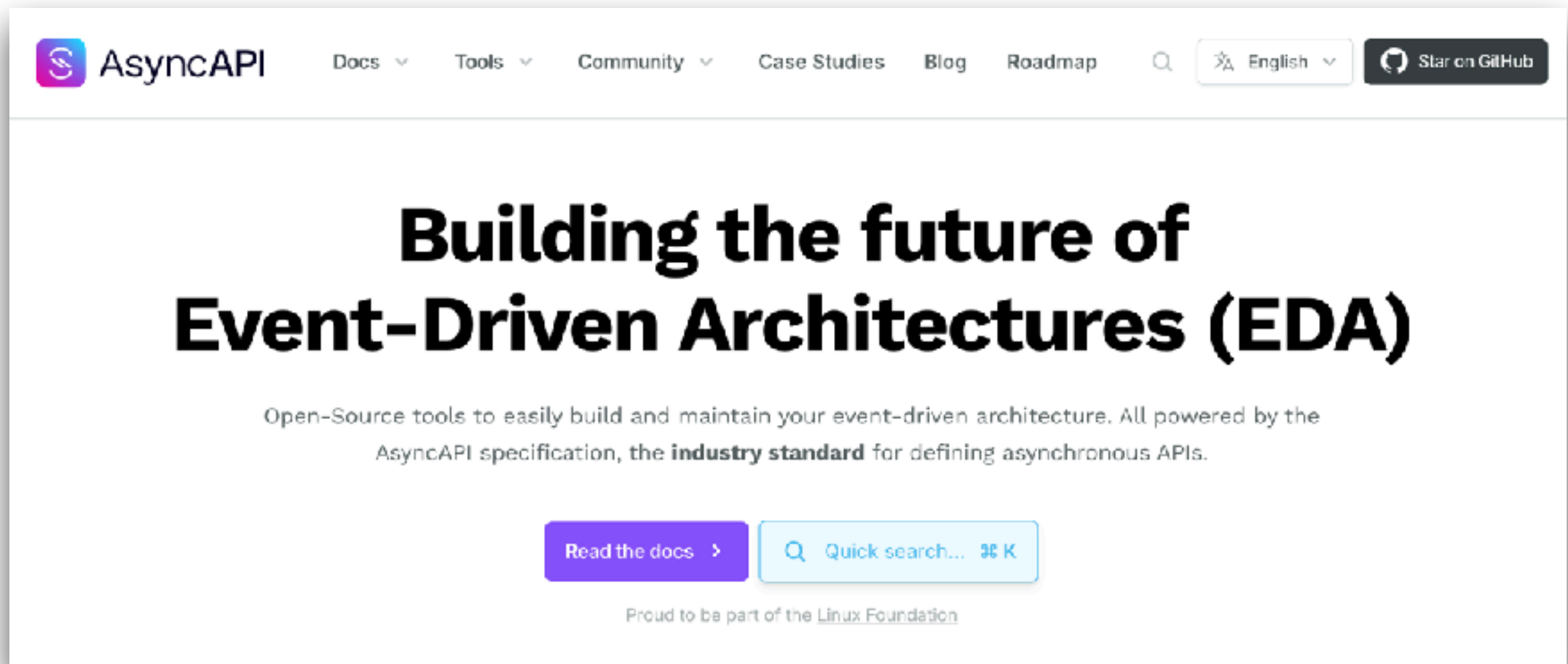
Method	Endpoint	Description
POST	/pet/{petId}/uploadImage	uploads an image
POST	/pet	Add a new pet to the store
PUT	/pet	Update an existing pet
GET	/pet/findByStatus	Finds Pets by status
GET	/pet/{petId}	Find pet by ID
POST	/pet/{petId}	Updates a pet in the store with form data
DELETE	/pet/{petId}	Deletes a pet

<https://swagger.io/>



AsyncAPI

Standardized specification for describe asynchronous and event-driven APIs (Kafka, AMQP, WebSocket)



<https://www.asyncapi.com/>



JSONSchema

Declarative language used to annotate and validate the structure, constraints and data type of JSON documents

Data validation

Documentation

Automated
testing

<https://json-schema.org/>



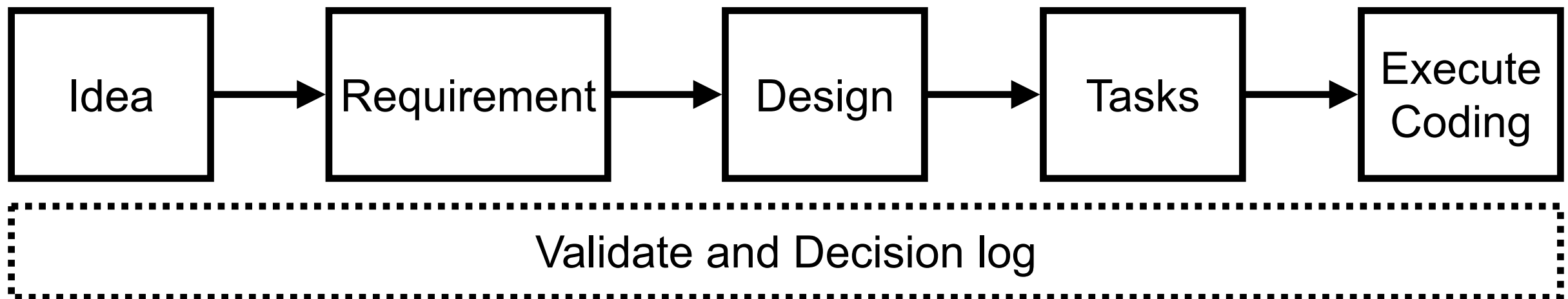
Specification-Driven Development (SDD)



SDD ?

Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code

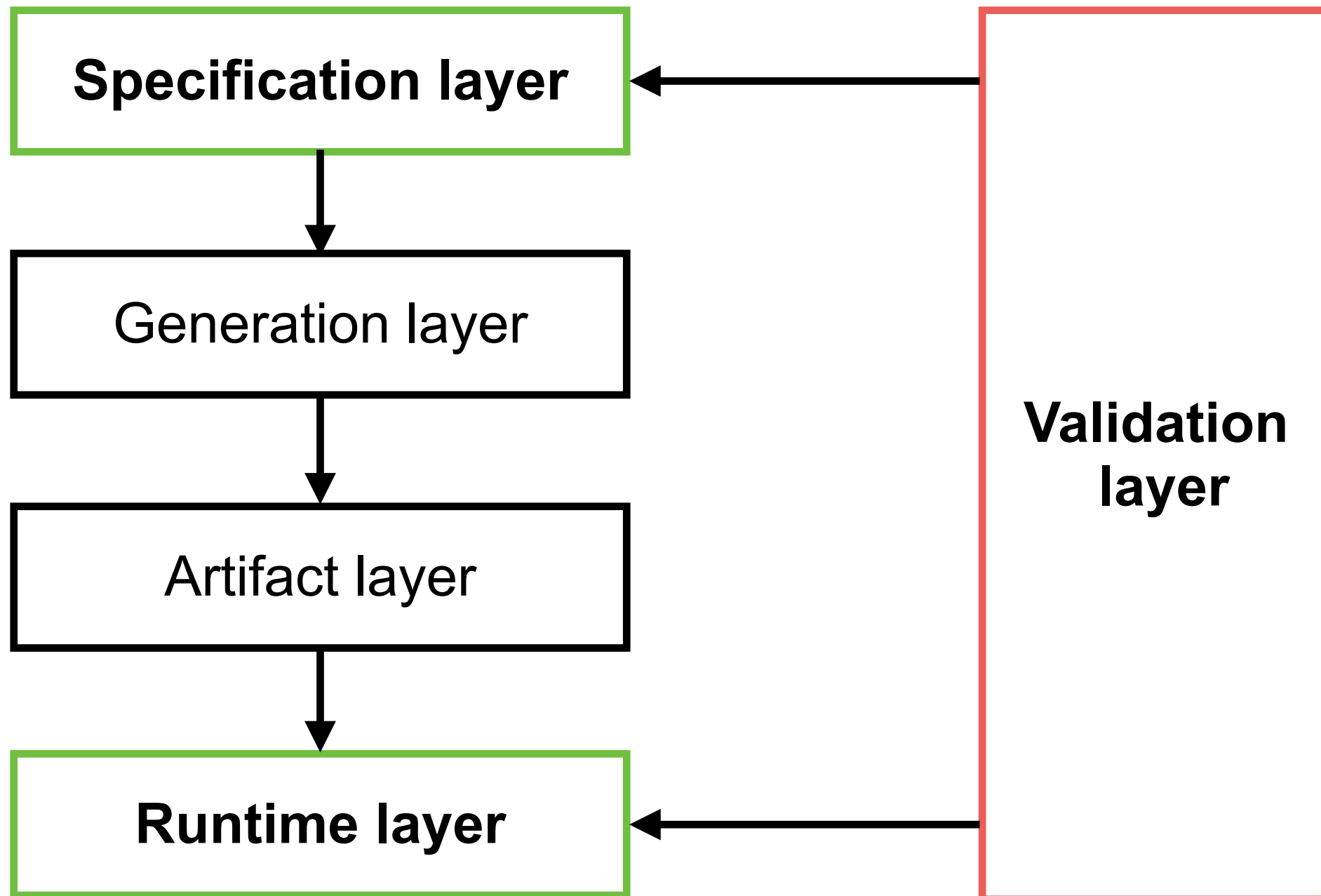


Types of Specification ?

User requirements
Design documents
Interface contracts (UI, API)
Data schemas
Validation rules
Data/business flows
Security policy and constraints
Resources and performance constraints



SSD Architecture



<https://www.infoq.com/articles/spec-driven-development>



MDD + SDD == Better result



When to use SDD ?

No one-size-fit-all !!

Low complexity

High complexity

<https://medium.com/google-cloud/benefits-and-challenges-of-spec-driven-development-and-how-antigravity-is-changing-the-game-3343a6942330>



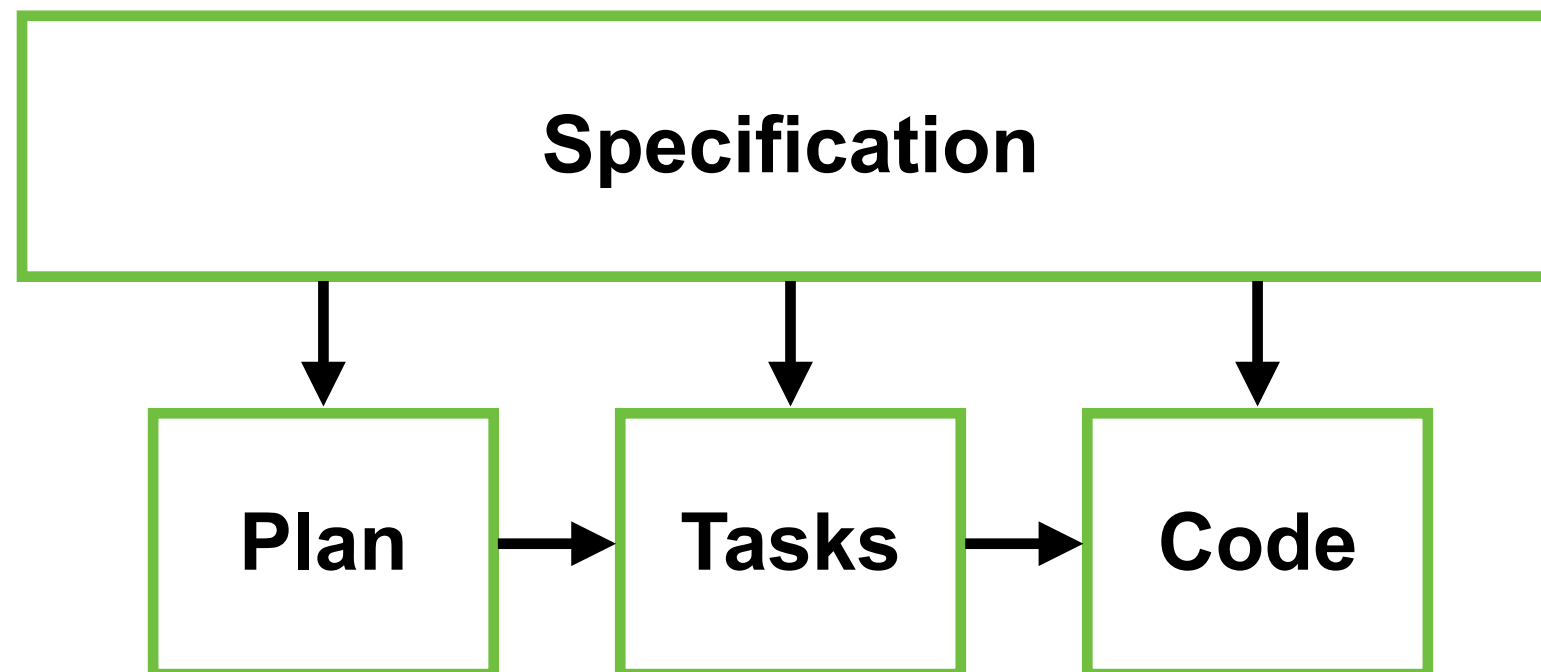
SDD Tools ?

GitHub Spec-kit

Open Spec

Kiro.dev

Tessl



Try by yourself !!





Frontend
Web application

Backend
RESTful API +
Database

Write specification

Coding

Run and Test !!



Frontend
Web application

Backend
RESTful API +
Database

Vercel React Skill

Write spec or feature

Prompt or command

Working with MCP



Frontend
Web application

Vercel React Skill

Write spec or feature

Prompt or command

Working with MCP

Backend
RESTful API +
Database

Global instruction

Write spec or feature

Prompt or command

Working with MCP

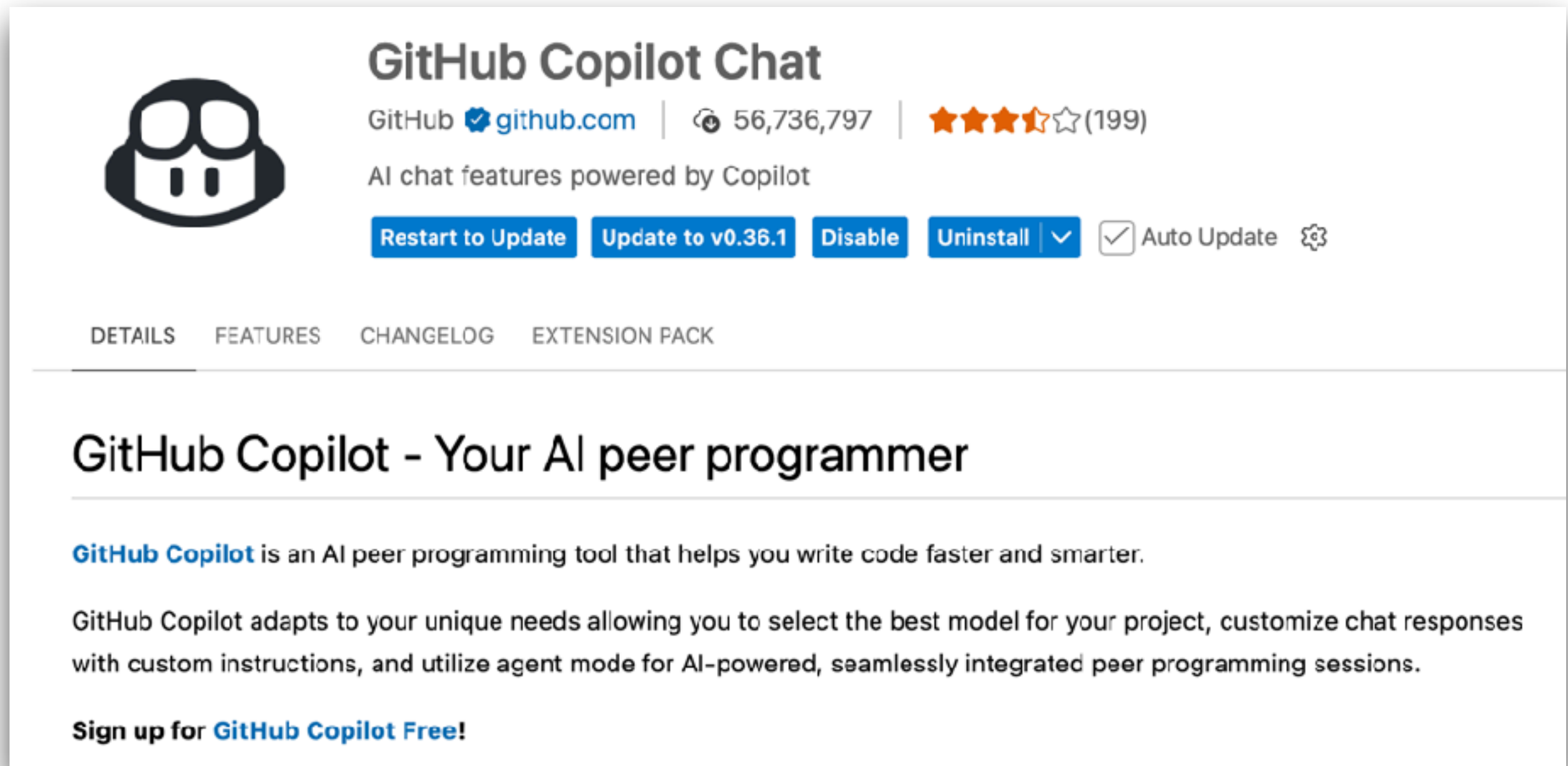


Workshop with VS Code + GitHub Copilot Chat

<https://code.visualstudio.com/docs/copilot/chat/getting-started-chat>



Install extension in VS Code



The image shows the GitHub Copilot Chat extension page in the Visual Studio Code marketplace. At the top, there's a header with the extension's name 'GitHub Copilot Chat', its publisher 'GitHub' with a verified badge and 'github.com', a download count of '56,736,797', and a star rating of '4.5 (199)'. Below this, a description states 'AI chat features powered by Copilot'. A row of buttons includes 'Restart to Update', 'Update to v0.36.1', 'Disable', 'Uninstall', and a dropdown menu. To the right of these buttons is a checked 'Auto Update' checkbox and a settings gear icon. A navigation bar below the buttons contains links for 'DETAILS', 'FEATURES', 'CHANGELOG', and 'EXTENSION PACK'. The main content area has a heading 'GitHub Copilot - Your AI peer programmer' followed by a paragraph describing the tool as an AI peer programming tool that helps write code faster and smarter. It mentions adapting to unique needs, selecting the best model, customizing chat responses, and using agent mode. At the bottom, there's a call to action: 'Sign up for GitHub Copilot Free!'.

GitHub Copilot Chat
GitHub  github.com |  56,736,797 |  (199)
AI chat features powered by Copilot

[Restart to Update](#) [Update to v0.36.1](#) [Disable](#) [Uninstall](#)  ☒ Auto Update 

[DETAILS](#) [FEATURES](#) [CHANGELOG](#) [EXTENSION PACK](#)

GitHub Copilot - Your AI peer programmer

GitHub Copilot is an AI peer programming tool that helps you write code faster and smarter.

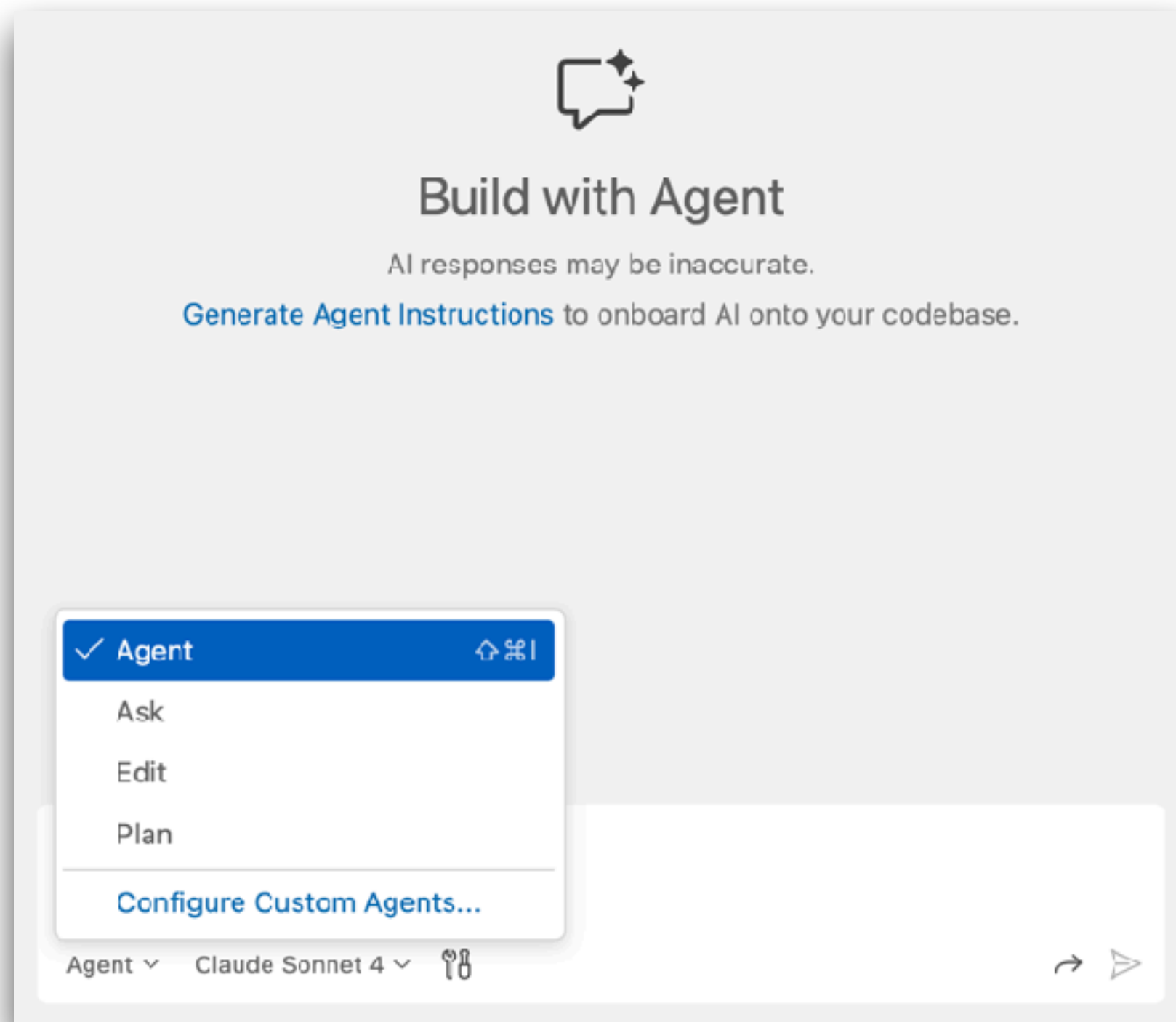
GitHub Copilot adapts to your unique needs allowing you to select the best model for your project, customize chat responses with custom instructions, and utilize agent mode for AI-powered, seamlessly integrated peer programming sessions.

Sign up for [GitHub Copilot Free!](#)

<https://marketplace.visualstudio.com/items?itemName=GitHub.copilot-chat>



Hello Copilot Chat in VS Code



Copilot Chat Modes

Agent

Ask

Edit

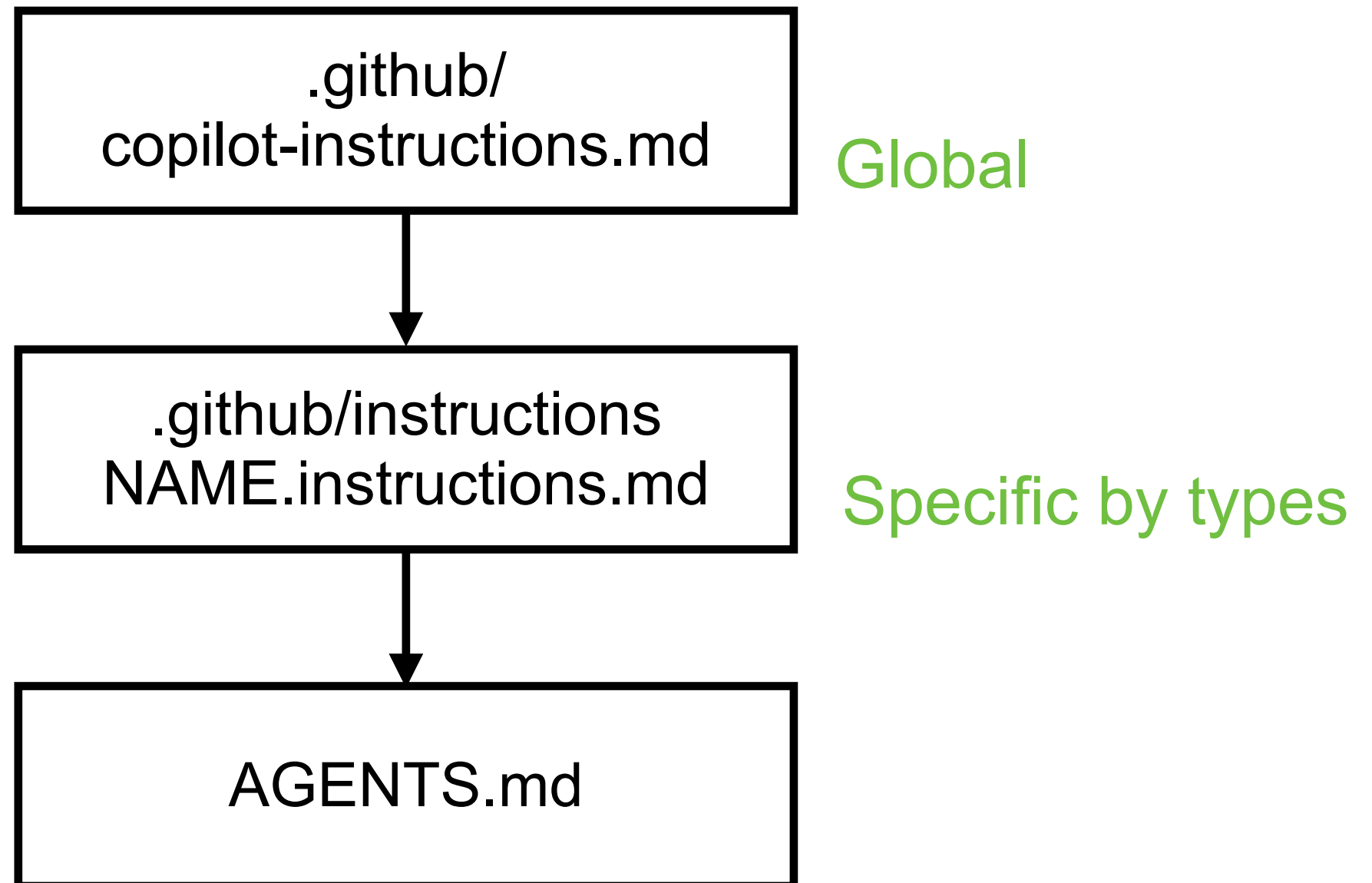
Plan

**Custom
Agent**

<https://code.visualstudio.com/docs/copilot/customization/custom-agents>



Custom instructions !!



Disable by default in VS Code

<https://docs.github.com/en/copilot/how-tos/configure-custom-instructions/add-repository-instructions>



Customization !!

Agent

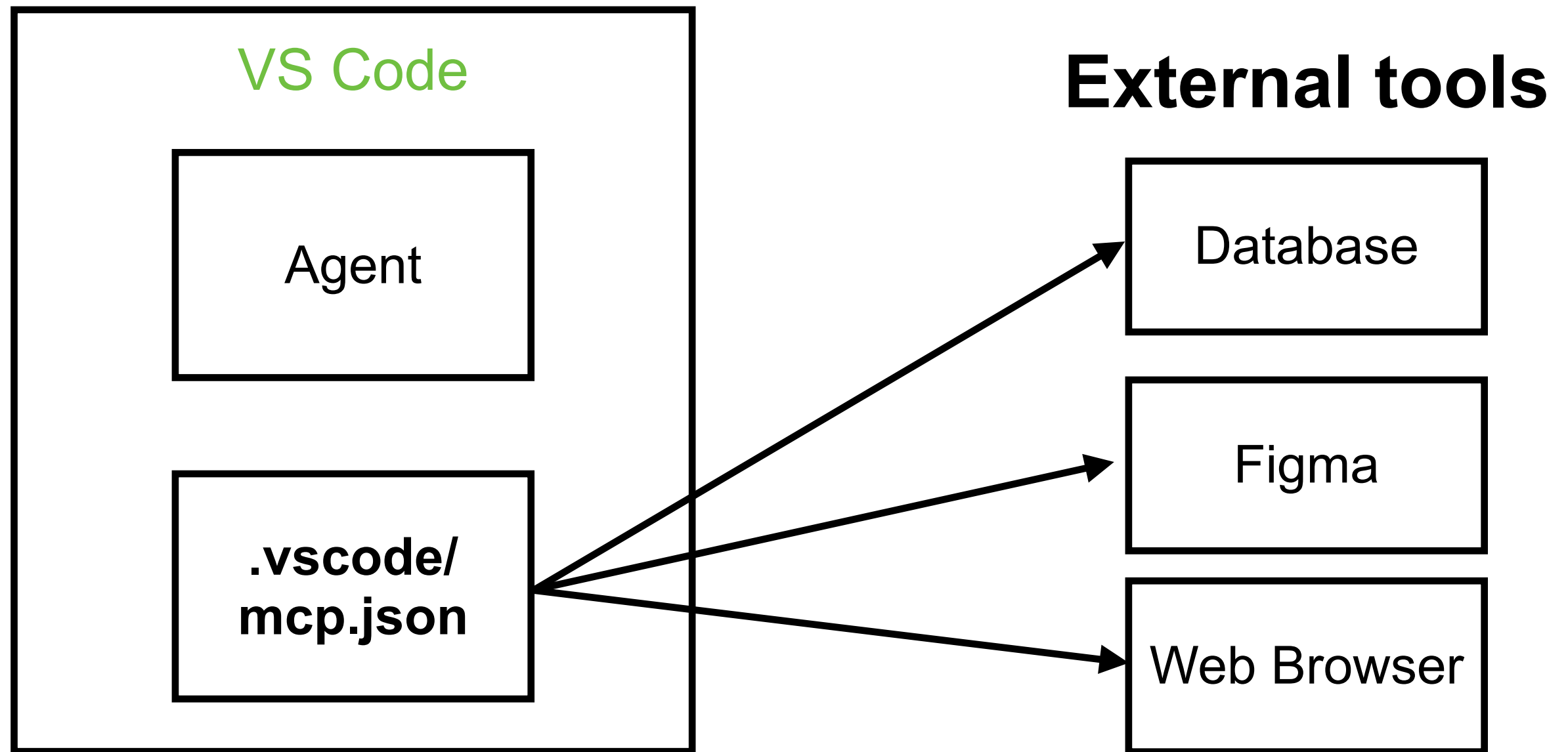
Instruction

Prompts

Skills



Working with MCP



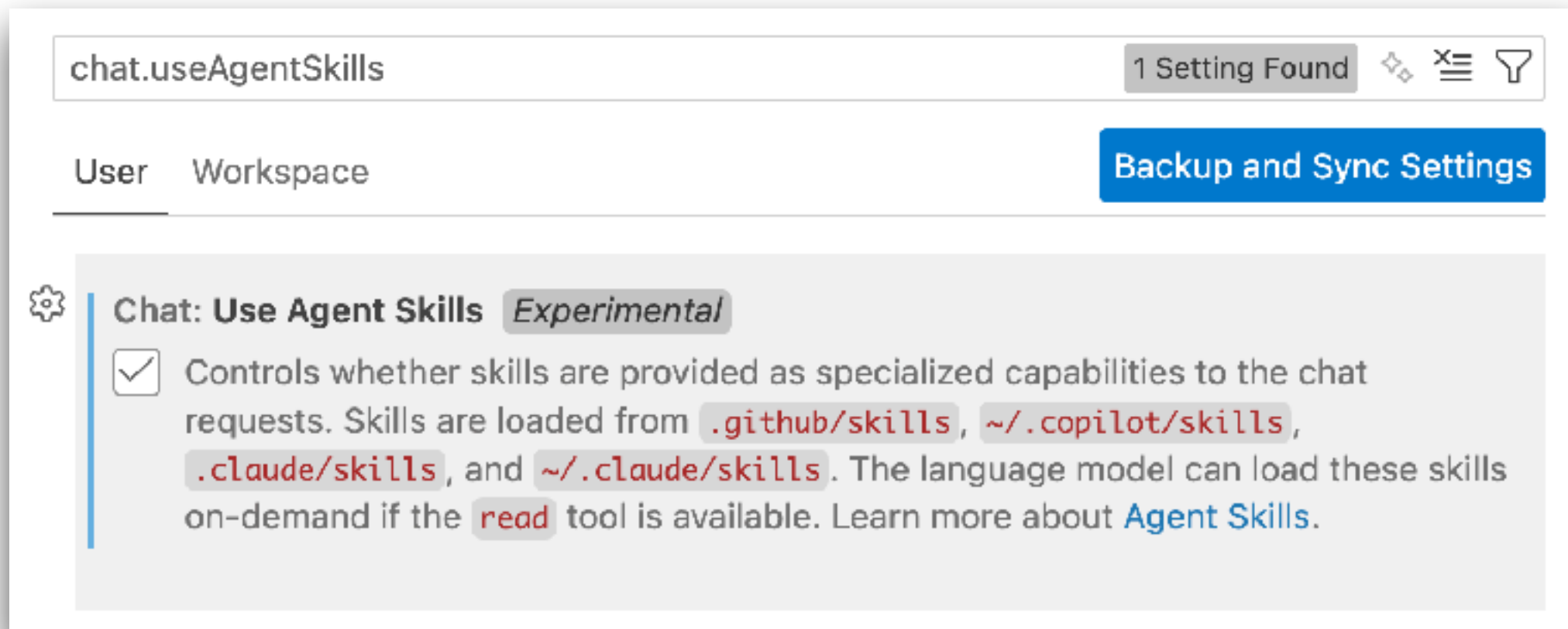
<https://code.visualstudio.com/docs/copilot/customization/custom-agents>



Enable Agent Skills in VS Code

Setting => “**chat.useAgentSkills**”

Create skills in “.github/skills/name/SKILL.md”



<https://code.visualstudio.com/docs/copilot/customization/agent-skills>



Workshop with Claude Code

<https://claude.com/product/claude-code>





`curl -fsSL https://claude.ai/install.sh | bash`



Workshop with VS Code + Cline

<https://cline.bot/>



Cline ?

Opensource AI coding agent
Development partner

Create/edit
files

Run in
terminal

Browse
website

Multi-step
tasks

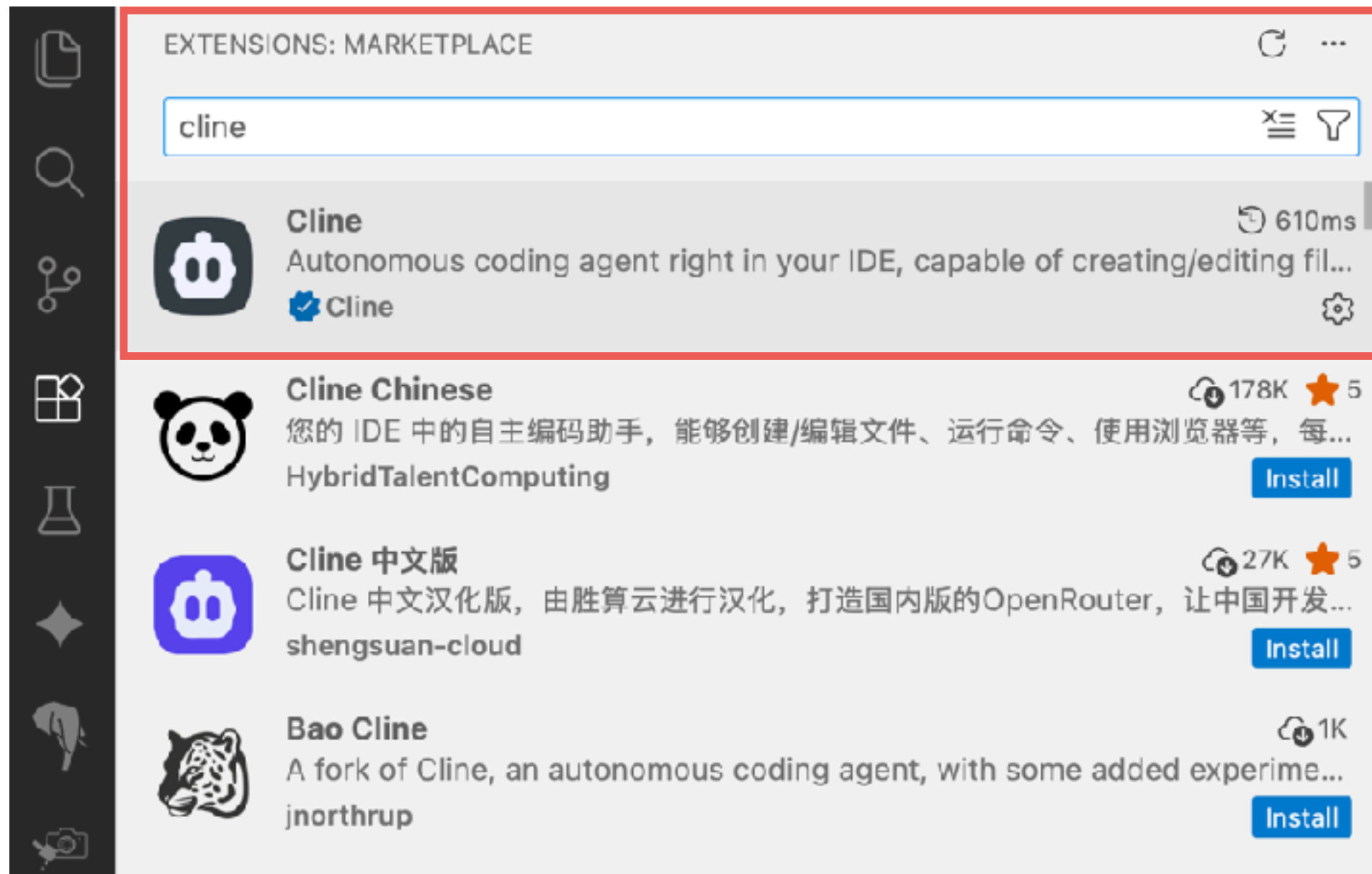
Human-in-the
loop

Support tools

<https://cline.bot/>



Install extension in VS Code



<https://marketplace.visualstudio.com/items?itemName=saoudrizwan.claude-dev>



Install extension in VS Code



Cline

Cline  [cline.bot](#) |  2,902,099 |  (264)

Autonomous coding agent right in your IDE, capable of creating/editing files, running commands,

[Disable](#)

[Uninstall](#)



Auto Update



[DETAILS](#)

[FEATURES](#)

[CHANGELOG](#)

Meet Cline, an AI assistant that can use your **CLI aNd Editor**.

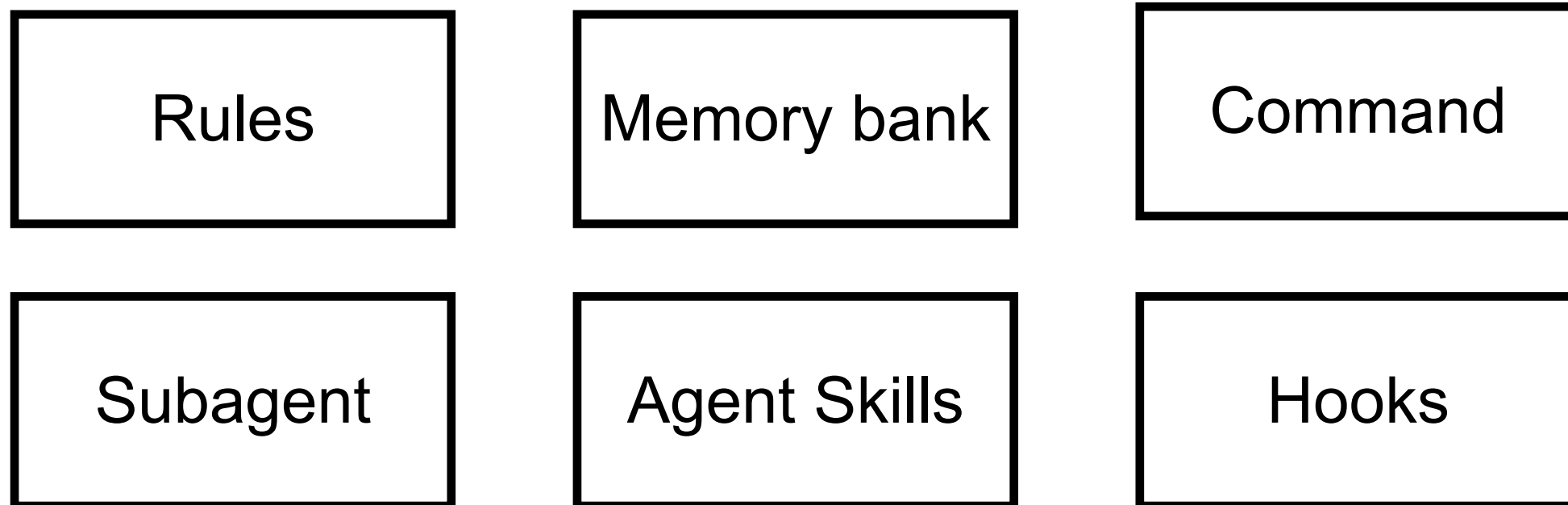
Thanks to [Claude Sonnet's agentic coding capabilities](#), Cline can handle complex software development tasks step-by-step. With tools that let him create & edit files, explore large projects, use the browser, and execute terminal commands (after you grant permission), he can assist you in ways that go beyond code completion or tech support. Cline can even use the Model Context Protocol (MCP) to create new tools and extend his own capabilities. While autonomous AI scripts traditionally run in sandboxed environments, this extension provides a human-in-the-loop GUI to approve every file change and terminal command, providing a safe and accessible way to explore the potential of agentic AI.

<https://marketplace.visualstudio.com/items?itemName=saoudrizwan.claude-dev>

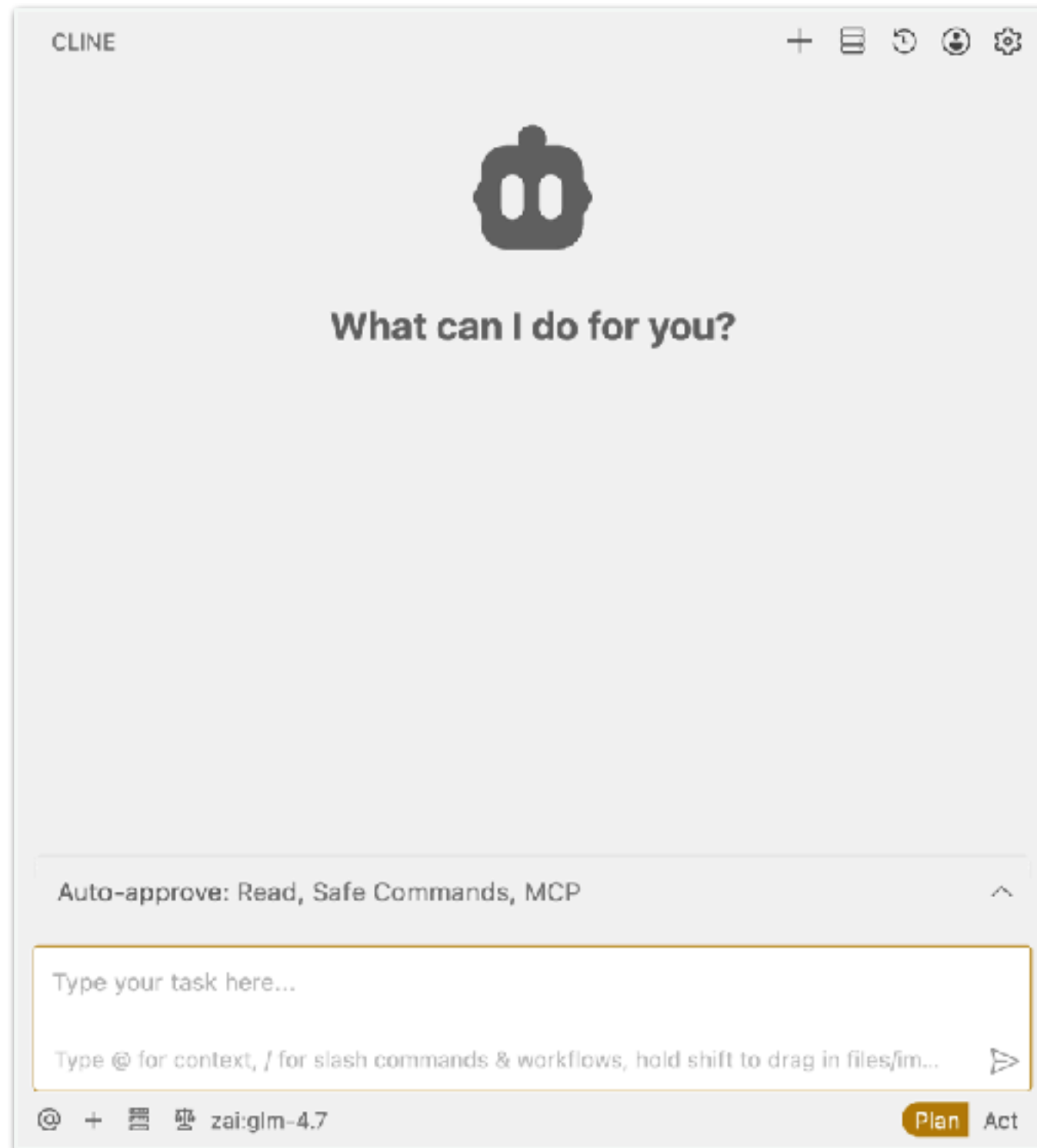


Working with Cline

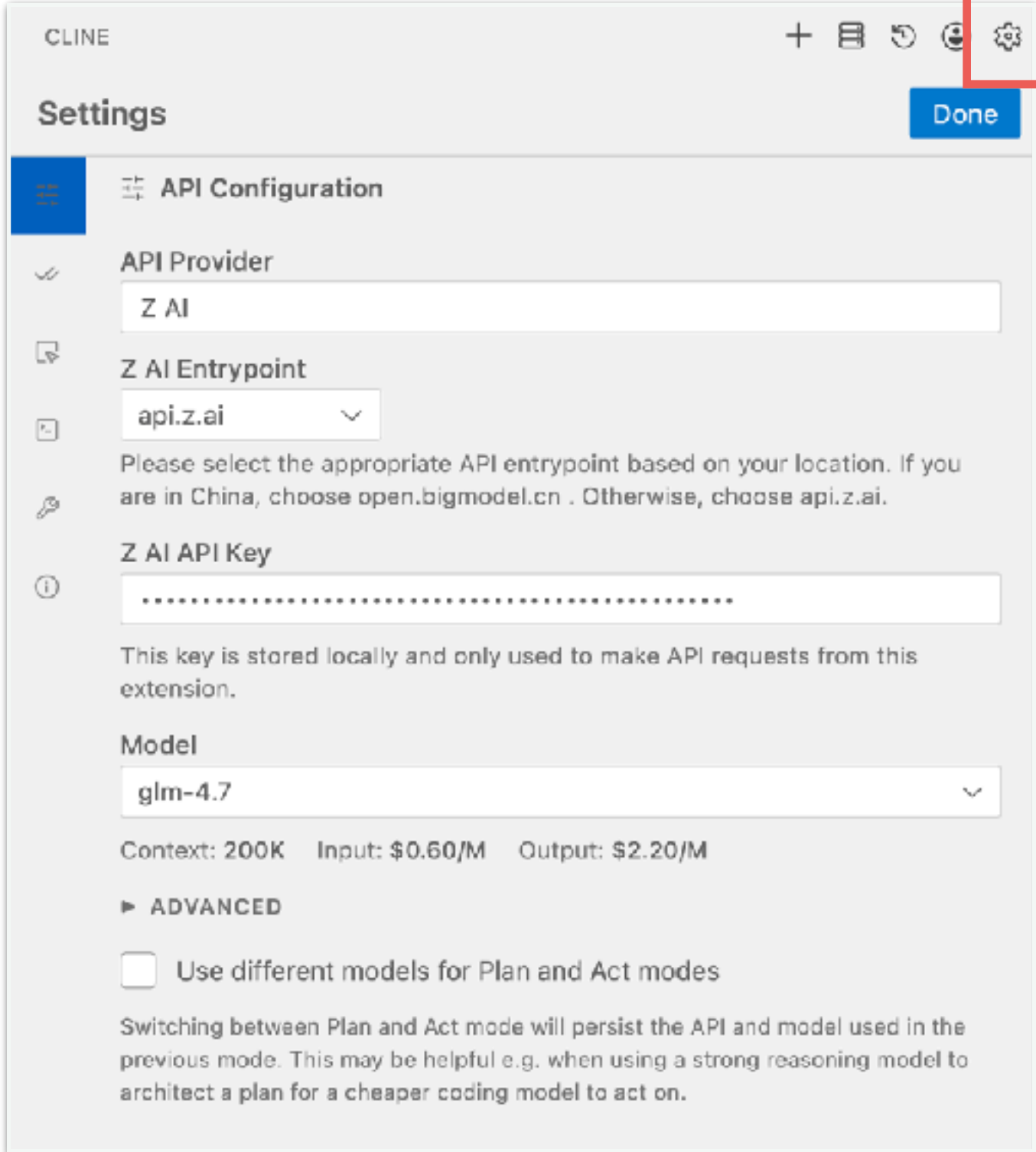
Plan and Act modes
Toggle-able rules (.clinerules)
Checkpoint management
Memory bank system



Chat with Cline



Select your model provider



CLINE

Settings

Done

API Configuration

API Provider

Z AI

Z AI Entrypoint

api.z.ai

Please select the appropriate API endpoint based on your location. If you are in China, choose open.bigmodel.cn . Otherwise, choose api.z.ai.

Z AI API Key

.....

This key is stored locally and only used to make API requests from this extension.

Model

glm-4.7

Context: 200K Input: \$0.60/M Output: \$2.20/M

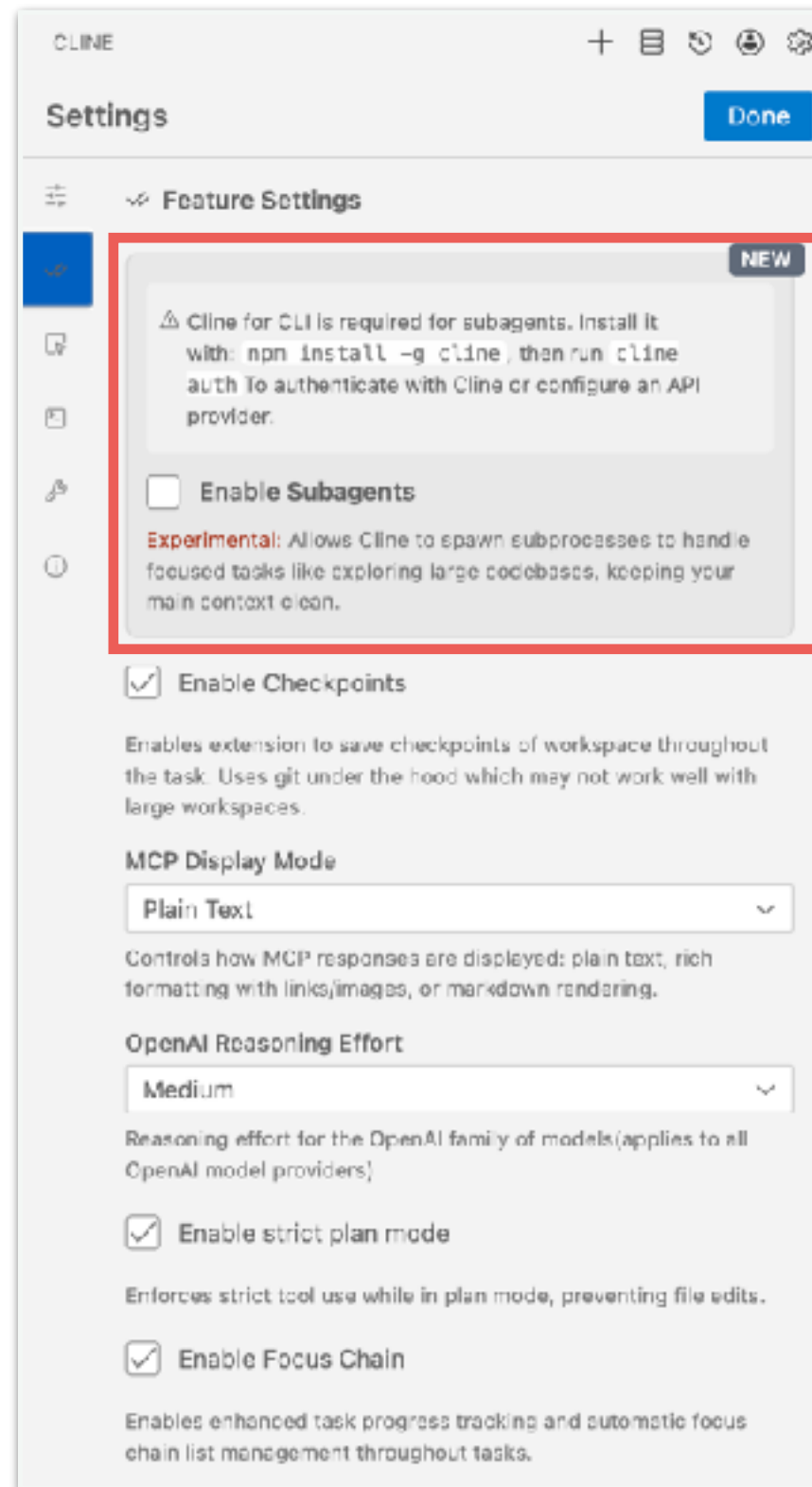
► ADVANCED

☐ Use different models for Plan and Act modes

Switching between Plan and Act mode will persist the API and model used in the previous mode. This may be helpful e.g. when using a strong reasoning model to architect a plan for a cheaper coding model to act on.



Feature Settings



Enable subagents
with Cline CLI



Cline CLI

`npm install -g cline
client --version`

```
Cline CLI  
Cline CLI Version: 1.0.10  
Cline Core Version: 3.52.0  
Commit: 66d7664  
Built: 2026-01-22T20:32:21Z  
Built by: runner  
Go version: go1.24.12  
OS/Arch: darwin/arm64
```

Support only Mac and Linux !!

<https://docs.cline.bot/cline-cli/overview>

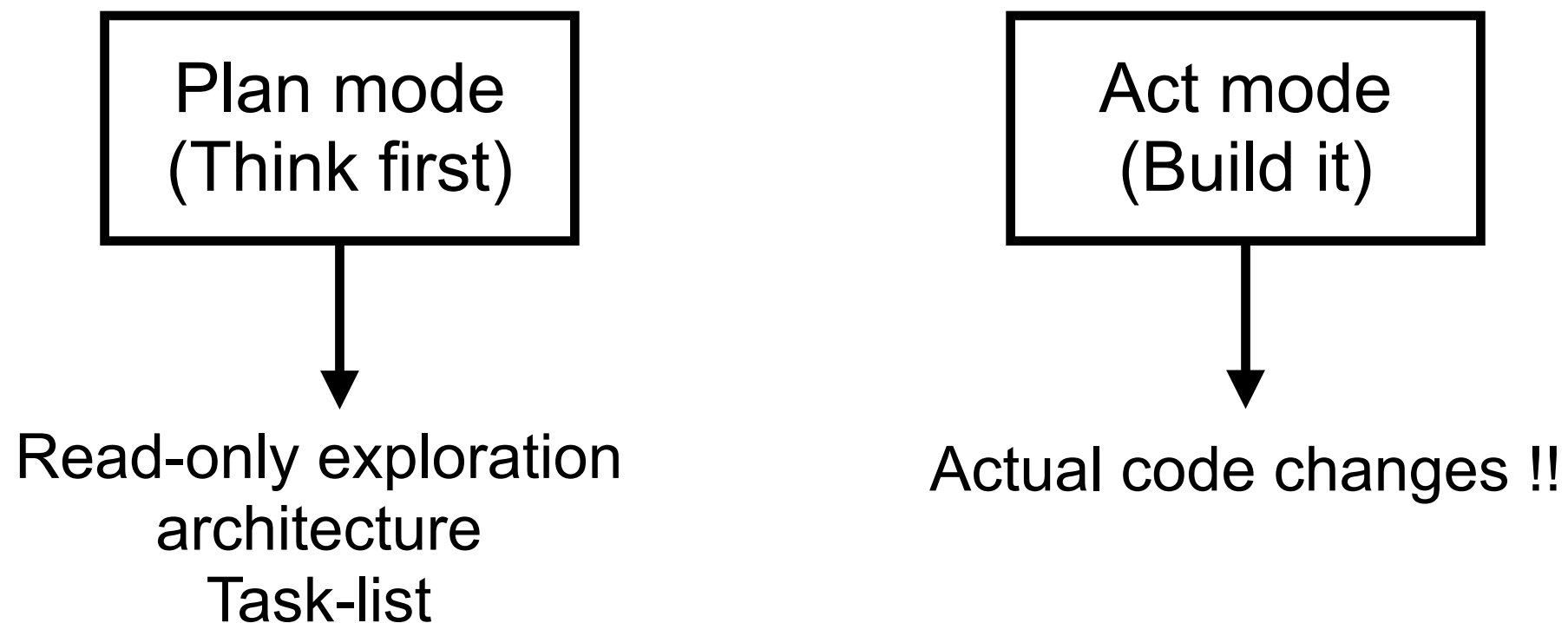


How to use Cline ?



Plan and Act mode

Separate strategic thinking from implementation

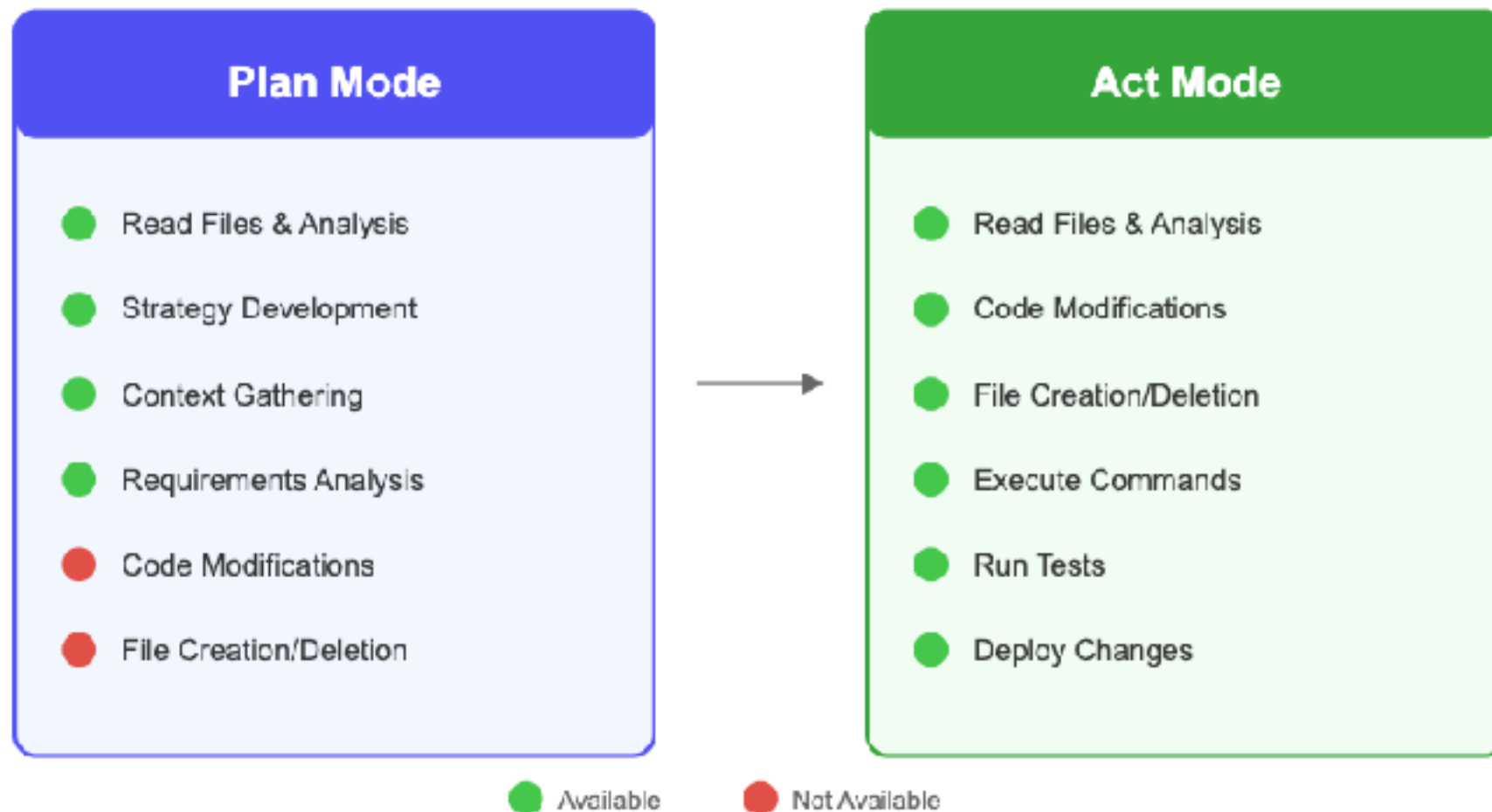


<https://docs.cline.bot/features/plan-and-act>



Plan and Act mode

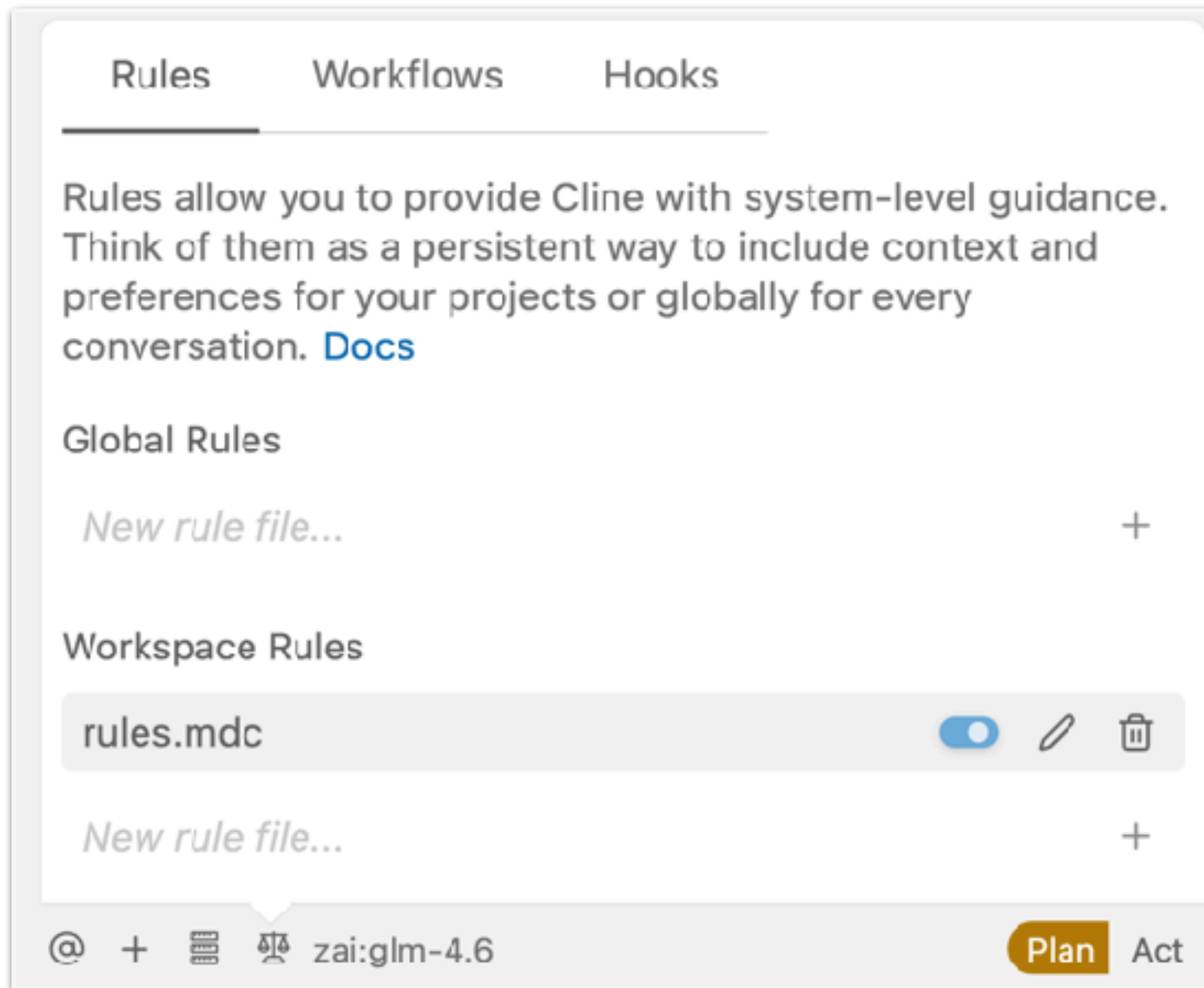
Plan vs Act Mode Capabilities



<https://docs.cline.bot/features/plan-and-act>

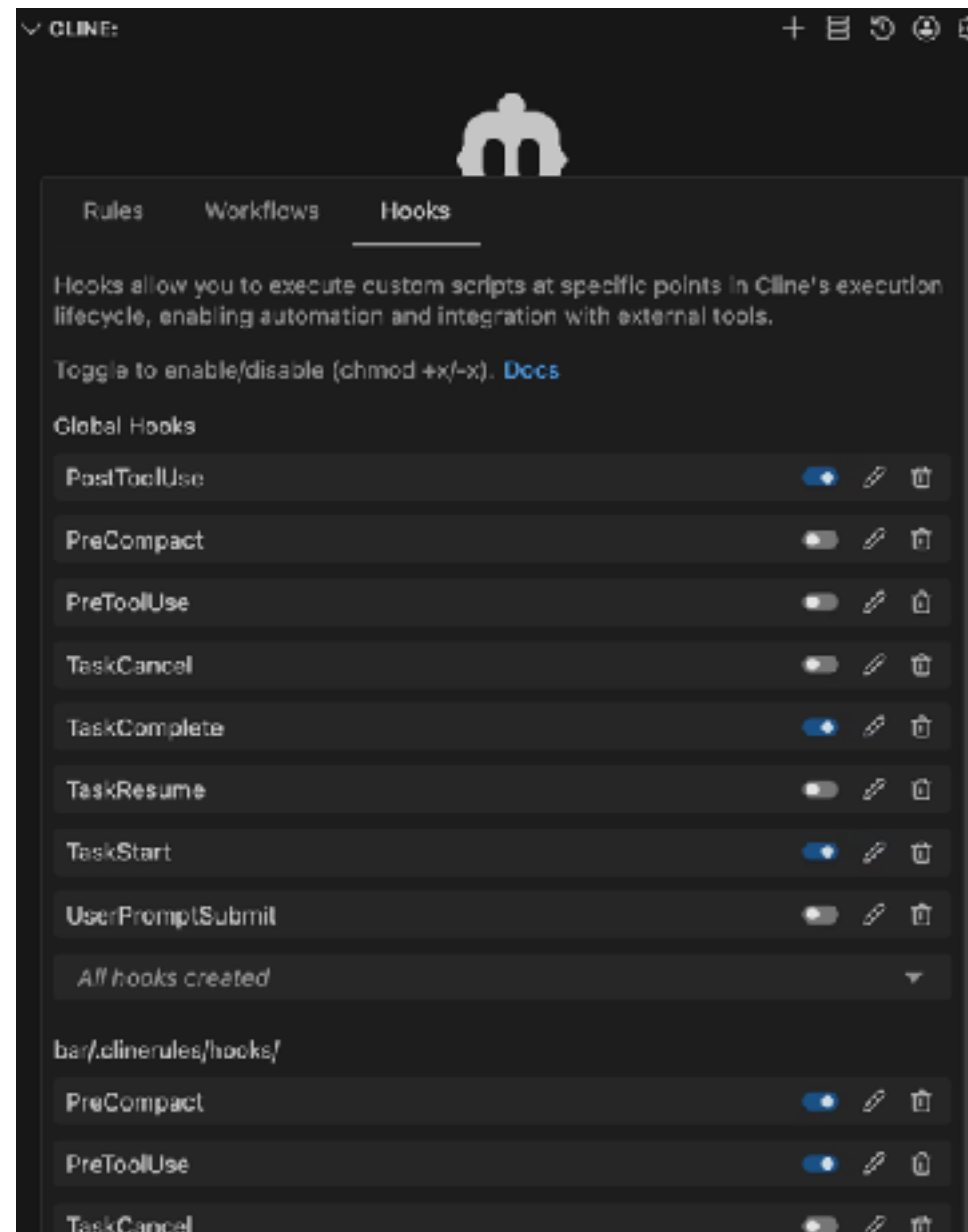


Manage rules/workflow/hooks



Hooks

Inject custom logics into Cline's workflow



<https://docs.cline.bot/features/hooks>



Auto-approve !!

Read and write files
Execute safe commands

Auto-approve: Read (all), Edit (all), All Commands, MCP

Let Cline take these actions without asking for approval. [Docs](#)

☒

 Read project files

☒

 Read all files

☒

 Edit project files

☒

 Edit all files

☒

 Execute safe commands

☒

 Execute all commands

☐

 Use the browser

☒

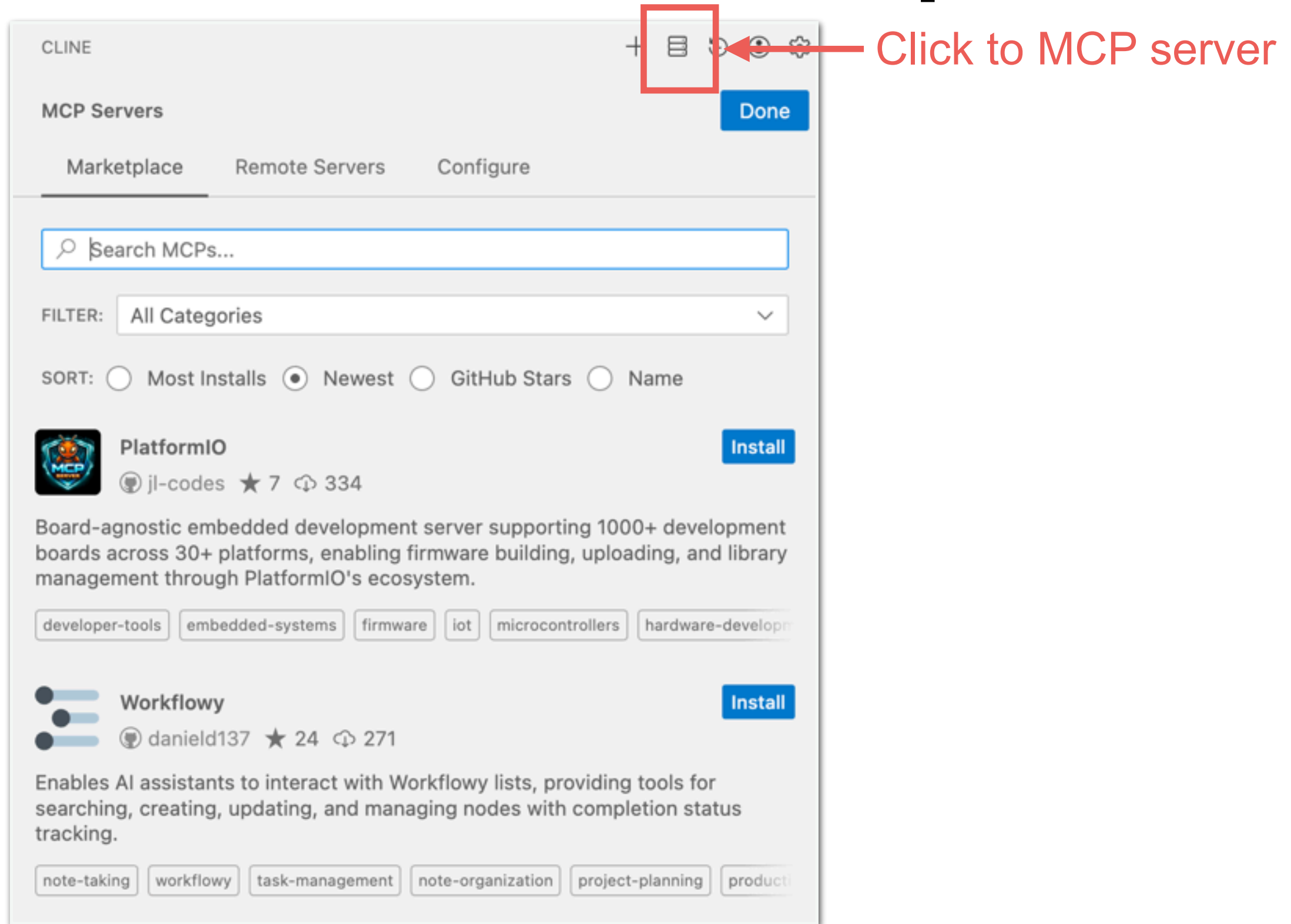
 Use MCP servers

☐

Enable notifications



Enable MCP in workspace

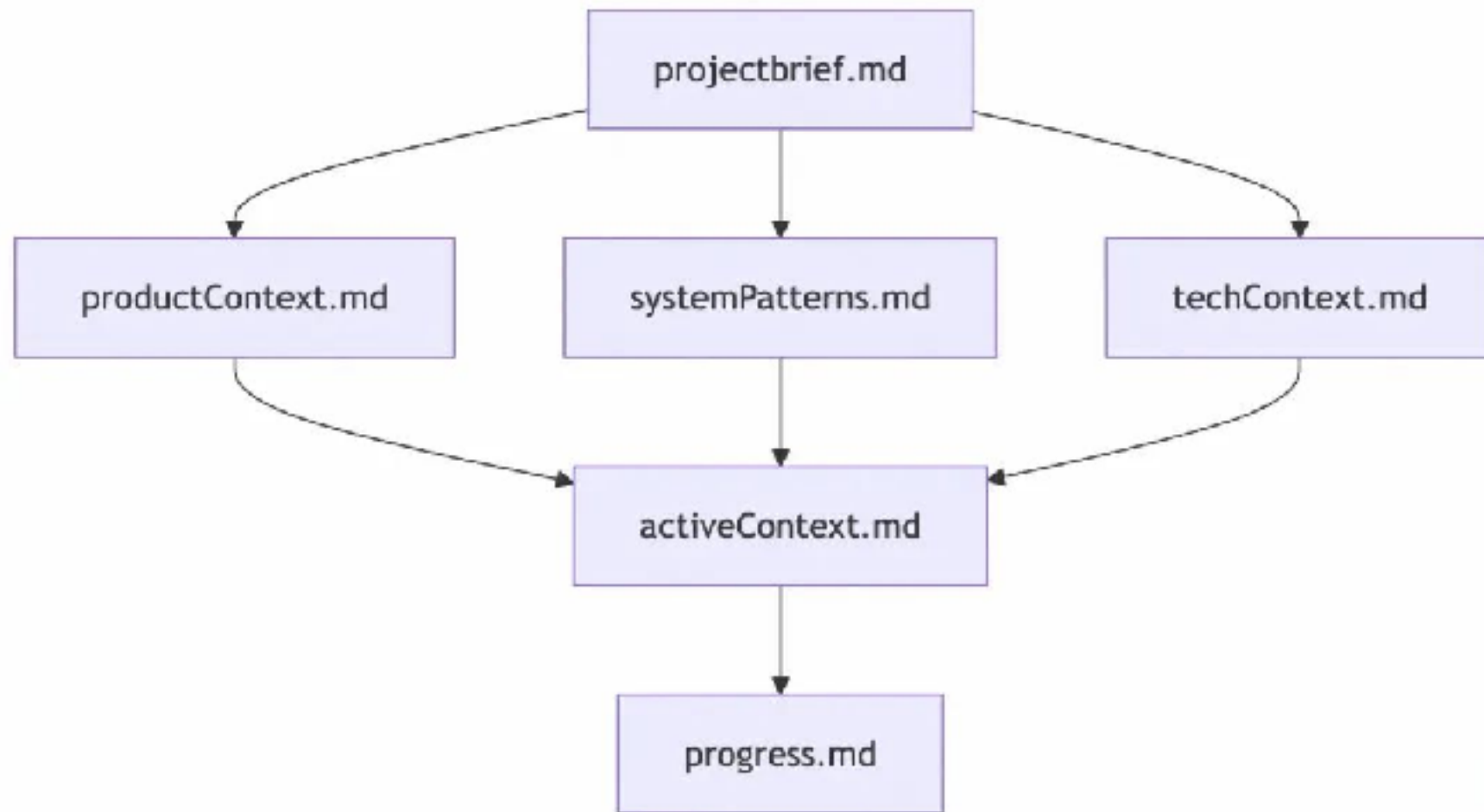


The screenshot shows the CLINE MCP Servers interface. At the top, there's a header with 'CLINE' and a menu icon (three horizontal lines) highlighted with a red box. A red arrow points to this icon with the text 'Click to MCP server'. Below the header, there's a 'MCP Servers' section with tabs for 'Marketplace', 'Remote Servers', and 'Configure'. The 'Marketplace' tab is active. It features a search bar labeled 'Search MCPs...', a filter dropdown set to 'All Categories', and sort options: 'Most Installs', 'Newest' (selected), 'GitHub Stars', and 'Name'. Two MCP servers are listed: 'PlatformIO' by 'jl-codes' (7 stars, 334 forks) and 'Workflowy' by 'danield137' (24 stars, 271 forks). Each server has an 'Install' button. The PlatformIO server description mentions it's a board-agnostic embedded development server. The Workflowy server description mentions it enables AI assistants to interact with Workflowy lists. Both servers have associated tags at the bottom.



Cline's Memory Bank

Manage context for AI across sessions
by using structured document



<https://docs.cline.bot/features/plan-and-act>



Agent Skills in Cline

☐ Enable Background Edit

Experimental: Allows editing files in background without opening the diff view in editor.

☐ Enable Multi-Root Workspace

Experimental: Allows cline to work across multiple workspaces.

☒ Enable Skills

Experimental: Enables Skills for reusable, on-demand agent instructions from `.cline/skills/` directories.

☐ Enable YOLO Mode

EXPERIMENTAL & DANGEROUS: This mode disables safety checks and user confirmations. Cline will automatically approve all actions without asking. Use with extreme caution.

Experimental feature !!



Q/A

